



Exploring the Applicability of Uncertainty Disentanglement for Medical Image Classification

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Contents

List of Figures v

List of Tables vi

List of Acronyms viii

Notation ix

Abstract 1

1 Motivation 1

2 Related Work 2

2.1 Uncertainty Disentanglement 2

2.2 Uncertainty Disentanglement in the Medical Domain 3

3 Theory of Uncertainty Disentanglement 4

3.1 Predictive Uncertainty 4

3.2 Aleatoric and Epistemic Uncertainty 5

3.3 Out-of-Distribution and Distribution-Shift Uncertainty 6

3.4 Label Ambiguity as an Aleatoric Probe 6

3.5 Label Noise as an Aleatoric-Oriented Probe 6

3.6 Data Scarcity as an Epistemic Probe 7

3.7 Uncertainty Disentanglement 7

3.8 Benchmark Evaluation of Disentanglement 8

3.9 Evaluation of Uncertainty Disentanglement 9

3.10 Relevance to Medical-Dataset Transfer 13

3.11 Synthetic Label-Noise Probe for PathMNIST 13

3.12 Evaluation Metrics 14

3.13 Experimental Axes and Uncertainty Sources 17

3.14 Calibration Versus Uncertainty 17

3.15 Implications for Medical Decision Support 18

3.16 Limitations of Disentanglement Experiments 18

4 Datasets **19**
4.1 PathMNIST . . . 19
4.2 MHIST . . . 20
4.3 Dataset Roles in the Thesis . . . 20

5 Uncertainty Methods **21**
5.1 Method Overview . . . 21
5.1.1 Taxonomy of Evaluated Methods . . . 22
5.1.2 Comparison of Method Characteristics . . . 22
5.2 Baseline and Calibration . . . 23
5.2.1 CE (Cross-Entropy) Baseline . . . 24
5.2.2 Temperature Scaling . . . 25
5.3 Prediction-Head Methods . . . 25
5.3.1 Correctness Prediction . . . 26
5.3.2 Loss Prediction . . . 27
5.3.3 Deep Correctness Prediction . . . 27
5.3.4 Deep Loss Prediction . . . 28
5.4 Ensembles and Sampling Methods . . . 28
5.4.1 Deep Ensemble . . . 29
5.4.2 Fast Deep Ensemble . . . 30
5.4.3 Shallow Ensemble . . . 30
5.4.4 MC (Monte Carlo) Dropout . . . 31
5.4.5 SWAG (Stochastic Weight Averaging-Gaussian) . . . 31
5.5 Density and Distance Methods . . . 32
5.5.1 DDU (Deep Deterministic Uncertainty) . . . 33
5.5.2 DUQ (Deterministic Uncertainty Quantification) . . . 33
5.5.3 Mahalanobis . . . 34
5.5.4 PostNet (Posterior Network) . . . 34
5.6 Bayesian and Posterior Methods . . . 35
5.6.1 Laplace . . . 36
5.6.2 SNGP (Spectral-normalized Neural Gaussian Process) . . . 37
5.7 Heteroscedastic and Evidential Methods . . . 37
5.7.1 HET (Heteroscedastic Classification) . . . 38
5.7.2 HET-XL (Extended Heteroscedastic Classification) . . . 38
5.7.3 HetClassNN (Heteroscedastic Classification Neural Network) . 39
5.7.4 Evidential Deep Learning . . . 40

6 CIFAR-10 Benchmark Reproduction **40**
6.1 Role in the Thesis . . . 40
6.2 Full-Data Reproduction . . . 41
6.3 Comparison with Published Findings . . . 42
6.4 Decomposition under Data Scarcity . . . 42
6.5 Corruption Evaluation . . . 43
6.6 Post-Hoc Results . . . 47

7 PathMNIST Experiments and Results **47**
7.1 Experimental Protocol . . . 47
7.2 Experimental Coverage . . . 48
7.3 Full-Data Training Results . . . 49
7.4 Uncertainty-Source Analysis . . . 49
7.5 Interpretation of PathMNIST Results . . . 59

8 MHIST Experiments and Results **59**
8.1 Experimental Scope . . . 60
8.2 Primary MHIST Results . . . 60
8.3 Post-Hoc and Supplementary Results . . . 62
8.4 Rater-Disagreement Analysis . . . 63
8.5 Data-Scarcity Analysis . . . 63
8.6 Integrated Disentanglement Analysis . . . 66
8.7 MHIST Findings . . . 67

9 Discussion **68**
9.1 Main Findings . . . 68
9.2 Sensitivity to the Decomposition Formula . . . 71
9.3 Comparison with Prior Work . . . 72
9.4 Implications for Medical Image Classification . . . 73
9.5 Limitations . . . 73
9.6 Outlook . . . 74

10 Conclusion **75**

A PathMNIST Method Parameters **76**

B CIFAR-10-to-MHIST Configuration Transfer **79**

C Software Availability **80**

D Selected Glossary **81**

Bibliography **83**

List of Figures

1 Experimental probes and uncertainty measures 17

2 Four training examples from PathMNIST. The samples show different tissue labels from the nine-class colorectal histology task. 19

3 MHIST sample-image panel omitted from the public-distribution edition. 20

4 CIFAR-10 scarcity response 45

5 CIFAR-10-C corruption response 46

6 PathMNIST paired component shifts 52

7 PathMNIST example-level component shifts 53

8 PathMNIST-C severity response 55

9 PathMNIST label-noise response 56

10 PathMNIST scarcity response 57

11 PathMNIST-C corruption response 58

12 MHIST rater-ambiguity response 64

13 MHIST scarcity response 65

14 Comparison of disentanglement experiments 70

List of Tables

1 Comparison of PathMNIST and MHIST for the thesis experiments. . 21

2 Taxonomy of uncertainty methods used in the PathMNIST adaptation. The grouping follows the mechanism that produces the uncertainty score, not only the final scalar metric. 22

3 Qualitative method comparison 23

4 CIFAR-10 full-data reproduction check against the archived original results. "Ref. best/final" and "R best/final" are the method-specific correctness-ranking metrics; "R acc." is reproduced test accuracy, and Δ is reproduction minus reference. Dashes indicate that no comparable reference metric was available. 41

5 CIFAR-10 data-scarcity reproduction for all rows with a like-for-like paper value. "Split" is the training-data fraction that was kept. AU and EU are information-theoretic decomposition means; Δ is reproduction minus paper, so a negative value indicates a lower reproduced mean. 43

6 Reproduced CIFAR-10C OOD-detection AUROC for full-data runs. "Total" uses total predictive uncertainty; IT EU and Br EU use the epistemic components of the information-theoretic and Bregman decompositions. Severity-2 values are averaged across corruptions at severity 2; all-severity values use every evaluated corruption and severity. 44

7 Additional CIFAR-10 post-hoc results. AU is the shared within-predictor entropy term; IT EU and Br EU denote the clean-test epistemic means from the information-theoretic and Bregman decompositions. A dash indicates that the method does not provide a measured epistemic component. OOD AUROC separates clean CIFAR-10 from CIFAR-10C, averaged across corruptions at severity 2 or across all evaluated severities. 47

8 PathMNIST experiment axes used in the thesis. 48

9 Interpretable full-data PathMNIST training results. "Best metric" is the highest method-specific validation correctness-ranking metric; "Best epoch" is the epoch at which it occurs. 49

10 PathMNIST post-hoc analysis 50

11 PathMNIST data-scarcity shifts 50

12 PathMNIST synthetic-label-noise shifts 51

13 PathMNIST-C method-level OOD AUROC 54

14 MHIST results with 100% of the training split. 61

15 MHIST results with 50% of the training split. 61

16 MHIST results with 10% of the training split. 61
17 MHIST post-hoc ambiguity results 62
18 MHIST post-hoc Bregman ambiguity results 62
19 Supplemental MHIST method results. AUROC refers to correctness-prediction AUROC using each method's uncertainty score. A dash in "Best epoch" denotes evaluation without additional training. 62
20 MHIST IT component comparison 66
21 MHIST Bregman component comparison 67
22 Cross-experiment decomposition comparison 71
23 Shared PathMNIST defaults. 76
24 PathMNIST direct-method overrides 77
25 Post-hoc and checkpoint-initialized PathMNIST method overrides. . . 78
26 Settings shared by the repeated MHIST transfer configurations. . . . 79
27 CIFAR-10 configurations transferred to the MHIST experiments. LR is learning rate, MC samples is the number of Monte Carlo samples, and SN denotes spectral normalization. Shared settings are shown in Table 26. 79

List of Acronyms

AUROC	Area under the receiver operating characteristic curve
CIFAR	Canadian Institute for Advanced Research
DDU	Deterministic Uncertainty Quantification
DUQ	Deterministic Uncertainty Quantification with radial basis functions
EDL	Evidential Deep Learning
MC	Monte Carlo
MHIST	Minimalist Histopathology Image Analysis Dataset
OOD	Out-of-distribution
SNGP	Spectral-normalized Neural Gaussian Process
SWAG	Stochastic Weight Averaging Gaussian

Notation

x	Input image.
y	Class label.
$p(y \mid x)$	Predictive class distribution for image x.
z	Logits, the raw class scores before the softmax.
$h_\theta(x)$	Feature representation produced by a network with parameters θ.
θ	Model parameters.
$\mathcal{D}_{\text{train}}$	Training dataset.
$\mathcal{D}_{\text{val}}$	Validation dataset.
$\mathcal{D}_{\text{test}}$	Test dataset.
$H[p(y \mid x)]$	Predictive entropy.
$I[y, \theta \mid x, \mathcal{D}]$	Mutual information between predictions and model parameters.
T	Temperature parameter for post-hoc calibration.

Abstract

This study evaluates uncertainty methods for medical image classification using an uncertainty-disentanglement benchmark. It asks whether uncertainty signals respond to their intended source of difficulty—label ambiguity, missing training evidence, or distribution shift—rather than only assigning high uncertainty to difficult images.

The CIFAR-10 benchmark workflow is first reproduced to check that the recovered configurations, post-hoc wrappers, corruption evaluation, and decomposition extraction behave comparably to the original benchmark (Mucsányi et al., 2024). The protocol is then transferred to PathMNIST, a nine-class medical histopathology benchmark. MHIST complements PathMNIST with image-level pathologist votes as a proxy for diagnostic ambiguity. The evaluated methods span deterministic and calibrated baselines, prediction heads, ensembles, sampling, density and distance methods, posterior approximations, and heteroscedastic or evidential models.

Several methods provide useful error-ranking or shift-detection signals, while MHIST shows only modest alignment between some aleatoric scores and rater disagreement. On PathMNIST, synthetic label noise primarily increases the component labelled aleatoric, whereas data scarcity does not produce a consistently dominant epistemic response. On PathMNIST-C, only a subset of auxiliary scores detects the corruption shift, and the decomposition scores do not respond consistently across five severity levels. Evidence for clean aleatoric–epistemic separation is therefore limited and depends on the dataset, probe, and decomposition formula. The findings support probe-specific reporting rather than a single global ranking of uncertainty methods.

1 Motivation

Medical image classification can benefit from uncertainty estimates because confident but wrong predictions may affect downstream decisions. Reviews of uncertainty quantification in medical imaging describe uncertainty as a way to make model predictions more transparent, to flag unreliable predictions, and to support clinical decision workflows rather than replacing clinical judgement (Lambert et al., 2024; Zou et al., 2023; Huang et al., 2024). A model may be uncertain because an image is ambiguous, a tissue pattern is rare in the training set, or staining, scanner, preparation, or acquisition changes move the image away from the training distribution. These cases reflect the common distinction between aleatoric uncertainty, epistemic uncertainty, and distribution shift (Kendall and Gal, 2017; Ovadia et al., 2019; Lambert et al., 2024). They also suggest different responses: expert review for an ambiguous image, more representative training data for a rare pattern, or a check of the validated use setting for a shifted image.

A scalar confidence score can flag a difficult histology patch without showing which response is appropriate. Combining several sources of difficulty in one number can

hide clinically relevant failure modes and distort method comparisons. Medical-imaging reviews likewise treat data noise, model uncertainty, distribution shift, calibration, and evaluation protocols as distinct parts of trustworthy uncertainty estimation (Lambert et al., 2024; Zou et al., 2023; Huang et al., 2024).

Mucsányi et al. (2024) frame this problem as a test of whether specialized uncertainty estimates respond to specialized uncertainty tasks. The present study transfers that idea to medical image data. PathMNIST provides a standardized histopathology benchmark and a PathMNIST-C corruption setting, while MHIST adds multiple pathologist votes per image (Yang et al., 2023; Di Salvo et al., 2024; Wei et al., 2021). Together, these datasets allow two complementary questions: whether methods react to scarcity and distribution shift, and whether their aleatoric scores relate to human label disagreement.

The research question is whether uncertainty methods that are useful for prediction also respond selectively to ambiguity, missing training evidence, and distribution shift when transferred to medical images. The benchmark implementation is first checked against recovered results and then applied to PathMNIST and MHIST, with evaluation focusing on the interpretation of each uncertainty signal rather than classifier accuracy alone. Clinical prediction reporting guidance similarly emphasises validation, intended use, and limitations (Collins et al., 2024).

2 Related Work

2.1 Uncertainty Disentanglement

A neural network can produce a confident prediction even when that prediction is unreliable. Calibration methods address one part of this problem by aligning predicted probabilities with observed frequencies. Temperature scaling is a common post-hoc example, but calibration alone does not identify the cause of uncertainty (Guo et al., 2017).

Several method families try to produce uncertainty signals that go beyond the maximum softmax probability. MC dropout treats dropout at test time as an approximate Bayesian procedure (Gal and Ghahramani, 2016). Deep ensembles compare independently trained networks and often give strong predictive uncertainty estimates (Lakshminarayanan et al., 2017). SWAG and Laplace approximations approximate uncertainty over network parameters after training (Maddox et al., 2019; Ritter et al., 2018). These methods can improve uncertainty estimation. Disentanglement asks the stricter question of whether a signal separates the source of uncertainty rather than only ranking unreliable predictions. Work on dataset shift shows why this stricter check matters: uncertainty quality can change when the input distribution changes (Ovadia et al., 2019).

The distinction between aleatoric and epistemic uncertainty gives the conceptual basis for disentanglement (Kendall and Gal, 2017). In practice, a single predictive

score can mix ambiguity, limited model knowledge, and distribution shift. Section 3.2 defines these uncertainty sources and their relationship in detail.

Mucsányi et al. (2024) formulate uncertainty disentanglement as a benchmark problem: specialized uncertainty estimates should respond to specialized uncertainty tasks. Their framework evaluates whether a method reacts to correctness errors, data scarcity, ambiguity, or out-of-distribution shift on natural-image benchmarks including CIFAR-10 and CIFAR-10-C. The medical transfer retains this controlled reference while introducing domain-specific sources of uncertainty.

Several studies also show why disentanglement should be treated as an empirical claim rather than assumed from the formula used to compute the score. Valdenegro-Toro and Saromo (2022) compare sampling-based uncertainty methods and report that aleatoric and epistemic estimates interact: aleatoric estimates can become unreliable under out-of-distribution inputs, some methods fail to express epistemic uncertainty, and ensembles give the strongest overall separation in their experiments. Wimmer et al. (2023) question the common information-theoretic split between conditional entropy and mutual information. They argue that the additive decomposition can violate desirable properties, and their computer-vision experiments support this concern. de Jong et al. (2024) propose tests that manipulate dataset size, out-of-distribution classes, and label noise while checking both uncertainty components at once. Their results show that apparent disentanglement depends on the decomposition and the evaluation setting; neither the information-theoretic approach nor Gaussian-logit variants give clean separation across all settings.

2.2 Uncertainty Disentanglement in the Medical Domain

Medical image analysis makes uncertainty estimation more than a model diagnostic. Clinical datasets can contain acquisition differences, staining variation, scanner effects, class imbalance, small subgroups, and disagreement between experts. Reviews of uncertainty quantification in medical imaging stress both the need for reliable uncertainty estimates and the difficulty of validating what an uncertainty score measures (Lambert et al., 2024; Zou et al., 2023; Huang et al., 2024).

The medical-domain literature often evaluates uncertainty through calibration, failure prediction, segmentation quality, or robustness under shift. These studies show whether uncertainty helps detect unreliable predictions, but they often stop short of establishing clean separation between aleatoric and epistemic causes. A high score on a difficult histology image could indicate expert disagreement, missing training evidence, or an input that differs from the training distribution. The experiment design must connect each score to a specific target, such as rater disagreement for label ambiguity or data scarcity for missing evidence (Mucsányi et al., 2024; Lambert et al., 2024; Zou et al., 2023).

Recent medical experiments point in the same direction. Lakara and Valdenegro-Toro (2022) study disentangled uncertainty in CycleGAN-based MRI image translation. Their out-of-distribution tests show that epistemic uncertainty is useful for

detecting Brain CT and natural-image inputs, while aleatoric uncertainty mainly reflects input noise and does not serve the same OOD role. Baur et al. (2025) adapt the benchmark idea to multi-label chest X-ray classification on MIMIC-CXR-JPG. They report method- and architecture-dependent behaviour: ensembles are the most reliable family in their comparison, while information-theoretic aleatoric and epistemic scores can fail to align with their intended tasks and should be used with caution.

PathMNIST and MHIST cover complementary parts of this problem. PathMNIST, as part of MedMNIST v2, offers a standardized histopathology classification task with fixed splits and enough images for scarcity experiments, while MedMNIST-C provides the corruption benchmark used for the PathMNIST-C shift evaluation (Yang et al., 2023; Di Salvo et al., 2024). It does not provide multiple rater votes per image. MHIST adds pathologist votes for colorectal polyp images, which makes it suitable for testing whether an uncertainty score tracks label disagreement (Wei et al., 2021). Together, the datasets connect the benchmark logic to a medical setting where expert disagreement can be observed directly.

3 Theory of Uncertainty Disentanglement

The same high uncertainty value can have different causes. A histology image may be ambiguous, differ from the training data, or lie in a region where too few examples support a stable decision boundary. Following Mucsányi et al. (2024), the comparison tests whether each method responds to the source suggested by its mechanism rather than seeking one universal score.

3.1 Predictive Uncertainty

For a classification model, predictive uncertainty describes how much doubt remains about the class label y for an input image x. The model returns a predictive distribution $p(y \mid x)$ over the possible classes. A sharp distribution puts most probability mass on one class. A diffuse distribution spreads probability across several classes. Common scalar summaries include the maximum softmax probability, predictive entropy, expected entropy, mutual information, and variance-based scores across model samples (Gal and Ghahramani, 2016; Kendall and Gal, 2017).

Predictive uncertainty can guide decisions without explaining its source. The same high-entropy prediction can arise from a noisy image, an ambiguous label, a shifted test image, or a sparse training region (Mucsányi et al., 2024; Wimmer et al., 2023). The following distinction defines the sources examined in the experiments.

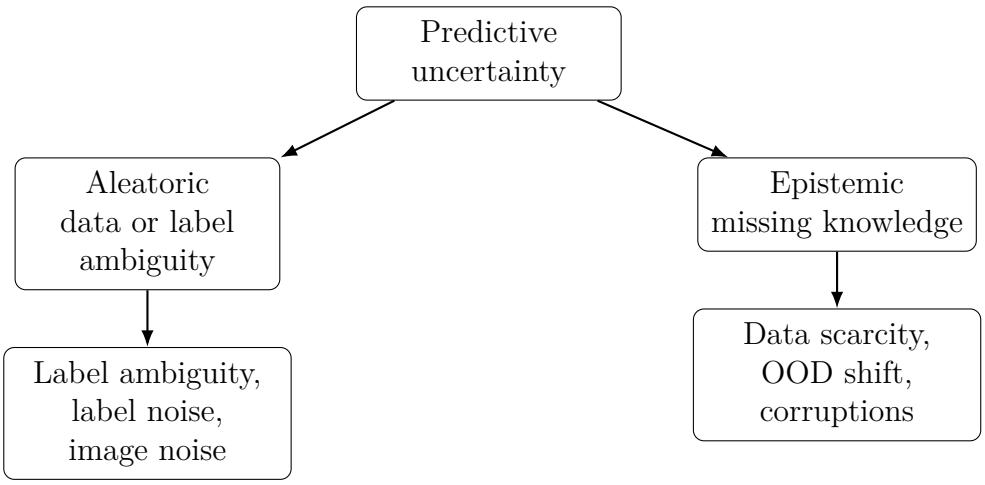
3.2 Aleatoric and Epistemic Uncertainty

The most common conceptual split is between aleatoric and epistemic uncertainty (Kendall and Gal, 2017).

Aleatoric uncertainty Aleatoric uncertainty refers to uncertainty that remains because the input–label relation is intrinsically ambiguous or noisy. It can arise from label noise, image noise, class overlap, or limited information in the image itself. In medical images, this can mean unclear tissue patterns, poor image quality, or borderline cases where the label remains difficult even for an expert. More training data cannot always remove aleatoric uncertainty because the observation or annotation process carries the ambiguity.

Epistemic uncertainty Epistemic uncertainty refers to uncertainty caused by limited model knowledge. It increases when the model has little relevant training evidence, for example for inputs far from the training distribution, rare subgroups, or small training sets. Representative training data, better coverage of the input space, or a better model can reduce this uncertainty.

The distinction depends on the experimental setting. Label noise is used here as an aleatoric-oriented proxy because the images remain fixed while the target labels become less reliable; it does not reproduce natural image ambiguity (Wimmer et al., 2023; de Jong et al., 2024). Corruptions are treated as distribution-shift probes when they move test images away from the clean training distribution, even though some real-world image degradations may also reflect aleatoric acquisition noise (Hendrycks and Dietterich, 2019; Di Salvo et al., 2024).



In Bayesian terms, aleatoric uncertainty concerns the label when the model is fixed, whereas epistemic uncertainty concerns uncertainty about the model. MC dropout, SWAG, Laplace approximations, and deep ensembles obtain several plausible predictions through sampling or independently trained models (Gal and Ghahramani,

2016; Maddox et al., 2019; Ritter et al., 2018; Lakshminarayanan et al., 2017). SNGP instead combines distance-aware features with a Gaussian-process output layer (Liu et al., 2020). Heteroscedastic methods target input-dependent aleatoric effects, while evidential methods use a Dirichlet-style evidence representation whose total evidence and class conflict can mix several sources (Collier et al., 2021; Sensoy et al., 2018). Section 5 describes the individual mechanisms.

3.3 Out-of-Distribution and Distribution-Shift Uncertainty

OOD detection is commonly treated as an epistemic or distributional uncertainty task because an OOD input is not well supported by the training distribution. Neural networks can still assign confident predictions under such shifts, so OOD evaluation tests whether an uncertainty method reacts to this missing support (Ovadia et al., 2019; Mucsányi et al., 2024).

Density and distance methods use this idea directly. They compare the test input's feature representation with the training feature distribution. If the representation lies far from known class regions, or has low density under a fitted feature-space model, the method treats the input as uncertain (Lee et al., 2018; van Amersfoort et al., 2020; Mukhoti et al., 2023; Charpentier et al., 2020). In the PathMNIST setting, corruption-based variants such as PathMNIST-C test whether methods detect image degradation as distribution shift rather than treating every difficult image as label ambiguity (Di Salvo et al., 2024).

3.4 Label Ambiguity as an Aleatoric Probe

Aleatoric uncertainty is harder to observe directly because the true ambiguity of an image is not available as a hidden variable. Multi-rater datasets provide one practical proxy: disagreement between annotators can be used as an observable target for label ambiguity (Mucsányi et al., 2024). In MHIST, disagreement between annotators is treated as a proxy for an ambiguous image-label relation: images with less agreement are taken to be more label-ambiguous (Wei et al., 2021). This probe is closest to the conceptual meaning of aleatoric uncertainty because it uses independent human judgements rather than changing the training procedure.

3.5 Label Noise as an Aleatoric-Oriented Probe

PathMNIST does not include several independent labels per image (Yang et al., 2023). Synthetic label noise is therefore used as a controlled aleatoric-oriented intervention rather than as a direct model of natural diagnostic ambiguity. Following prior disentanglement studies, a fixed fraction of training labels is replaced while the images remain unchanged (Wimmer et al., 2023; de Jong et al., 2024). The intervention tests whether aleatoric-oriented scores react to unreliable supervision; unlike MHIST rater disagreement, it does not provide an observed ambiguity target.

3.6 Data Scarcity as an Epistemic Probe

Data scarcity experiments reduce the training set while keeping validation and test sets fixed. This creates a controlled epistemic stress test, because reducing the training set removes evidence that could otherwise constrain the learned decision boundary (Mucsányi et al., 2024; de Jong et al., 2024). With fewer examples, the model has less evidence for the task. Methods that capture model uncertainty should then flag regions where the decision boundary is weakly learned.

This probe differs from OOD evaluation. OOD evaluation asks whether a trained model recognises unfamiliar inputs at test time. Data scarcity asks whether a model trained with limited evidence expresses uncertainty even on otherwise in-distribution data. Both settings are used in prior disentanglement studies because they stress different epistemic mechanisms (Mucsányi et al., 2024; de Jong et al., 2024).

3.7 Uncertainty Disentanglement

Uncertainty disentanglement separates uncertainty signals by the phenomenon they measure. The benchmark perspective asks methods to do more than assign large values to hard cases. Each method should respond most to the uncertainty type that matches its mechanism (Mucsányi et al., 2024). A density method, for example, should be strong under OOD shift, while a heteroscedastic method should help when image-dependent ambiguity dominates.

Experiments evaluate disentanglement through proxies. They create different uncertainty conditions, then compare how methods rank examples, separate test groups, or change their decomposed uncertainty scores (Mucsányi et al., 2024; de Jong et al., 2024). In this context, an "improvement" means a measurable gain on the target of the probe, not a general increase in uncertainty:

- In OOD or corruption experiments, improvement means that the uncertainty score separates shifted images from clean in-distribution images better, for example through higher OOD AUROC, lower false-positive rate at fixed recall, or stronger corruption-severity ranking. A selective gain here suggests epistemic or distributional uncertainty.
- In ambiguous-label experiments, improvement means that the aleatoric score increases with the ambiguity target, such as rater-vote entropy, minority-vote fraction, or synthetic label noise. A useful method should show a stronger relation for aleatoric uncertainty than for epistemic uncertainty.
- In data-scarcity experiments, improvement means that epistemic uncertainty rises when the training set is reduced, while aleatoric uncertainty stays more stable on the same test images. This is reported through the paired shifts ΔEU and ΔAU.

Correctness AUROC is still reported in the empirical chapters because it answers a useful but different question: do high-uncertainty examples tend to be wrong? It is used as an error-ranking and model-selection measure, not as direct evidence that aleatoric and epistemic uncertainty are separated. It supports disentanglement only indirectly, for example by establishing that a score ranks errors before asking whether the same score responds specifically to ambiguity, scarcity, or distribution shift (Mucsányi et al., 2024).

3.8 Benchmark Evaluation of Disentanglement

Mucsanyi et al. do not use one identical disentanglement computation for every method. Their benchmark separates two ideas (Mucsányi et al., 2024). First, distributional methods can produce several plausible class-probability vectors for the same image, for example through ensembles, Monte Carlo samples, posterior samples, or a Dirichlet distribution. For these methods, the benchmark applies information-theoretic and Bregman decomposition formulas. Their definitions, equations, and interpretations are given once in Section 3.9. In the experiments, agreement between the two decompositions is treated as evidence that an interpretation is less dependent on one decomposition formula.

Second, the paper checks whether those components, and also single scalar uncertainty scores from deterministic methods, solve specialized uncertainty tasks. This is important because many methods do not output a meaningful two-component aleatoric–epistemic pair. Correctness prediction, loss prediction, temperature scaling, Mahalanobis, DDU, and DUQ are mainly scalar estimators. They can be useful for a specialized task, but they do not by themselves provide a full aleatoric–epistemic decomposition (Mucsányi et al., 2024).

Benchmark question	Operational target	Main interpretation
Do decomposition formulas separate components?	Rank correlation between the estimated aleatoric and epistemic components	Low internal correlation is necessary for disentanglement; high correlation suggests both components measure the same difficulty signal.
Does an estimate capture epistemic uncertainty?	OOD detection AUROC on corrupted data, with uncertainty higher for shifted inputs	Strong performance suggests sensitivity to missing training support or distribution shift.
Does an estimate capture aleatoric uncertainty?	Rank correlation with human soft-label entropy from ImageNet-ReaL or CIFAR-10H	Strong performance suggests sensitivity to label ambiguity or human disagreement.
Does an estimate capture predictive uncertainty?	Correctness-prediction AUROC and abstained-prediction metrics	Strong performance means the score ranks likely errors, but it does not identify the source of uncertainty.

The main empirical result is negative for standard decomposition formulas: the estimated aleatoric and epistemic components are often strongly correlated. The paper argues that a useful disentangled system may need specialized estimators for specialized tasks rather than relying only on a formula that splits one predictive distribution into two components. Their CIFAR-10 data-scarcity experiment appears in Appendix H.9. It trains models from scratch on 10%, 50%, and 100% of the training data and studies how mean uncertainty changes with training evidence. It supports the same caution: both aleatoric and epistemic components can change together under scarcity, so a scarcity response alone does not prove clean disentanglement (Mucsányi et al., 2024).

3.9 Evaluation of Uncertainty Disentanglement

Evaluation asks whether the appropriate part of an uncertainty signal reacts to the intended experimental cause. It separates the model output, the component computed from that output, and the external probe used as a target (Mucsányi et al., 2024; de Jong et al., 2024).

Step 1: Obtain one or several predictive distributions. For an image x, a deterministic classifier returns one class-probability vector $p(y \mid x)$. Sampling and ensemble methods return several vectors $p_1(y \mid x), \ldots, p_M(y \mid x)$, for example from different ensemble members, dropout masks, posterior samples, SWAG samples, or

Laplace samples (Gal and Ghahramani, 2016; Lakshminarayanan et al., 2017; Maddox et al., 2019; Ritter et al., 2018). Methods such as evidential learning or Posterior Networks may instead parameterize a distribution over categorical predictions (Sensoy et al., 2018; Charpentier et al., 2020). In all cases the goal is to describe how uncertain the prediction is for the same input image.

Step 2: Split total predictive uncertainty when a split is available. For methods with several plausible predictive distributions, the information-theoretic split used by Mucsanyi et al. starts with the averaged prediction

$$\bar{p}(y \mid x) = \frac{1}{M} \sum_{m=1}^{M} p_m(y \mid x).$$

Predictive uncertainty is the entropy of this average,

$$PU(x) = H[\bar{p}(y \mid x)].$$

The aleatoric component is the average entropy inside the individual predictions,

$$AU(x) = \frac{1}{M} \sum_{m=1}^{M} H[p_m(y \mid x)].$$

The epistemic component is the remaining disagreement between plausible predictions,

$$EU(x) = PU(x) - AU(x).$$

This equation is often described as predictive entropy decomposed into expected entropy and mutual information (Gal and Ghahramani, 2016; Kendall and Gal, 2017; Mucsányi et al., 2024). It has an intuitive reading. If every sampled model is uncertain in the same way, *AU* is high. If sampled models make different confident predictions, *EU* is high. If all sampled models agree on one class, both components are small.

The Bregman decomposition uses the same collection of plausible predictive distributions, but changes the way disagreement is measured. Instead of measuring epistemic uncertainty as the gap between the entropy of the arithmetic average and the average entropy, it compares each plausible prediction with a dual Bayesian model average. In the implementation used here, this dual average is the normalised geometric mean of the predictive distributions (Mucsányi et al., 2024),

$$p^{\star}(y \mid x) = \operatorname{softmax}\left(\frac{1}{M} \sum_{m=1}^{M} \log p_m(y \mid x) \right).$$

In the experiments, the Bregman components are estimated from the finite set of predictive distributions. The Bregman aleatoric component keeps the within-predictor entropy term,

$$AU^{\mathrm{B}}(x) = \frac{1}{M} \sum_{m=1}^{M} H[p_m(y \mid x)],$$

while the Bregman epistemic component is the average KL divergence from the dual average to the individual plausible predictions,

$$EU^{\mathrm{B}}(x) = \frac{1}{M} \sum_{m=1}^{M} D_{\mathrm{KL}}(p^{\star}(y \mid x) \,\|\, p_m(y \mid x)) \,.$$

The corresponding Bregman total predictive uncertainty is

$$PU^{\mathrm{B}}(x) = AU^{\mathrm{B}}(x) + EU^{\mathrm{B}}(x).$$

The intuition is similar to the information-theoretic split, but the reference prediction is different. The information-theoretic split centres disagreement on the arithmetic average prediction, while the Bregman split centres it on the dual average and measures divergence to each sampled prediction. The decompositions can thus rank examples differently from the same model outputs. Bregman values serve as a sensitivity check: conclusions that change strongly between the information-theoretic and Bregman estimates should not be attributed to a stable uncertainty source without further evidence.

Several methods do not provide this split. A calibrated softmax model, correctness-prediction head, loss-prediction head, or distance score may return only one uncertainty value. That value can still rank hard examples or detect shift, but it cannot by itself assign uncertainty to aleatoric and epistemic parts. For these methods the thesis reports task performance and treats any source-specific interpretation as mechanism-based rather than as a measured two-component decomposition (Mucsányi et al., 2024).

Step 3: Choose an experimental target for each uncertainty source. The true source of uncertainty is not observed directly, so the benchmark uses controlled targets. A rater-disagreement target is used as an aleatoric proxy, because disagreement between experts can reflect an ambiguous image-label relation. A reduced training set is used as an epistemic probe, because it removes training evidence while the test images stay fixed. A corruption or out-of-distribution target is also epistemic or distributional, because it tests whether the model recognises inputs that are poorly supported by the training distribution (Mucsányi et al., 2024; Ovadia et al., 2019; Hendrycks and Dietterich, 2019). These targets are proxy targets rather than direct observations of the true uncertainty source. They turn the thesis question into testable comparisons: if a component is interpreted as aleatoric, it should respond most clearly to ambiguity probes; if it is interpreted as epistemic, it should respond most clearly to missing evidence or shift probes. The desired pattern is a source-specific response, not merely high uncertainty on difficult images.

The metric follows the type of target. Controlled training interventions use changes in mean AU and EU because the question is how each component changes relative to the corresponding clean, full-data model. MHIST uses Spearman correlation because rater-vote entropy is a continuous ordinal ambiguity target and uncertainty scales differ between methods. Corruption experiments use OOD AUROC because

the evaluation target is binary: an image is clean or corrupted. Here, AUROC measures how often an AU- or EU-based score ranks a corrupted image above a clean image; evaluating it separately at each severity retains the severity trajectory.

Uncertainty source	**Observable probe**	**Expected selective response**
Aleatoric ambiguity	Rater-vote entropy in MHIST or synthetic label noise in PathMNIST	AU should increase or rank ambiguous examples higher than EU.
Epistemic missing evidence	Training on 50% or 10% of the data	EU should increase more than AU when training evidence is removed.
Distribution shift	PathMNIST-C or OOD/corruption evaluation	An epistemic, density, or disagreement score should separate shifted images from clean images.
Predictive difficulty	Correct versus wrong predictions	Any useful uncertainty score should rank likely errors above likely correct predictions.

Step 4: Compare intended and off-target components. A high correlation with a target is useful only if the off-target component does not show the same relation. For MHIST, let r_A denote the Spearman correlation between AU and rater-vote entropy, and let r_E denote the same correlation for EU. The ordering expected under aleatoric specialization has r_A positive and larger than r_E. If both correlations are similar, the method may be ranking generally difficult images rather than isolating label ambiguity. This intended-versus-off-target comparison follows the benchmark logic of Mucsányi et al. (2024) and de Jong et al. (2024).

For data scarcity, raw means are less informative because each method has its own scale. Each method is compared with itself. If $\overline{AU}_{1.0}$ and $\overline{EU}_{1.0}$ are the mean uncertainties after full-data training, and $\overline{AU}_q$ and $\overline{EU}_q$ are the means after training on fraction q of the data, then

$$\Delta AU_q = \overline{AU}_q - \overline{AU}_{1.0}, \qquad \Delta EU_q = \overline{EU}_q - \overline{EU}_{1.0}.$$

The ordering expected under epistemic specialization has ΔEU_q positive and larger than the corresponding change in AU. If AU changes as much as, or more than, EU, the result is interpreted as cross-sensitivity rather than clean epistemic separation. The label-noise probe in PathMNIST uses the same logic with the roles reversed: an aleatoric response should show a larger ΔAU than ΔEU under the noisy-label intervention.

Step 5: Interpret the pattern across probes. Clean disentanglement requires an aleatoric response to ambiguity, an epistemic response to scarcity or shift, and only small cross-sensitivity. Three common outcomes are distinguished:

- **Task usefulness without disentanglement:** the method ranks wrong predictions or hard images well, but one scalar score carries the whole signal.
- **Partial specialization:** one component reacts to its intended target, but the other component reacts too.
- **Cleaner disentanglement:** the intended component reacts strongly and the off-target component stays relatively small across more than one probe.

A method can be useful for error detection while failing to separate why an image is difficult. Specialized uncertainty estimates must therefore be judged against specialized tasks (Mucsányi et al., 2024).

3.10 Relevance to Medical-Dataset Transfer

The medical transfer tests whether benchmark uncertainty mechanisms remain interpretable when they are moved from natural images to histopathology. In this domain, uncertainty can serve as a signal for cases that merit caution or review, but its meaning and reliability require explicit validation (Lambert et al., 2024; Zou et al., 2023; Huang et al., 2024).

The adaptation from CIFAR-10 to PathMNIST asks two related questions. The first is practical: can the recovered implementation reproduce the original CIFAR-10 benchmark closely enough to serve as a transfer basis? The second is scientific: when the same uncertainty mechanisms move to a medical dataset, do their strengths and weaknesses still match the uncertainty type they are meant to measure?

PathMNIST cannot provide a direct reference target for aleatoric label ambiguity because it does not include several independent labels per image. A model can still output an aleatoric-like score, for example predictive entropy or expected entropy, but the dataset cannot validate whether this score follows human disagreement. PathMNIST supports predictive, scarcity, and distribution-shift probes; MHIST adds the missing rater-disagreement axis (Yang et al., 2023; Wei et al., 2021).

3.11 Synthetic Label-Noise Probe for PathMNIST

PathMNIST uses a controlled synthetic label-noise experiment as an aleatoric proxy. The images remain unchanged, but a fixed fraction of the training labels is replaced by a wrong class sampled uniformly from the other eight PathMNIST classes. Validation labels, clean test labels, and PathMNIST-C labels remain unchanged. This design isolates label-level noise from data scarcity: the number of training images stays constant, while the reliability of part of the training signal is reduced.

The experiment uses noise fractions of 10%, 20%, and 40%, with a fixed selection seed. For an image with original class $y_i \in \{1, \ldots, 9\}$, a noisy training label $\tilde{y}_i$ is created only if the sample is selected for corruption:

$$\tilde{y}_i = \begin{cases} y_i, & \text{if sample } i \text{ is not selected,} \\ \text{Uniform}\left(\{1, \ldots, 9\} \setminus \{y_i\}\right), & \text{otherwise.} \end{cases}$$

The same selected indices are reused for every method at a given noise fraction, so method differences cannot be explained by different noisy examples.

This probe is not a replacement for multi-rater ambiguity. It is an intervention that asks whether methods with aleatoric-oriented mechanisms react to corrupted labels more selectively than epistemic-oriented components. Wimmer et al. (2023) and de Jong et al. (2024) use label-noise manipulations to study aleatoric uncertainty and uncertainty disentanglement in classification. Northcutt et al. (2021) also frame noisy labels as a dataset-label uncertainty problem and estimate label errors through confident learning.

The evaluation mirrors the MHIST analysis. First, each method is trained on clean PathMNIST and on the noisy-label variants. Second, the trained model is evaluated on the unchanged validation and test splits. Third, the analysis compares the change in aleatoric and epistemic uncertainty relative to the clean run:

$$\Delta AU_\eta = \overline{AU}_\eta - \overline{AU}_0, \qquad \Delta EU_\eta = \overline{EU}_\eta - \overline{EU}_0,$$

where η is the label-noise fraction. A method supports aleatoric specialization if ΔAU_η is positive and larger than the off-target epistemic shift. If ΔEU_η changes by a similar or larger amount, the method is interpreted as treating noisy labels as generic difficulty, model uncertainty, or both.

3.12 Evaluation Metrics

No single metric covers classifier quality, uncertainty usefulness, and source separation. Accuracy and proper scoring rules assess predictive validity. AUROC and AUPR test whether uncertainty ranks a target event above a non-target event, such as wrong predictions above correct ones. Paired component comparisons ask whether the intended component reacts more strongly than its off-target counterpart. Only this third role directly addresses uncertainty disentanglement (Mucsányi et al., 2024).

Classifier quality provides necessary context for disentanglement, but the link is indirect. A classifier that performs close to chance gives unstable uncertainty estimates: many samples are wrong, decision boundaries are poorly learned, and high uncertainty may reflect general failure rather than a specific uncertainty source. Classifier quality is thus necessary for meaningful interpretation, but can also introduce bias. If one method has much lower accuracy than another, its higher uncertainty under scarcity or corruption may come from weaker classification rather than better epistemic awareness. For this reason, accuracy, negative log-likelihood, and Brier score

are reported beside the uncertainty metrics. They show whether a disentanglement result comes from a competent classifier. Strong classifier quality is not sufficient: a high-accuracy model can still compress ambiguity, missing evidence, and shift into one confidence score (Mucsányi et al., 2024).

The tables distinguish training-trajectory summaries from final evaluation metrics. A best metric is the largest value reached on the validation or evaluation split used for checkpoint selection; it is useful for describing how a run was selected, but it can be optimistic when the same split is repeatedly checked during training (Cawley and Talbot, 2010). A final metric is computed at the final or selected checkpoint and more closely reflects the reported result under the common protocol. When tables report finite sample counts, non-finite uncertainty values such as `NaN` or infinity have been excluded before means, correlations, or component shifts are computed. This avoids numerical failures from unstable methods being silently averaged into the uncertainty summaries.

Accuracy Accuracy reports the fraction of correctly classified images. It is not an uncertainty metric. It is reported because uncertainty estimates are difficult to interpret if the underlying classifier fails the task. A more accurate model can still be overconfident, poorly calibrated, or unable to distinguish uncertainty sources.

AUROC The area under the receiver operating characteristic curve measures how well a score separates positive from negative cases across thresholds. It is threshold independent and widely used for binary detection tasks such as OOD detection or error detection (Fawcett, 2006). In this thesis, correctness AUROC answers: do uncertain samples tend to be wrong? OOD AUROC answers: do uncertain samples tend to be shifted or corrupted? In both cases the positive class is the event that should be flagged by uncertainty: an incorrect prediction for correctness AUROC, and a corrupted or shifted image for OOD AUROC. A value of 0.5 corresponds to chance ranking; a value below 0.5 means that the chosen score ranks the target event lower rather than higher. Equivalently, an OOD AUROC of 0.7 means that a randomly selected shifted image receives a higher uncertainty score than a randomly selected clean image in 70% of pairwise comparisons. These are specialized usefulness tests. They support a disentanglement claim only when the score being tested is tied to the intended uncertainty source and the off-target scores do not show the same effect.

AUPR The area under the precision–recall curve helps when the positive class is rare, for example when few examples are failures or OOD samples (Davis and Goadrich, 2006). Like AUROC, AUPR measures separation for one chosen target. It does not explain why the uncertainty score separates that target.

Calibration error Calibration asks whether predicted probabilities match empirical frequencies. Expected calibration error groups predictions into confidence bins and compares bin confidence with bin accuracy (Naeini et al., 2015; Guo

et al., 2017). Calibration is useful because it makes confidence values numerically meaningful. It does not by itself identify whether uncertainty comes from ambiguity, missing evidence, or distribution shift.

NLL and Brier score Negative log-likelihood penalises probability assigned to the true class and is particularly sensitive to overconfident mistakes. The Brier score measures squared error across the full probability forecast (Brier, 1950). Both assess probabilistic prediction quality and are reported as context rather than as source-separation metrics.

Correlations Spearman or Pearson correlations compare an uncertainty score with an external target such as rater-vote entropy. These correlations are closer to disentanglement than correctness AUROC because the target represents a proposed uncertainty source. They still need an off-target comparison: if both AU and EU correlate with rater disagreement, the method may detect difficult images rather than aleatoric ambiguity.

Component comparisons Rater correlations, data-scarcity shifts, and label-noise shifts provide the most direct disentanglement evidence when AU and EU are both available. They compare an intended response with an off-target response. Source-specific behaviour requires the intended component to be clearly larger than the other component.

Metric group	**What it answers**	**Relation to disentanglement**
Accuracy, NLL, Brier score	Is the classifier and probability forecast useful?	Necessary context, but no source separation.
Calibration error	Are confidence values numerically reliable?	Helps interpret probabilities, but does not identify the uncertainty source.
Correctness AUROC/AUPR	Does uncertainty rank likely errors?	Shows predictive usefulness; source remains unclear.
OOD or corruption AUROC/AUPR	Does uncertainty detect shifted inputs?	Supports epistemic or distributional sensitivity for that score.
Rater correlations	Does uncertainty follow label ambiguity?	Supports aleatoric sensitivity if ρ_A exceeds ρ_E.
Component shifts	Does the intended component react more than the off-target component?	Direct operational evidence for or against disentanglement.

Predictive entropy, expected entropy, mutual information, and the Bregman components are defined in Section 3.9.

3.13 Experimental Axes and Uncertainty Sources

The experimental design combines the probes introduced in Sections 3.4–3.6 with clean in-distribution evaluation. Figure 1 summarises how these probes map onto the empirical chapters; their expected component-level responses are specified in Section 3.9.

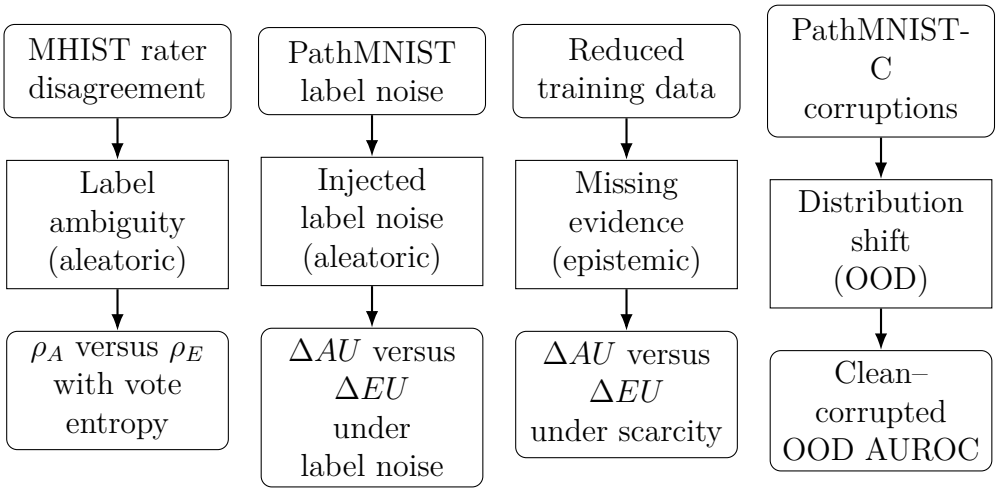
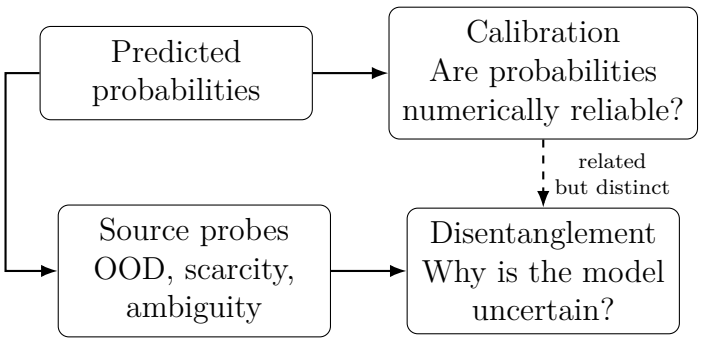


Figure 1: Experimental probes, their intended uncertainty sources, and the corresponding empirical measures. The arrows encode the tested operational hypotheses; they do not assume that a probe affects only one uncertainty component.

3.14 Calibration Versus Uncertainty

Calibration and uncertainty are related, but they answer different questions. A calibrated classifier assigns probabilities that match long-run empirical frequencies; for example, predictions made with confidence 0.8 should be correct about 80% of the time (Guo et al., 2017). This property supports probability estimates, but it does not explain why a prediction is uncertain.

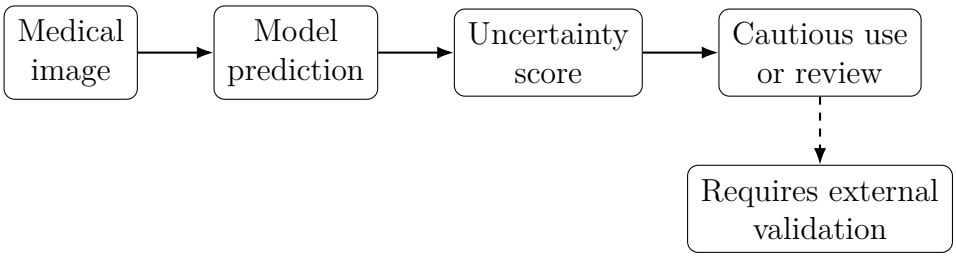


An uncertainty method can benefit from calibration, and a calibrated model can still fail to disentangle uncertainty sources. Temperature scaling illustrates this point: it

can improve probability calibration, but it does not create a separate model of epistemic uncertainty, aleatoric uncertainty, or feature-space novelty (Guo et al., 2017; Mucsányi et al., 2024). For this reason, calibration metrics are interpreted alongside OOD, data-scarcity, and error-detection metrics rather than as replacements for them.

3.15 Implications for Medical Decision Support

In a medical-image setting, uncertainty estimates can identify cases for closer review, but their value depends on the application and evaluation setting (Lambert et al., 2024; Zou et al., 2023; Huang et al., 2024). They do not establish clinical safety. Reporting standards for AI-based clinical prediction models emphasise transparent descriptions of model development, validation, intended use, and limitations (Collins et al., 2024). PathMNIST supports controlled benchmarking rather than real-world clinical use.



Here, uncertainty is used as a method-comparison tool. The experiments test whether uncertainty mechanisms behave as expected when moved from CIFAR-10-style natural-image benchmarks to a medical image benchmark. Clinical interpretation would require external data, additional validation, and workflow-specific evaluation (Collins et al., 2024; Lambert et al., 2024).

3.16 Limitations of Disentanglement Experiments

Uncertainty disentanglement is difficult because experiments rarely observe the true uncertainty source. Instead, they use proxies: corruptions approximate distribution shift, reduced training sets approximate missing knowledge, and error labels approximate difficult examples. These proxies help, but they do not measure epistemic or aleatoric uncertainty directly (Mucsányi et al., 2024; de Jong et al., 2024).

Method rankings also depend on architecture, training recipe, hyperparameters, dataset size, preprocessing, and metric choice. Ovadia et al. (2019) show that uncertainty quality can degrade under dataset shift, so good in-distribution behaviour does not guarantee reliable shifted-data behaviour. The benchmark framing of Mucsányi et al. (2024) remains comparative and task-specific: it asks which methods fit which uncertainty tasks, not which method is best in general.

4 Datasets

The experiments use PathMNIST as the primary medical image benchmark and MHIST as a complementary multi-rater histopathology dataset. PathMNIST supports a controlled transfer from CIFAR-10 to a larger medical classification task; MHIST provides individual votes as an observable proxy for rater disagreement.

4.1 PathMNIST

PathMNIST belongs to MedMNIST v2, a standardized collection of small biomedical image classification benchmarks (Yang et al., 2023). It contains colorectal histology patches derived from NCT-CRC-HE-100K and CRC-VAL-HE-7K (Kather et al., 2019). The task has nine tissue classes and 107,180 images in total, split into 89,996 training images, 10,004 validation images, and 7,180 test images. MedMNIST stores the original $3 \times 224 \times 224$ patches as $3 \times 28 \times 28$ images. The experiments can resize them again when a backbone expects a larger input.

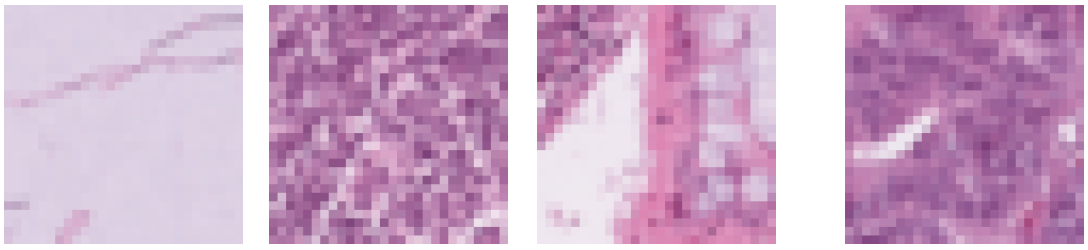


Figure 2: Four training examples from PathMNIST. The samples show different tissue labels from the nine-class colorectal histology task.

PathMNIST fits the main experiment because it is medical, standardized, and large enough for repeated method comparisons. It also belongs to histopathology, where visual ambiguity, staining variation, tissue heterogeneity, and scanner differences can affect predictive uncertainty. The external test source gives an additional reason to study distribution shift rather than only clean in-distribution accuracy (Yang et al., 2023; Kather et al., 2019; Lambert et al., 2024).

The released labels do not provide several independent ratings for each image, so PathMNIST cannot measure label disagreement as an aleatoric proxy. Synthetic label corruption instead provides a controlled aleatoric-oriented probe, while data scarcity and PathMNIST-C corruptions probe missing training evidence and distribution shift. PathMNIST can therefore test method behaviour, but not its relationship with rater disagreement (Yang et al., 2023; Wimmer et al., 2023; Di Salvo et al., 2024).

4.2 MHIST

MHIST is a histopathology dataset for colorectal polyp classification (Wei et al., 2021). It contains 3,152 de-identified hematoxylin and eosin stained images from Dartmouth-Hitchcock Medical Center. Each image has size 224×224 pixels. The task is binary classification between hyperplastic polyp and sessile serrated adenoma. This distinction matters clinically: the two serrated-polyp categories carry different malignancy and surveillance implications, with sessile serrated lesions treated as higher-risk precursor lesions than typical hyperplastic polyps (Jaravaza et al., 2020; Wei et al., 2021).

Seven pathologists label each MHIST image. The released annotation file gives the majority-vote label and the number who selected sessile serrated adenoma. This count can be converted into a soft label, for example 6/7 versus 4/7, and supplies an image-level measure of diagnostic disagreement that PathMNIST lacks (Wei et al., 2021).

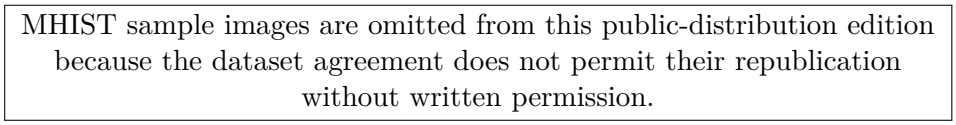


Figure 3: MHIST sample-image panel omitted from the public-distribution edition.

MHIST is suitable for the experimental design because it is an image-classification dataset rather than a segmentation or detection benchmark. Its images already match the 224×224 input size used by the transferred models, and its histopathology domain is shared with PathMNIST (Wei et al., 2021; Yang et al., 2023).

MHIST has fewer images than PathMNIST and only two classes, so it complements rather than replaces the nine-class benchmark. Its pathologist votes allow model uncertainty to be compared with diagnostic disagreement: an aleatoric score that captures ambiguity should be higher for low-agreement images. Epistemic sensitivity still requires separate assessment through data scarcity or shifted evaluation data (Wei et al., 2021; Mucsányi et al., 2024).

4.3 Dataset Roles in the Thesis

PathMNIST tests transfer to a larger medical histology task. MHIST tests responses to human label disagreement. Together, they provide probes for distribution shift, limited model knowledge, and rater disagreement.

Table 1: Comparison of PathMNIST and MHIST for the thesis experiments.

Property	PathMNIST	MHIST
Medical domain	Colorectal histopathology patches from tissue slides	Colorectal-polyp histopathology images
Task	Nine-class tissue classification	Binary classification between hyperplastic polyp and sessile serrated adenoma
Dataset size	107,180 images	3,152 images
Official split	89,996 train, 10,004 validation, 7,180 test	2,175 train and 977 test; no separate official validation split
Released resolution	$3 \times 28 \times 28$ RGB images in MedMNIST v2	224×224 RGB images
Original image scale	Derived from $3 \times 224 \times 224$ colorectal histology patches	Released at the model-ready 224×224 image scale
Label source	Single released class label per image	Majority-vote class label with seven pathologist votes per image
Rater disagreement	Not available in the released benchmark labels	Available through the number of pathologists voting for sessile serrated adenoma
Main thesis role	Larger medical benchmark for method transfer, data scarcity, and corruption or shift experiments	Additional medical dataset for testing whether uncertainty follows expert disagreement
Main limitation	No image-level rater disagreement target	Smaller dataset, binary task, and no validated MHIST-C or medical OOD protocol in this thesis

5 Uncertainty Methods

This chapter groups the uncertainty methods by the mechanism that produces the uncertainty signal. The grouping follows prior uncertainty-disentanglement work and uses PathMNIST from MedMNIST as the first medical-image adaptation target (Yang et al., 2023; Mucsányi et al., 2024). Each family is introduced before the individual methods, so the reader can compare methods by mechanism rather than by name alone.

5.1 Method Overview

The methods are grouped by mechanism before their individual descriptions: calibration baselines, supervised prediction heads, sampling methods, density or dis-

tance methods, posterior approximations, and heteroscedastic or evidential models. The empirical chapters use the same grouping to relate observed behaviour to each mechanism.

5.1.1 Taxonomy of Evaluated Methods

Mechanism group	Methods
Baseline and calibration	CE Baseline; Temperature Scaling
Prediction-head methods	Correctness Prediction; Loss Prediction; Deep Correctness Prediction; Deep Loss Prediction
Ensembles and sampling	Deep Ensemble; Fast Deep Ensemble; Shallow Ensemble; MC Dropout; SWAG
Density and distance methods	DDU; DUQ; Mahalanobis; PostNet
Bayesian and posterior methods	Laplace; SNGP
Heteroscedastic and evidential methods	HET; HET-XL; HetClassNN; Evidential Deep Learning

Table 2: Taxonomy of uncertainty methods used in the PathMNIST adaptation. The grouping follows the mechanism that produces the uncertainty score, not only the final scalar metric.

5.1.2 Comparison of Method Characteristics

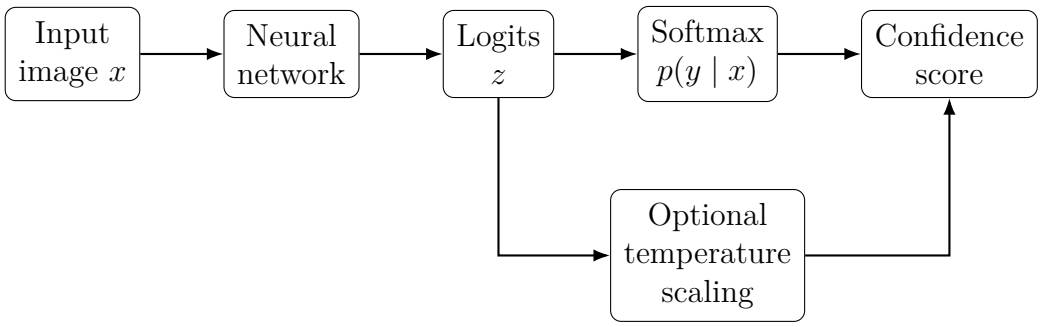
Table 3 is a qualitative reading guide, not an experimental result. The entries come from the mechanism of each method and from the role assigned to the method in previous uncertainty work. “Epistemic signal” means that the method has a mechanism that can react to missing model knowledge, for example posterior variation, disagreement between predictors, or low support in feature space. This interpretation follows Bayesian dropout, deep ensembles, SWAG, Laplace approximations, SNGP, and density- or distance-aware methods (Gal and Ghahramani, 2016; Lakshminarayanan et al., 2017; Maddox et al., 2019; Ritter et al., 2018; Liu et al., 2020; van Amersfoort et al., 2020; Mukhoti et al., 2023; Lee et al., 2018; Charpentier et al., 2020). “Aleatoric signal” means that the method can react to image-dependent ambiguity, label noise, or irreducible class overlap. Heteroscedastic and evidential methods target this more directly than calibration or distance-only methods (Kendall and Gal, 2017; Collier et al., 2021; Sensoy et al., 2018). “OOD use” describes whether the method has a mechanism that should separate inputs unsupported by the training distribution; this is supported most directly by distance-aware, density-aware, ensemble, and posterior approaches (Lee et al., 2018; van Amersfoort et al., 2020; Mukhoti et al., 2023; Lakshminarayanan et al., 2017; Ovadia et al., 2019). “Cost” summarizes the extra computation or storage needed at training or inference time,

especially extra heads, density models, checkpoints, posterior samples, or repeated forward passes. The benchmark of Mucsányi et al. (2024) uses these method families to test whether such mechanism-based expectations match specialized uncertainty tasks.

Method family	**Additional mechanism**	**Multiple passes**	**Epistemic signal**	**Aleatoric signal**	**OOD use**	**Cost**
CE Baseline	No	No	Low	Indirect	Weak	Low
Temperature Scaling	No	No	Low	Indirect	Weak	Low
Prediction-head methods	Yes	No	Medium	Medium	Medium	Medium
Deep / Fast Ensembles	Yes	Yes	High	Medium	Strong	High
Shallow Ensemble	Yes	Yes	Medium	Medium	Medium	Medium
MC Dropout	No	Yes	Medium	Medium	Medium	Medium
SWAG	Posterior approx.	Yes	High	Medium	Strong	High
Laplace	Posterior approx.	Yes	High	Medium	Medium	Medium
SNGP	GP head	Often yes	High	Indirect	Strong	Medium
DDU / DUQ / Mahalanobis	Density or distance	No	Medium–High	Low	Strong	Medium
PostNet	Density model	Often yes	High	Medium	Strong	Medium
HET / HET-XL / HetClassNN	Uncertainty head	Yes	Low–Medium	High	Medium	Medium–High
EDL	Evidence head	No	Medium	Medium	Medium	Low–Medium

Table 3: Qualitative comparison matrix for thesis-level interpretation. "Additional mechanism" states whether the method adds an auxiliary head, density or distance model, posterior approximation, or ensemble beyond the base classifier; "No" denotes no such addition. Low, medium, high, weak, and strong are mechanism-based expectations from the cited literature, not measured scores.

5.2 Baseline and Calibration



Baseline and calibration methods use the trained classifier itself as the uncertainty source. The model maps an input image to class scores and then converts those scores into probabilities. No extra density model, ensemble, or posterior approximation is added; the uncertainty estimate is read from the final predictive distribution.

Calibration methods keep the classifier fixed and only adjust how sharp these probabilities are, as in temperature scaling for modern neural-network calibration (Guo et al., 2017). In the benchmark framing, this family provides the reference point for more specialised uncertainty mechanisms (Mucsányi et al., 2024).

Logits z The raw, unnormalised class scores produced by the neural network before probabilities are computed. Larger logits indicate stronger model preference for a class, but logits are not probabilities.

Softmax $p(y \mid x)$ The transformation that converts logits into a probability distribution over classes for image x. The notation $p(y \mid x)$ means the model's predicted probability that the label is y given the input image.

Optional temperature scaling A post-hoc calibration step that divides logits by a learned scalar temperature before the softmax. Higher temperatures make predictions less sharp; lower temperatures make them more confident (Guo et al., 2017).

Confidence score A scalar summary of the predictive distribution, typically the maximum softmax probability. Low confidence indicates high uncertainty, although this baseline mainly reflects predictive uncertainty rather than a clean separation of epistemic and aleatoric uncertainty.

5.2.1 CE (Cross-Entropy) Baseline

Method mechanism The cross-entropy baseline trains one deterministic classifier. For an image x, the network produces logits $z_\theta(x) \in \mathbb{R}^K$ for K classes. The softmax function converts these logits into class probabilities,

$$p_\theta(y = k \mid x) = \frac{\exp z_{\theta,k}(x)}{\sum_{j=1}^{K} \exp z_{\theta,j}(x)}.$$

Symbols: x is the input image, y the class label, k and j class indices, K the number of classes, $z_{\theta,k}$ the logit for class k, and θ the classifier parameters. Training minimises the negative log-likelihood, usually written as cross entropy,

$$\mathcal{L}_{\mathrm{CE}}(\theta) = -\log p_\theta(y \mid x).$$

Symbols: $\mathcal{L}_{\mathrm{CE}}$ is the cross-entropy loss, and $p_\theta(y \mid x)$ is the probability assigned to the true class. This is the standard discriminative image-classification objective used with architectures such as ResNet (He et al., 2016). At test time the method does not add a separate uncertainty model. It reads uncertainty from the softmax distribution, for example with confidence $\max_k p_\theta(y = k \mid x)$, predictive entropy $H[p_\theta(y \mid x)]$, or the negative maximum class probability. Guo et al. (2017) show that modern neural networks can be miscalibrated even when accuracy is high, so this softmax score is a baseline rather than a validated uncertainty decomposition. Mucsányi et al. (2024) use the same kind of deterministic classifier as a reference point for specialised uncertainty methods.

Interpretation The baseline does not disentangle aleatoric and epistemic uncertainty. On PathMNIST, increased uncertainty under image corruption indicates sensitivity to aleatoric degradation, while increased uncertainty under data scarcity or PathMNIST-C shift serves only as an indirect epistemic signal. This interpretation remains weak because both cases use the same softmax score.

5.2.2 Temperature Scaling

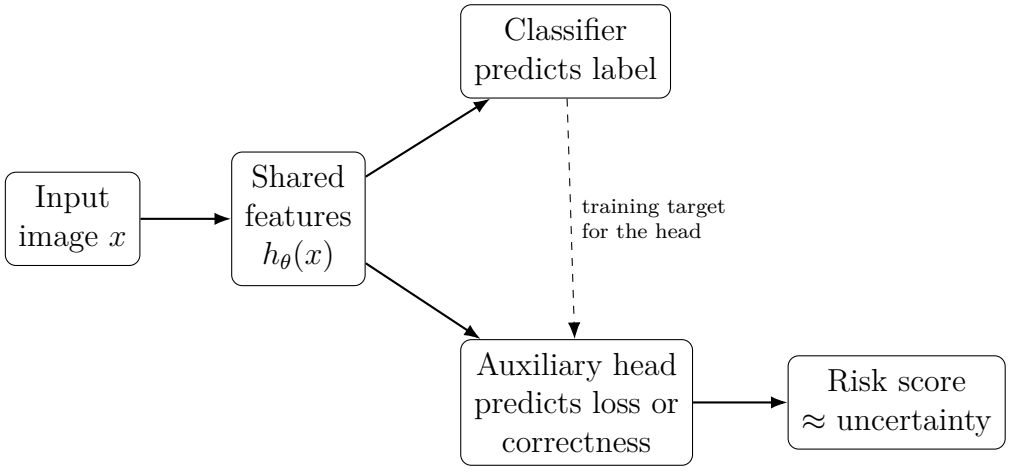
Method mechanism Temperature scaling is a post-hoc calibration method. The trained classifier stays fixed, and only a scalar temperature $T > 0$ is learned on validation data. If $z_\theta(x)$ are the logits, the calibrated probabilities are

$$p_T(y = k \mid x) = \frac{\exp(z_{\theta,k}(x)/T)}{\sum_{j=1}^{K} \exp(z_{\theta,j}(x)/T)}.$$

Symbols: T is the learned temperature, $z_{\theta,k}$ is the logit for class k, and p_T is the calibrated probability after rescaling logits. Guo et al. (2017) choose T by minimising validation negative log-likelihood. A larger temperature smooths the softmax distribution and lowers overconfident probabilities; a smaller temperature sharpens it. Since all logits are divided by the same scalar, the predicted class usually stays the same, and calibration changes the probability scale rather than the learned decision boundary. This is why the method is useful as a calibration control but does not introduce a new epistemic or aleatoric mechanism.

Interpretation The method can improve calibration but does not create new uncertainty information. It preserves the ranking of class logits and cannot by itself distinguish aleatoric from epistemic uncertainty. It serves as a calibration baseline against which OOD separation and scarcity sensitivity can be assessed independently.

5.3 Prediction-Head Methods



Prediction-head methods keep the usual image classifier, but attach an additional head that learns whether the classifier is likely to fail. Instead of relying only on the maximum softmax probability, they train a separate uncertainty signal from supervised targets such as the realised loss or the correctness indicator. This follows the failure-prediction view of confidence learning and the loss-prediction view used in active learning (Corbière et al., 2019; Yoo and Kweon, 2019). The motivation is simple: internal features may contain warning signs about difficult examples even when the final softmax distribution is overconfident.

Shared features $h_\theta(x)$ The representation produced by the backbone network for input image x. The symbol θ denotes the learned network parameters.

Classifier predicts label The standard classification branch that maps shared features to class logits and a predicted PathMNIST class.

Auxiliary head A second prediction branch trained to estimate a risk quantity, for example the classification loss or whether the classifier prediction is correct (Corbière et al., 2019; Yoo and Kweon, 2019).

Head target The dashed arrow indicates that the auxiliary head is supervised using information derived during training, such as the true loss or correctness of the classifier output.

Risk score ≈ uncertainty The auxiliary output serves as an uncertainty score: high predicted loss or low predicted correctness marks the prediction as uncertain.

5.3.1 Correctness Prediction

Method mechanism Correctness prediction augments the classifier with an auxiliary head that predicts whether the main classifier will be correct. Let $\hat{y}(x) = \arg\max_k p_\theta(y = k \mid x)$ be the predicted class. The target for the auxiliary head is the binary variable

$$c(x, y) = \mathbb{1}[\hat{y}(x) = y].$$

Symbols: $c(x, y)$ is the binary correctness target, $\mathbb{1}[\cdot]$ is the indicator function, $\hat{y}(x)$ is the predicted class, and y is the true class. The head receives a network representation $h_\theta(x)$ and predicts $\hat{c}_\phi(x) \in [0, 1]$, often trained with binary cross entropy between $\hat{c}_\phi(x)$ and $c(x, y)$. Corbière et al. (2019) formulate this as confidence estimation for failure prediction: the model should assign low confidence to samples it is likely to misclassify. In the disentanglement benchmark, this learned correctness score becomes an uncertainty score, with uncertainty increasing as predicted correctness decreases (Mucsányi et al., 2024).

Interpretation The method estimates predictive failure rather than a probabilistic uncertainty component. On PathMNIST, high predicted error can arise from ambiguous histology, label noise, or distribution shift. It serves as a pragmatic risk predictor, not as an aleatoric–epistemic decomposition. Experiments disentangle its signal by comparing the same score across corruption severity and training-data scarcity.

5.3.2 Loss Prediction

Method mechanism Loss prediction adds an auxiliary head that estimates the loss the main classifier will incur on an input. If the task loss is $\ell_\theta(x, y) = -\log p_\theta(y \mid x)$, the auxiliary head predicts a scalar $\hat{\ell}_\phi(x)$ from the model features. Yoo and Kweon (2019) introduce this idea for active learning: samples with high predicted loss are expected to be informative because the current model handles them poorly. Their learning-loss framework trains the auxiliary module from intermediate features and uses a loss-ranking objective so that the predicted ordering of sample losses matches the observed ordering. In uncertainty benchmarking, the same predicted loss is read as a risk score: high $\hat{\ell}_\phi(x)$ means the model expects a difficult or error-prone prediction (Mucsányi et al., 2024).

Interpretation The predicted loss mixes several causes of difficulty. Ambiguous tissue morphology, mislabeled examples, rare classes, and out-of-distribution inputs can all produce high expected loss. It is treated as a task-risk estimator rather than a separate uncertainty component. Source sensitivity is assessed experimentally under label noise, reduced training evidence, and distribution shift.

5.3.3 Deep Correctness Prediction

Method mechanism Deep correctness prediction uses the same binary failure-prediction target as correctness prediction, but exposes several internal feature levels to the auxiliary head. For backbone representations $h_\theta^{(1)}(x), \ldots, h_\theta^{(L)}(x)$, the auxiliary predictor estimates

$$\hat{c}_\phi(x) = g_\phi\Big(h_\theta^{(1)}(x), \ldots, h_\theta^{(L)}(x)\Big).$$

Symbols: $\hat{c}_\phi(x)$ is predicted correctness, g_ϕ the auxiliary head, $h_\theta^{(\ell)}(x)$ the feature representation at layer ℓ, and L the number of used feature levels. This follows the confidence-learning idea of Corbiere et al., where the auxiliary model estimates whether the classifier will be correct (Corbière et al., 2019). The deep variant used in the benchmark extends that idea by allowing the risk head to inspect lower-level texture features as well as higher-level semantic features (Mucsányi et al., 2024). This matters for histology because artifacts can affect early visual features, while class ambiguity can appear in later representations.

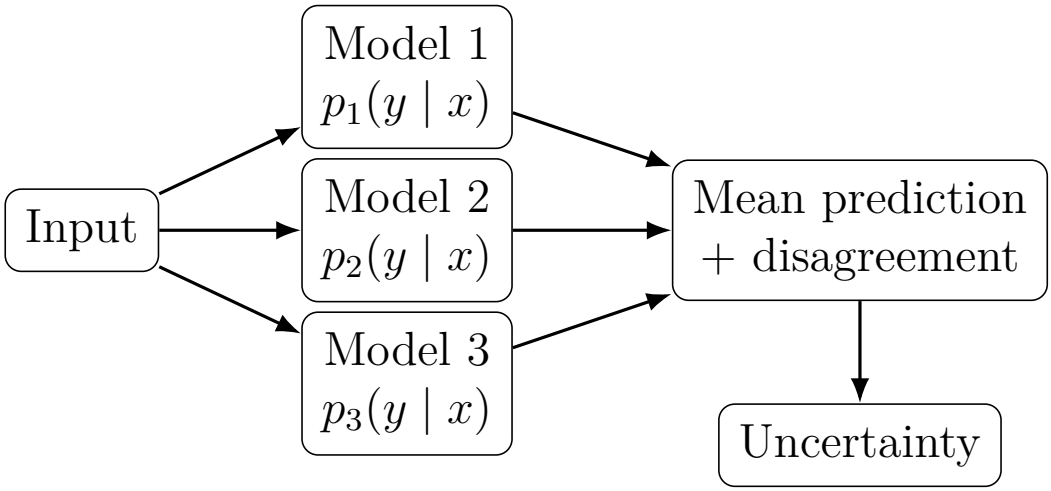
Interpretation Intermediate features can make failure prediction more sensitive to local texture degradation and representation drift. For uncertainty disentanglement, this matters because low-level corruption and high-level class ambiguity may affect different layers. The score remains a supervised failure proxy, not a Bayesian posterior quantity.

5.3.4 Deep Loss Prediction

Method mechanism Deep loss prediction extends loss prediction by drawing information from several feature stages. Instead of estimating $\hat{\ell}_\phi(x)$ only from the final representation, the head can combine features $h_\theta^{(1)}(x), \ldots, h_\theta^{(L)}(x)$ and predict the expected task loss from this multi-level representation. This follows the learning-loss approach of Yoo and Kweon, who use intermediate network features to predict which unlabeled samples would have high loss if labeled (Yoo and Kweon, 2019). In the benchmark adaptation, the predicted loss is used as a scalar uncertainty score rather than only as an active-learning acquisition value (Mucsányi et al., 2024). For medical images this design is relevant because low-level stain or blur effects and high-level class confusion can both contribute to loss.

Interpretation The resulting uncertainty score remains a learned proxy for expected task loss. It can react to low-level degradation, such as blur or staining artifacts, and to higher-level class confusion. It is therefore interpreted as a risk score rather than as a calibrated posterior variance.

5.4 Ensembles and Sampling Methods



Ensembles and sampling methods estimate uncertainty by comparing several plausible predictions for the same input. The predictions may come from independently trained networks, stochastic dropout masks, or sampled SWAG weights (Lakshminarayanan et al., 2017; Gal and Ghahramani, 2016; Maddox et al., 2019). Agreement

suggests stable evidence. Disagreement suggests that the learned decision boundary still depends on which plausible model is used, so the signal is mainly epistemic.

Model i One member of the ensemble or one stochastic forward pass. In a deep ensemble this is a separately trained model; in MC dropout or SWAG it is a sampled version of the same underlying model (Lakshminarayanan et al., 2017; Gal and Ghahramani, 2016; Maddox et al., 2019).

$p_i(y \mid x)$ The class-probability distribution predicted by member or sample i for input image x.

Mean prediction The average of the individual predictive distributions, used as the final class prediction.

Disagreement The spread between member predictions. Large variation means the chosen plausible model or sample changes the result.

Uncertainty Summarised with predictive entropy, expected entropy, mutual information, or variance across ensemble members.

5.4.1 Deep Ensemble

Method mechanism A deep ensemble trains M neural networks independently, usually with different random initialisations and data-ordering noise. Each member produces a predictive distribution $p_{\theta_m}(y \mid x)$. The ensemble predictive distribution is the arithmetic mean,

$$\bar{p}(y \mid x) = \frac{1}{M} \sum_{m=1}^{M} p_{\theta_m}(y \mid x).$$

Symbols: M is the number of ensemble members, m indexes a member, p_{θ_m} is member m's predictive distribution, and $\bar{p}$ is the ensemble mean. Lakshminarayanan et al. (2017) show that this simple averaging gives strong predictive uncertainty and robustness because different trained solutions can disagree on poorly supported inputs. For sampled or ensemble predictions, the total predictive uncertainty is often measured by $H[\bar{p}(y \mid x)]$. The disagreement component can be measured with mutual information,

$$I[y, \theta \mid x, D] \approx H[\bar{p}(y \mid x)] - \frac{1}{M} \sum_{m=1}^{M} H[p_{\theta_m}(y \mid x)],$$

Symbols: $I[y, \theta \mid x, D]$ is mutual information between the predicted label and model parameters, D is the training data, and $H[\cdot]$ denotes entropy. which Mucsányi et al. (2024) use as one way to analyse uncertainty components in benchmark experiments.

Interpretation Deep ensembles are strong epistemic estimators because disagreement between independently trained members reflects uncertainty about the decision boundary. The expected entropy of members captures residual class ambiguity, while the gap between ensemble entropy and expected member entropy estimates epistemic mutual information. On PathMNIST this requires training several CE member checkpoints before running the ensemble wrapper.

5.4.2 Fast Deep Ensemble

Method mechanism Fast deep ensemble variants keep the ensemble predictive rule but obtain the M members with less training cost than M fully independent runs. The prediction is still

$$\bar{p}(y \mid x) = \frac{1}{M} \sum_{m=1}^{M} p_{\theta_m}(y \mid x),$$

Symbols: M is the number of fast-ensemble members, p_{θ_m} is the prediction of member m, and $\bar{p}$ is the averaged predictive distribution. and disagreement is still measured from the spread of member predictions. The difference is how the members are produced: they may come from saved checkpoints, a compressed ensemble construction, or a benchmark-specific efficient training procedure. The interpretation therefore inherits the deep-ensemble mechanism of Lakshminarayanan et al., but the members are usually more correlated than independently trained networks (Lakshminarayanan et al., 2017; Mucsányi et al., 2024). This reduces cost and can also reduce the epistemic disagreement signal.

Interpretation The uncertainty interpretation matches deep ensembles, but weakens when members correlate. Member disagreement is an epistemic signal only to the extent that the members explore different hypotheses. Fast Deep Ensemble is therefore reported separately from a full deep ensemble because its computational savings can reduce posterior diversity.

5.4.3 Shallow Ensemble

Method mechanism A shallow ensemble shares most of the feature extractor and attaches several classification heads. If the shared backbone is $h_\theta(x)$ and the heads are $g_{\phi_1}, \ldots, g_{\phi_M}$, then each member predicts

$$p_m(y \mid x) = \mathrm{softmax}(g_{\phi_m}(h_\theta(x))).$$

Symbols: $h_\theta(x)$ is the shared backbone feature, g_{ϕ_m} is head m, and $p_m(y \mid x)$ is the class distribution predicted by that head. The final prediction averages the head probabilities, as in a deep ensemble. This preserves the ensemble readout proposed by Lakshminarayanan et al. (2017) but removes independent feature extractors,

so the members explore less of the function space. In the benchmark this makes shallow ensembles a cheaper comparison point for ensemble-style uncertainty, with lower cost and often lower diversity (Mucsányi et al., 2024).

Interpretation Head disagreement provides a limited epistemic signal. Because the feature extractor is shared, the ensemble diversity is smaller than in independent deep ensembles, so the method may underestimate epistemic uncertainty. The method serves as a compute-efficient PathMNIST comparison when full ensembles are too expensive.

5.4.4 MC (Monte Carlo) Dropout

Method mechanism MC Dropout keeps dropout active at inference time and evaluates the network T times with different dropout masks ω_t. Gal and Ghahramani interpret these stochastic forward passes as approximate variational samples from a Bayesian neural network (Gal and Ghahramani, 2016). The predictive distribution is approximated by

$$p(y \mid x, D) \approx \frac{1}{T} \sum_{t=1}^{T} p_{\theta,\omega_t}(y \mid x).$$

Symbols: T is the number of stochastic forward passes, ω_t is dropout mask t, D is the training data, and p_{θ,ω_t} is the prediction under that mask. The predictive mean gives the class prediction, while variation across stochastic passes is used as a model-uncertainty proxy. As with ensembles, predictive entropy describes total uncertainty and the gap between total entropy and expected entropy estimates disagreement between stochastic subnetworks. Kendall and Gal use this Monte Carlo view to discuss epistemic uncertainty in vision models, while aleatoric uncertainty requires either data-dependent likelihood noise or the expected-entropy part of a sampled predictive distribution (Kendall and Gal, 2017; Mucsányi et al., 2024).

Interpretation The variation between stochastic passes is mainly an epistemic proxy: if the learned representation is unsupported by the training data, different subnetworks may disagree. The expected entropy of individual passes captures residual class ambiguity. The method permits the standard decomposition of total predictive entropy into expected entropy and mutual information, although the quality of the decomposition depends strongly on the dropout placement and dropout probability.

5.4.5 SWAG (Stochastic Weight Averaging-Gaussian)

Method mechanism SWAG approximates the posterior over weights by fitting a Gaussian to points along a stochastic-gradient-descent trajectory. After collecting

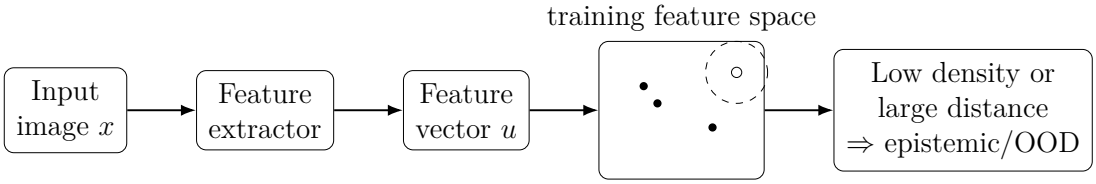
K weight vectors $w_1, \ldots, w_K$, SWAG estimates a mean $\bar{w}$ and a covariance with diagonal and low-rank components,

$$q(w) = \mathcal{N}(w; \bar{w}, \Sigma_{\text{SWAG}}) .$$

Symbols: w denotes network weights, $q(w)$ is the Gaussian weight posterior approximation, $\bar{w}$ is the SWAG mean, and Σ_{SWAG} is the estimated covariance. Maddox et al. (2019) construct Σ_{SWAG} from the empirical second moment of the trajectory and a low-rank matrix of deviations from the running mean. At test time, weights are sampled from this Gaussian and predictions are averaged. This makes SWAG a posterior-sampling method in practice: uncertainty comes from how much sampled plausible weights change the prediction. The benchmark then analyses the same entropy and mutual-information quantities used for other sampled predictive distributions (Mucsányi et al., 2024).

Interpretation Because SWAG samples weights from an approximate posterior, it supports the standard predictive decomposition into total entropy, expected entropy, and mutual information. The mutual information term is the main epistemic estimate. On PathMNIST, SWAG requires either a checkpoint trajectory or continued training from a checkpoint while collecting model states.

5.5 Density and Distance Methods



Density and distance methods move the uncertainty question from output probabilities into representation space. The classifier maps each image to a feature vector, and the method asks whether this vector lies in a familiar training region. Points far from known class clusters, or in regions with low estimated density, become epistemically uncertain cases and often potential OOD samples. Mahalanobis OOD scoring, DUQ, DDU, and Posterior Networks all use this kind of reasoning (Lee et al., 2018; van Amersfoort et al., 2020; Mukhoti et al., 2023; Charpentier et al., 2020).

Feature extractor The trained network up to an intermediate or penultimate representation layer.

Feature vector u The numerical representation of image x in the learned feature space. Similar images ought to have nearby feature vectors.

Training feature space The collection of feature vectors observed for training images. The filled dots sketch known training examples or class regions.

Open query point The new test example under evaluation. Its distance to training regions, or density under a fitted model, becomes the uncertainty cue (Lee et al., 2018; van Amersfoort et al., 2020; Mukhoti et al., 2023).

Low density or large distance A signal that the input is poorly supported by the training distribution, so these methods target epistemic and OOD uncertainty.

5.5.1 DDU (Deep Deterministic Uncertainty)

Method mechanism DDU fits a density model to the feature representation of a deterministic classifier. Let $u = h_\theta(x)$ be the penultimate feature vector. After classifier training, DDU fits class-conditional Gaussian or Gaussian-mixture statistics in this feature space and estimates the marginal feature density,

$$p(u) = \sum_{c=1}^{K} \pi_c \mathcal{N}(u; \mu_c, \Sigma_c).$$

Symbols: u is the feature vector, c indexes classes, π_c is the class mixture weight, and μ_c, Σ_c are the class mean and covariance. Mukhoti et al. (2023) argue that a good deterministic uncertainty method should combine a discriminative classifier with a feature space where density remains meaningful. The uncertainty score is then the negative log density $-\log p(h_\theta(x))$ or an equivalent low-density score. High uncertainty means the test feature is poorly supported by the training feature distribution, even if the softmax output is sharp. Mucsányi et al. (2024) include DDU as a density-based method for disentanglement tests.

Interpretation DDU is a deterministic epistemic method: low feature density indicates that the input is not well supported by the training distribution. This suits PathMNIST-C and cross-domain medical shifts. It does not separate irreducible class ambiguity unless ambiguous samples systematically occupy low-density or overlapping regions.

5.5.2 DUQ (Deterministic Uncertainty Quantification)

Method mechanism DUQ learns class prototypes in feature space and scores an input by its radial-basis-function similarity to those prototypes. In simplified notation, for feature representation $f_\theta(x)$ and class centroid e_c, the class score has the form

$$K_c(x) = \exp\left(-\frac{\|f_\theta(x) - e_c\|_2^2}{2\sigma^2}\right).$$

Symbols: $K_c(x)$ is the similarity to class c, $f_\theta(x)$ is the learned feature vector, e_c is the class centroid, and σ controls the RBF length scale. van Amersfoort et al. (2020) combine these distance-aware class scores with centroid updates and a gradient

penalty that encourages smooth changes of the output with respect to the input. The prediction is confident near a learned class centroid and uncertain far away from all centroids. This differs from a plain softmax classifier because distance from the training representation is part of the score itself. The benchmark uses this distance-aware uncertainty as an epistemic and OOD-oriented method (Mucsányi et al., 2024).

Interpretation DUQ targets epistemic uncertainty by assigning low confidence to points far from learned class prototypes. This suits OOD detection and data-scarcity experiments. It addresses aleatoric uncertainty less directly: ambiguous images can become uncertain if they lie between prototypes, but irreducible label noise lacks an explicit model. DUQ is therefore treated as a distance-aware epistemic method; smaller batch sizes handle its practical memory constraints.

5.5.3 Mahalanobis

Method mechanism The Mahalanobis method fits class-conditional Gaussian statistics to hidden representations of a trained network. For representation $u_\ell(x)$ at layer ℓ, class mean $\mu_{\ell,c}$, and shared covariance Σ_ℓ, the class score is based on the negative Mahalanobis distance,

$$s_{\ell,c}(x) = -\left(u_\ell(x) - \mu_{\ell,c}\right)^\top \Sigma_\ell^{-1} \left(u_\ell(x) - \mu_{\ell,c}\right).$$

Symbols: $s_{\ell,c}$ is the layer-class score, $u_\ell(x)$ the representation at layer ℓ, $\mu_{\ell,c}$ the class mean, and Σ_ℓ the shared covariance. Lee et al. (2018) use the maximum class score, optionally combined across layers and calibrated with a held-out set, for OOD and adversarial detection. A test sample far from all class-conditional feature distributions receives a low confidence score. In the benchmark this makes Mahalanobis a distance-based epistemic/OOD method rather than a method that models label ambiguity directly (Mucsányi et al., 2024).

Interpretation Mahalanobis mainly targets epistemic and OOD uncertainty. Inputs far from class-conditional feature clouds receive high uncertainty, which matches the desired behavior under distribution shift. It does not model aleatoric uncertainty directly, although ambiguous images may appear between class-conditional densities. Its PathMNIST evaluation consequently focuses on OOD AUROC and scarcity sensitivity rather than label noise.

5.5.4 PostNet (Posterior Network)

Method mechanism Posterior Networks estimate uncertainty by combining a learned latent representation with class-conditional density modelling. The network maps x to a latent vector z, and a density estimator models $p(z \mid y = c)$ for each

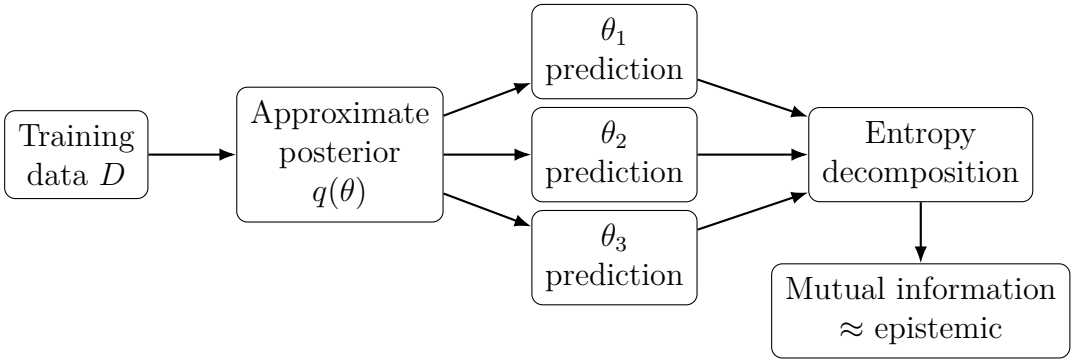
class. Charpentier et al. (2020) convert these class-conditional densities into pseudo-counts or evidence for a Dirichlet distribution over class probabilities. A simplified view is

$$\alpha_c(x) = 1 + \beta_c \, p(z \mid y = c),$$

Symbols: α_c is the Dirichlet concentration for class c, β_c is an evidence scaling factor, z is the latent representation, and $p(z \mid y = c)$ is the class-conditional latent density. where α_c is the Dirichlet concentration for class c and β_c scales the evidence. The posterior mean $\alpha_c / \sum_j \alpha_j$ gives the class probability, while the total concentration $\sum_j \alpha_j$ expresses how much support the latent density provides. Low density gives low evidence and therefore epistemic uncertainty; competing class evidence can express ambiguity. Mucsányi et al. (2024) include this method because it can expose both density support and class conflict in a single posterior-predictive object.

Interpretation The method distinguishes class ambiguity from lack of support in latent space. Dense regions near known classes support confident predictions, while low-density regions indicate epistemic uncertainty. If an image lies between class densities, the posterior can express ambiguity. For PathMNIST, rare tissue patterns and corrupted images can appear as low-density or mixed-density cases.

5.6 Bayesian and Posterior Methods



Bayesian and posterior methods represent uncertainty over the model parameters themselves. Instead of assuming that training produced one fixed best network, they approximate a distribution over plausible parameter settings. Several parameter samples produce predictions, and their disagreement reflects epistemic uncertainty. Laplace approximations implement this parameter-posterior approach in deep networks (Ritter et al., 2018). SNGP follows a related but distinct route: it combines a deterministic, distance-aware network with a Gaussian-process output layer rather than approximating a posterior over all network parameters (Liu et al., 2020).

Training data D The dataset that constrains which model parameters remain plausible.

Approximate posterior $q(\theta)$ A tractable approximation to the Bayesian posterior over network parameters. It expresses which parameter settings remain plausible after training (Ritter et al., 2018).

θ_i prediction A prediction made with one sampled or approximated parameter setting from $q(\theta)$.

Entropy decomposition This split separates expected data uncertainty from disagreement between parameter samples.

Mutual information ≈ epistemic Mutual information between predictions and parameters measures how much the prediction would change if the true parameter setting were known, so it serves as an epistemic uncertainty proxy in Bayesian uncertainty decompositions (Gal and Ghahramani, 2016; Kendall and Gal, 2017).

5.6.1 Laplace

Method mechanism Laplace approximations fit a Gaussian posterior around a trained maximum-a-posteriori solution θ_{MAP}. Around this optimum, the negative log posterior is approximated by a quadratic expansion. This gives

$$q(\theta) = \mathcal{N}\left(\theta_{\text{MAP}}, \Lambda^{-1}\right), \qquad \Lambda \approx H + \lambda I,$$

Symbols: $q(\theta)$ is the approximate posterior, θ_{MAP} the fitted parameters, Λ the posterior precision, H a curvature matrix, λ prior precision, and I the identity matrix. where H is a Hessian or generalized Gauss–Newton curvature approximation and λI represents prior precision or weight decay. Ritter et al. (2018) describe scalable neural-network Laplace approximations using structured curvature, such as Kronecker-factored forms. Prediction requires integrating over this approximate posterior,

$$p(y \mid x, D) \approx \int p_\theta(y \mid x) q(\theta)\, d\theta,$$

Symbols: $p(y \mid x, D)$ is the posterior predictive distribution, $p_\theta(y \mid x)$ the prediction for one parameter setting, and $d\theta$ indicates integration over parameters. which is usually estimated with samples or a local linear approximation. Since uncertainty comes from posterior spread over parameters, Laplace is mainly an epistemic method in the benchmark (Mucsányi et al., 2024).

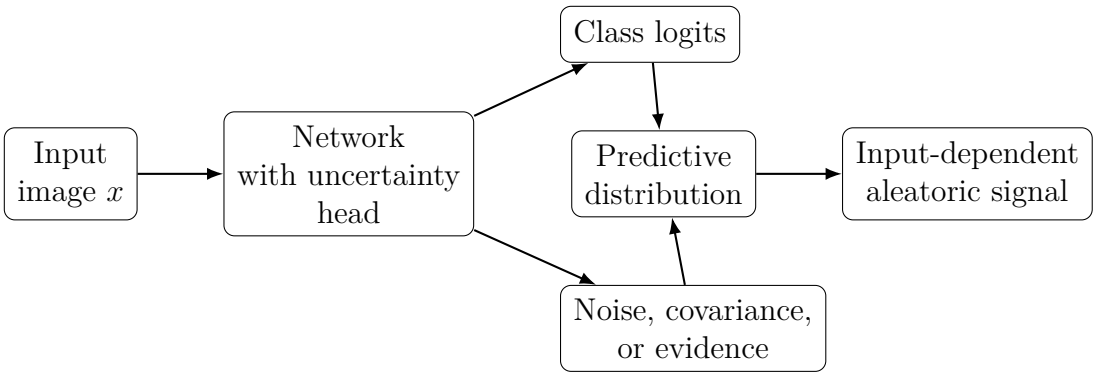
Interpretation Laplace is an epistemic posterior approximation. Predictive samples from the approximate posterior decompose into expected entropy and mutual information. In the experiments it is interpreted as a post-hoc epistemic method, with numerical stabilization applied to the predictive covariance where required.

5.6.2 SNGP (Spectral-normalized Neural Gaussian Process)

Method mechanism SNGP combines a spectrally normalized feature extractor with a Gaussian-process-style output layer. Spectral normalization constrains layer Lipschitz constants, which helps preserve distance information in the learned representation. Liu et al. (2020) then replace the final linear classifier with a random-feature Gaussian-process approximation. In simplified notation, features $h_\theta(x)$ are mapped to random Fourier features $\phi(h_\theta(x))$, and the output layer keeps a covariance estimate for its weights. The predictive variance increases when a test point is far from the training features that updated this covariance. The method is deterministic during training but produces a distance-aware posterior-like uncertainty estimate at the output, so the benchmark treats it as an epistemic/OOD method (Mucsányi et al., 2024).

Interpretation SNGP is primarily epistemic: uncertainty ought to increase when a PathMNIST input lies far from the training feature manifold or when training evidence is scarce. This makes it relevant for PathMNIST-C or cross-domain medical shifts. It lacks a direct model for irreducible label ambiguity, so aleatoric uncertainty comes indirectly from predictive entropy.

5.7 Heteroscedastic and Evidential Methods



Heteroscedastic and evidential methods make the classifier output more than a single deterministic class-probability vector. Heteroscedastic classifiers predict image-dependent noise or covariance parameters alongside class logits. This targets aleatoric uncertainty because ambiguous or noisy images can receive broader predictive distributions even when the model parameters stay fixed. Evidential methods instead predict evidence for a Dirichlet distribution over class probabilities; this can expose low support or class conflict, but it should not be read as a clean aleatoric mechanism without empirical checks (Kendall and Gal, 2017; Collier et al., 2021; Sensoy et al., 2018).

Class logits The usual class scores used to form a prediction over PathMNIST classes.

Noise/evidence Additional quantities predicted by the network that describe uncertainty in the output distribution rather than only the most likely class (Collier et al., 2021; Sensoy et al., 2018).

Predictive distribution The final class distribution after combining class evidence with the predicted uncertainty parameters.

Aleatoric signal The uncertainty component that changes with the particular image, for example because tissue appearance is ambiguous, corrupted, or inherently difficult to label.

5.7.1 HET (Heteroscedastic Classification)

Method mechanism Heteroscedastic classification methods predict image-dependent uncertainty in logit or latent space. Instead of using fixed logits, the model predicts parameters of a logit distribution. One common form is

$$z(x, \epsilon) = \mu_\theta(x) + L_\theta(x)\epsilon,$$
$$\epsilon \sim \mathcal{N}(0, I),$$

Symbols: $z(x, \epsilon)$ is a sampled logit vector, $\mu_\theta(x)$ the predicted mean logits, $L_\theta(x)$ a covariance factor, and ϵ standard Gaussian noise. where $\mu_\theta(x)$ is the mean logit vector and $L_\theta(x)L_\theta(x)^\top$ defines an input-dependent covariance. The predictive distribution is estimated by Monte Carlo integration,

$$p(y \mid x) \approx \frac{1}{S}\sum_{s=1}^{S} \mathrm{softmax}(z(x, \epsilon_s)).$$

Symbols: S is the number of Monte Carlo samples, ϵ_s is sample s of the logit noise, and $p(y \mid x)$ is the averaged predictive distribution. Kendall and Gal frame such input-dependent noise as aleatoric uncertainty in vision, and Collier et al. adapt heteroscedastic modelling to classification with structured logit uncertainty (Kendall and Gal, 2017; Collier et al., 2021). Mucsányi et al. (2024) include HET-style models to test whether this input-dependent predictive spread behaves like aleatoric uncertainty under benchmark stressors.

Interpretation The input-dependent noise term targets aleatoric uncertainty. If a PathMNIST image is noisy or genuinely ambiguous, the heteroscedastic head can increase predictive dispersion. Epistemic uncertainty is less direct because the method does not necessarily maintain a posterior over weights. Experiments must therefore assess epistemic sensitivity separately using data scarcity and OOD evaluation.

5.7.2 HET-XL (Extended Heteroscedastic Classification)

Method mechanism HET-XL is the larger heteroscedastic variant used in the benchmark code. It keeps the same stochastic-logit idea as HET, but uses a more

expressive structured covariance for the logit distribution. Rather than storing a full $K \times K$ covariance matrix without structure, the model uses a low-rank or factorised parameterisation so that samples can be written as

$$z(x, \epsilon) = \mu_\theta(x) + L_\theta(x)\epsilon.$$

Symbols: $z(x, \epsilon)$ is a sampled logit vector, $\mu_\theta(x)$ the mean logits, $L_\theta(x)$ the low-rank or factorised covariance term, and ϵ random noise. More Monte Carlo samples or a larger uncertainty head can represent richer class correlations in the predictive distribution. This follows the heteroscedastic classification formulation of Collier et al., building on the aleatoric uncertainty view of Kendall and Gal (Kendall and Gal, 2017; Collier et al., 2021). The benchmark treats HET-XL as a more expressive aleatoric model and tests whether this extra expressivity improves separation from epistemic stressors (Mucsányi et al., 2024).

Interpretation Like HET, HET-XL is mainly an aleatoric method because the noise distribution is conditioned on the input. The low-rank structure increases expressivity without forming a full covariance matrix. On PathMNIST, the method is expected to be most informative under label-noise or tissue-ambiguity probes, while epistemic behavior must be checked separately via scarcity and OOD tests.

5.7.3 HetClassNN (Heteroscedastic Classification Neural Network)

Method mechanism HetClassNN combines heteroscedastic classification with stochastic sampling over class logits. The network predicts a distribution over logits rather than a single logit vector, and repeated samples are passed through the softmax before averaging. This gives

$$p(y \mid x) \approx \frac{1}{S} \sum_{s=1}^{S} \mathrm{softmax}(\mu_\theta(x) + \eta_s(x)) \,,$$

Symbols: S is the number of stochastic logit samples, $\mu_\theta(x)$ the mean logits, $\eta_s(x)$ sample s of input-dependent logit noise, and $p(y \mid x)$ the averaged prediction. where $\eta_s(x)$ denotes input-dependent stochastic logit noise. This mechanism follows the same aleatoric motivation as Kendall and Gal and the heteroscedastic classification treatment of Collier et al. (Kendall and Gal, 2017; Collier et al., 2021). In the benchmark implementation, HetClassNN is evaluated as a stochastic predictive classifier whose repeated samples permit entropy-based uncertainty summaries (Mucsányi et al., 2024).

Interpretation The heteroscedastic component targets aleatoric uncertainty, while repeated stochastic samples may also respond to epistemic instability. Because the architecture does not separate these effects by itself, the PathMNIST evaluation reports predictive entropy and, when samples are available, expected entropy and mutual information.

5.7.4 Evidential Deep Learning

Method mechanism Evidential deep learning predicts evidence parameters for a Dirichlet distribution over class probabilities. The network outputs non-negative evidence $e_k(x) \geq 0$ for each class, usually through a non-negative activation. Dirichlet concentrations are then

$$\alpha_k(x) = e_k(x) + 1, \qquad S(x) = \sum_{k=1}^{K} \alpha_k(x).$$

Symbols: $e_k(x)$ is evidence for class k, $\alpha_k(x)$ the Dirichlet concentration, $S(x)$ total concentration, and K the number of classes. The predictive mean is $\mathbb{E}[p_k] = \alpha_k/S$, and Sensoy et al. (2018) define an uncertainty mass that decreases as total evidence increases, often written as K/S for K classes. The training objective encourages correct classes to receive evidence while penalising unsupported evidence. In the benchmark, this makes EDL a method whose uncertainty depends on both class conflict and total evidence, so experiments must test whether those effects align with aleatoric ambiguity or epistemic lack of support (Mucsányi et al., 2024).

Interpretation Low total evidence indicates uncertainty. Ambiguous PathMNIST images tend to yield diffuse evidence across classes, while unfamiliar or shifted images tend to yield low evidence because the model has little support for a confident decision. In practice, evidential methods may mix aleatoric and epistemic effects unless the training objective penalizes unjustified evidence on difficult samples.

6 CIFAR-10 Benchmark Reproduction

Before the uncertainty-disentanglement protocol was transferred to medical images, its implementation was checked on CIFAR-10. This natural-image benchmark verifies that the recovered hyperparameters, corruption evaluation, and decomposition analysis behave similarly to the original benchmark study (Mucsányi et al., 2024). It serves as a reproduction check rather than a medical target dataset.

6.1 Role in the Thesis

The reproduction follows the original benchmark in three ways. First, the full-data runs compare method-specific correctness-ranking metrics against recovered original run summaries. Second, the data-scarcity runs repeat the 100%, 50%, and 10% training-fraction protocol and compare the information-theoretic decomposition values with Appendix H.9 of the paper. Third, the CIFAR-10C evaluations test whether the implementation can evaluate corruption or distribution-shift behaviour in the same spirit as Appendix G. These checks use different metrics because they answer different validation questions. Correctness AUROC checks whether the

recovered uncertainty score ranks wrong predictions above correct predictions. Decomposition means check whether the AU and EU estimates have the same scale and training-fraction behaviour as the paper. CIFAR-10C OOD AUROC checks whether uncertainty scores separate corrupted images from clean in-distribution images. Together, the three axes check predictive usefulness, decomposition behaviour, and distribution-shift sensitivity rather than accuracy alone.

This design separates validation from transfer. CIFAR-10 checks whether the benchmark machinery can be reproduced at all. PathMNIST then asks whether the same machinery transfers to a larger medical histopathology benchmark. MHIST adds the separate question that CIFAR-10 and PathMNIST cannot answer directly: whether uncertainty estimates relate to human label disagreement.

6.2 Full-Data Reproduction

Table 4 reports the full-data reproduction check. The reference values come from archived summaries of the paper experiments, and the reproduced values use the corresponding configurations. The metric is the method-specific correctness AUROC used for checkpoint selection or final reporting in the original run records. A positive Δ means that the local reproduction is larger than the recovered original value.

Table 4: CIFAR-10 full-data reproduction check against the archived original results. "Ref. best/final" and "R best/final" are the method-specific correctness-ranking metrics; "R acc." is reproduced test accuracy, and Δ is reproduction minus reference. Dashes indicate that no comparable reference metric was available.

Method	Ref. best	Ref. final	R best	R final	R acc.	Δ best
SNGP (SN off)	0.9498	0.9211	0.9509	0.9300	0.9581	+0.0011
EDL	0.9338	0.9188	0.9355	0.9299	0.9354	+0.0017
DDU	0.9486	0.9337	0.9345	0.9297	0.9550	-0.0141
PostNet	0.9334	0.9166	0.9298	0.9127	0.9575	-0.0036
Loss prediction	–	–	0.8214	0.7144	0.9426	–
HET	0.9490	0.9462	0.9376	0.9253	0.9540	-0.0114
SNGP (SN on)	0.9476	0.9283	0.9422	0.9337	0.9570	-0.0054
Shallow Ensemble	0.9438	0.9239	0.9462	0.9339	0.9573	+0.0024
HET-XL	0.9441	0.9309	0.9433	0.9433	0.9509	-0.0008
HetClassNN	0.9532	0.9518	0.9460	0.9307	0.9603	-0.0072
CE baseline	0.9469	0.9374	0.9391	0.9379	0.9553	-0.0078
Correctness prediction	0.9259	0.9193	0.9176	0.9039	0.9433	-0.0083
MC Dropout	0.9495	0.9385	0.9444	0.9384	0.9588	-0.0051

Most full-data runs with recovered reference metrics differ from the originals by a few 10^{-3} to about $2 \cdot 10^{-2}$, providing the required check of comparable benchmark behaviour rather than a bitwise reproduction. Loss Prediction is listed without a reference delta because the recovered summary did not contain a comparable original best/final value. The remaining gaps are also informative. DDU, HET, HetClassNN, and the corrected correctness-prediction run show the larger deviations and are interpreted more cautiously in the transfer experiments.

6.3 Comparison with Published Findings

The reproduced CIFAR-10 results are consistent with the qualitative interpretation of Mucsanyi et al. The authors ask whether an uncertainty value value responds to the intended uncertainty source. They report that aleatoric and epistemic components from common decompositions are often strongly correlated, that specialized uncertainty tasks are harder than generic predictive uncertainty, and that task-specific estimators can outperform estimators that look theoretically appropriate from the decomposition alone (Mucsányi et al., 2024). Most reproduced full-data correctness-ranking metrics differ from the recovered runs by less than approximately $2 \cdot 10^{-2}$, supporting the implementation check without establishing source separation.

Under data scarcity, several decomposition means match the paper closely, but AU often changes more than EU, including for EDL and HET-XL. The medical experiments accordingly report paired changes rather than treating either component as ground truth. CIFAR-10C is similarly method-dependent: Mahalanobis has a stronger dedicated OOD score than its entropy-style score. Medical corruption results are thus treated as task-specific shift evidence.

6.4 Decomposition under Data Scarcity

The data-scarcity reproduction is closer to the disentanglement question than plain accuracy. The benchmark reports information-theoretic decomposition means for different training fractions. Predictive entropy is split into an expected entropy term, interpreted as aleatoric uncertainty, and a mutual-information term, interpreted as epistemic uncertainty. Table 5 summarizes the reproduced comparison to Appendix H.9, Table H.2 of the paper. Only rows with like-for-like paper values are included.

Several decomposition means are close to the paper values, particularly for EDL, HET-XL at 10%, HetClassNN, and SNGP. This partial agreement checks those specific calculations but does not validate every method or condition. The main outlier is PostNet at 10% training data, where the reproduced aleatoric mean is 0.222 rather than the paper value 0.480. This difference is reported because it is a reason to avoid treating PostNet as a cleanly reproduced reference when discussing scarcity behaviour.

The Bregman decomposition was also evaluated for all 30 CIFAR-10 scarcity conditions. It uses the same within-predictor AU term reported in the IT comparison, while its EU increases from full data to 10% data by 0.2986 for EDL, 0.1053 for HetClassNN, 0.0162 and 0.0173 for the two SNGP configurations, and 0.0123 for HET-XL. The remaining Bregman EU changes are below 0.008. Thus, the Bregman decomposition assigns a larger EU scarcity response to EDL and HetClassNN than the IT comparison. The size of this difference depends on the formula, and neither decomposition produces a uniform method-level response.

Table 5: CIFAR-10 data-scarcity reproduction for all rows with a like-for-like paper value. "Split" is the training-data fraction that was kept. AU and EU are information-theoretic decomposition means; Δ is reproduction minus paper, so a negative value indicates a lower reproduced mean.

Method	Split	Paper AU	Repro AU	Δ AU	Paper EU	Repro EU	Δ EU
EDL	100%	0.716	0.678	-0.038	0.083	0.077	-0.006
EDL	50%	0.869	0.911	+0.042	0.106	0.113	+0.007
EDL	10%	1.701	1.657	-0.044	0.296	0.287	-0.009
HET	100%	0.088	0.072	-0.016	0.001	0.000	-0.001
HET	50%	0.108	0.078	-0.030	0.001	0.001	0.000
HET	10%	0.200	0.158	-0.042	0.004	0.007	+0.003
HET-XL	100%	0.079	0.051	-0.028	0.004	0.001	-0.003
HET-XL	50%	0.104	0.144	+0.040	0.007	0.009	+0.002
HET-XL	10%	0.214	0.213	-0.001	0.014	0.013	-0.001
HetClassNN	100%	0.080	0.084	+0.004	0.034	0.034	0.000
HetClassNN	50%	0.082	0.080	-0.002	0.047	0.047	0.000
HetClassNN	10%	0.184	0.187	+0.003	0.109	0.110	+0.001
PostNet	100%	0.118	0.102	-0.016	0.001	0.001	0.000
PostNet	50%	0.109	0.131	+0.022	0.001	0.002	+0.001
PostNet	10%	0.480	0.222	-0.258	0.008	0.002	-0.006
Shallow	100%	0.076	0.052	-0.024	0.000	0.000	0.000
Shallow	50%	0.093	0.084	-0.009	0.000	0.000	0.000
Shallow	10%	0.228	0.238	+0.010	0.001	0.000	-0.001
SNGP	100%	0.058	0.056	-0.002	0.000	0.000	0.000
SNGP	50%	0.082	0.081	-0.001	0.001	0.001	0.000
SNGP	10%	0.201	0.214	+0.013	0.016	0.016	0.000

6.5 Corruption Evaluation

CIFAR-10C provides the reproduction check for corruption and distribution-shift evaluation. The evaluated method set covers the CIFAR-10C corruption protocol. Table 6 reports the reproduced OOD-detection AUROC. Severity 2 matches the paper's Appendix G, Figure G.1 setting, but exact paper values were unavailable as machine-readable numbers in the local extraction files. The table reports only reproduced values.

The Bregman AU ranking is numerically aligned with total uncertainty in this evaluation: its mean AUROC differs from the reported total column by less than 10^{-3} for every full-data row. The two EU formulas also agree closely; their largest full-data all-severity difference is 0.0117 for Shallow Ensemble.

Mean severity-2 OOD AUROC drops from 0.673 at full data to 0.667 at 50% training data and 0.593 at 10% training data. The score separates clean and corrupted images less effectively as training data is reduced. Corruption AUROC remains a task-specific shift measure, not a direct measure of epistemic uncertainty.

Figures 4 and 5 compare all available training fractions and corruption severities with the same coordinate system. For scarcity, a source-specific epistemic response requires a positive EU change that exceeds the AU change. Most configurations do not meet both conditions; only 2 of the 16 method–fraction comparisons have a larger EU than AU response. EDL has the largest absolute response, but it is dominated by AU (Figure 4). For CIFAR-10-C, a useful epistemic shift signal requires EU AUROC

Table 6: Reproduced CIFAR-10C OOD-detection AUROC for full-data runs. "Total" uses total predictive uncertainty; IT EU and Br EU use the epistemic components of the information-theoretic and Bregman decompositions. Severity-2 values are averaged across corruptions at severity 2; all-severity values use every evaluated corruption and severity.

Method	S2 total	S2 IT EU	S2 Br EU	All total	All IT EU	All Br EU
DDU	0.681	0.500	0.500	0.725	0.500	0.500
EDL	0.669	0.669	0.669	0.715	0.715	0.715
HET	0.685	0.672	0.672	0.728	0.717	0.718
HET-XL	0.645	0.644	0.644	0.696	0.695	0.695
HetClassNN	0.685	0.685	0.684	0.730	0.729	0.727
Loss prediction	0.661	0.500	0.500	0.705	0.500	0.500
PostNet	0.655	0.622	0.622	0.700	0.655	0.654
Shallow Ensemble	0.650	0.605	0.595	0.699	0.649	0.637
SNGP (SN off)	0.700	0.691	0.691	0.748	0.741	0.742
SNGP (SN on)	0.697	0.693	0.692	0.745	0.741	0.740

to be above chance and higher than AU AUROC. Although EU-based detection is above chance in all 40 displayed method–severity comparisons, EU exceeds AU in only 7. Most points remain on the wrong side of the equal-response diagonal, and PostNet is clearly AU-dominated (Figure 5).

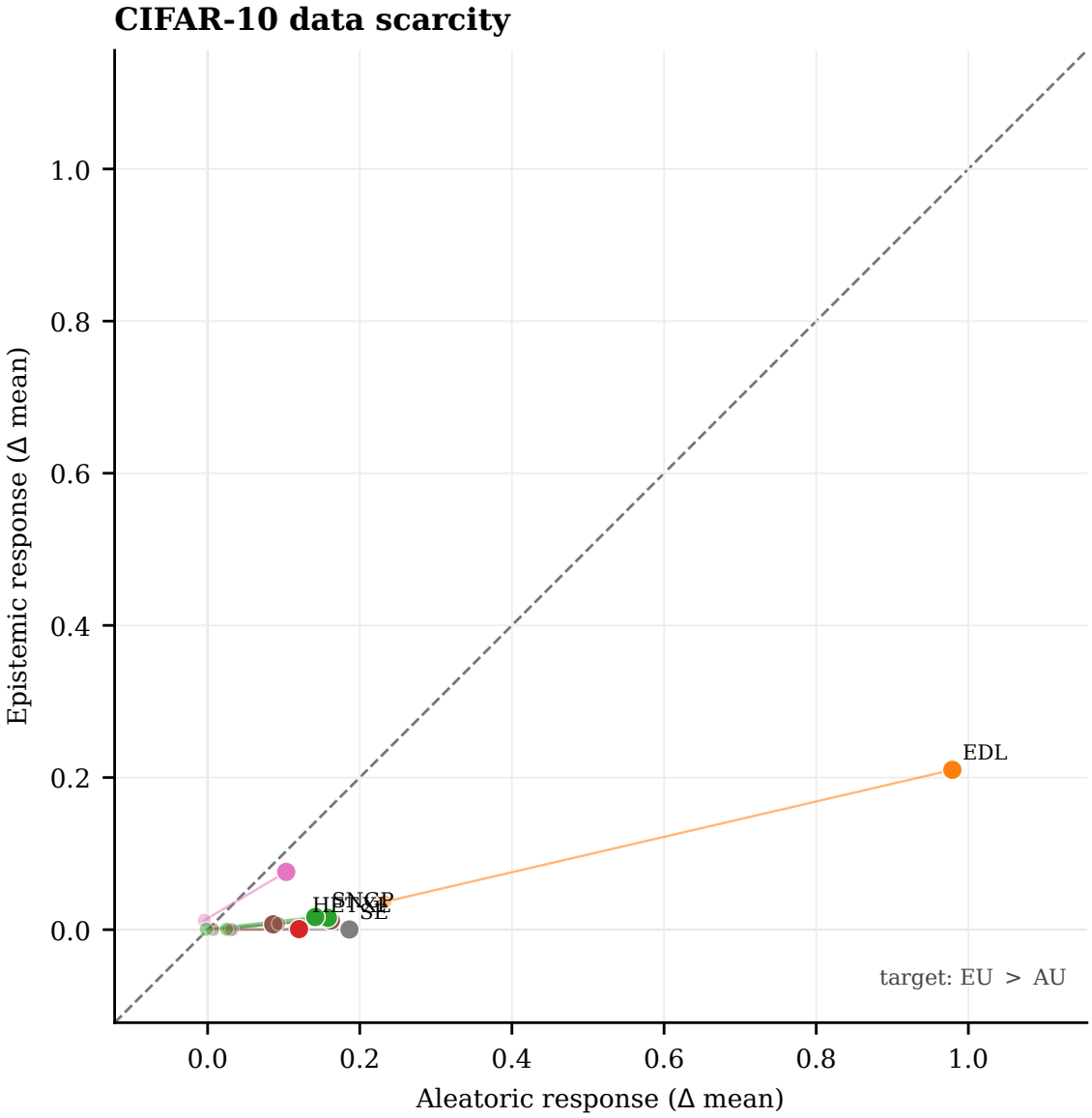


Figure 4: Information-theoretic AU and EU responses under CIFAR-10 data scarcity. Lines connect the 50% and 10% training fractions; larger, darker points indicate stronger scarcity. The intended epistemic response has a positive ΔEU that exceeds ΔAU.

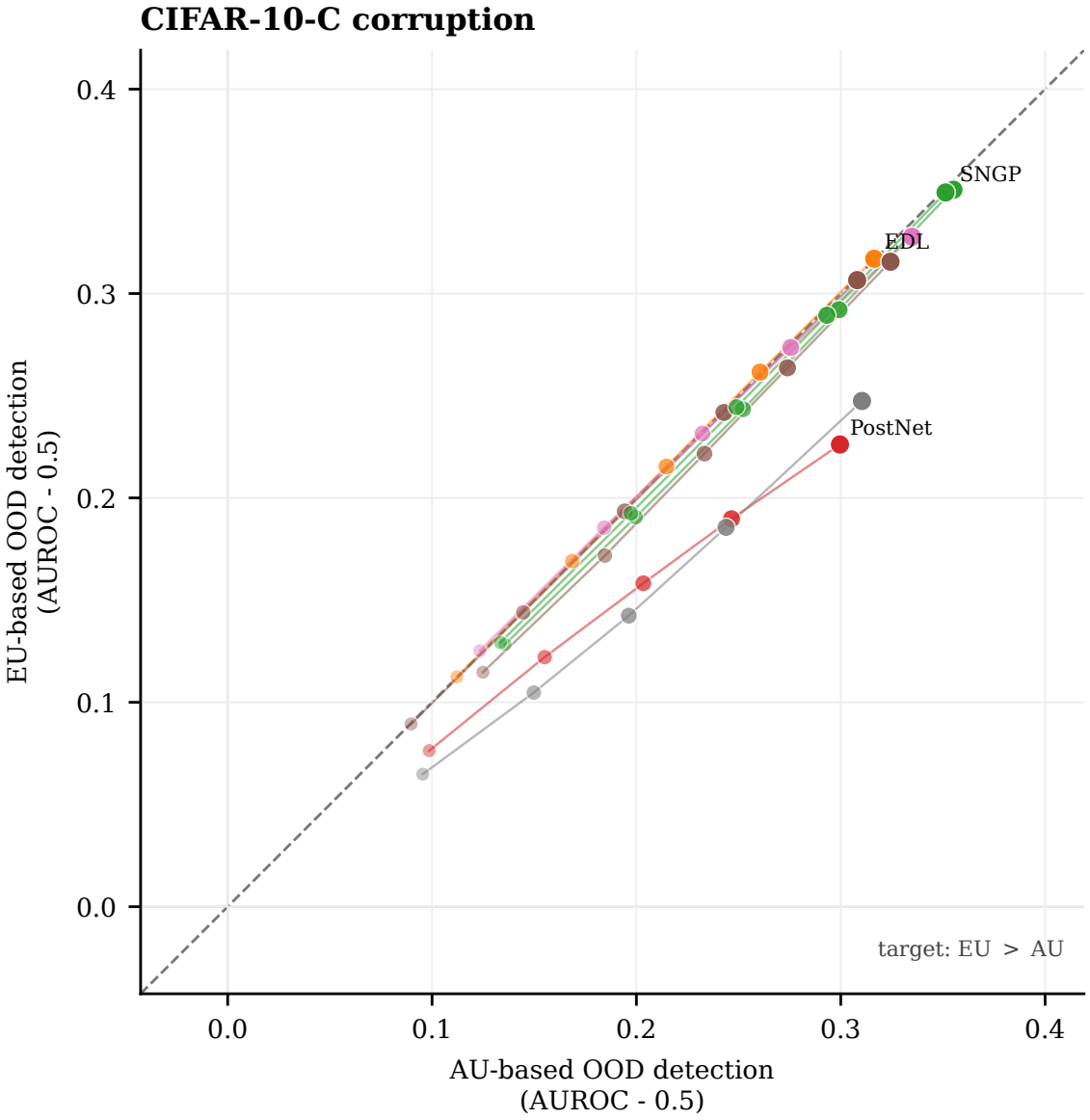


Figure 5: Information-theoretic AU and EU OOD-detection responses across CIFAR-10-C corruption severities 1–5. Larger, darker points indicate stronger corruption. The intended epistemic response has higher EU than AU AUROC; values above zero denote above-chance detection.

6.6 Post-Hoc Results

The post-hoc methods use a common set of five cross-entropy member checkpoints. Deep Ensemble, Temperature Scaling, Mahalanobis, Laplace, and SWAG are evaluated from this shared base-model configuration. This common basis controls variation due to the base classifiers and allows comparison of the uncertainty wrappers and their decomposition outputs.

Table 7: Additional CIFAR-10 post-hoc results. AU is the shared within-predictor entropy term; IT EU and Br EU denote the clean-test epistemic means from the information-theoretic and Bregman decompositions. A dash indicates that the method does not provide a measured epistemic component. OOD AUROC separates clean CIFAR-10 from CIFAR-10C, averaged across corruptions at severity 2 or across all evaluated severities.

Method	AU	IT EU	Br EU	Severity-2 OOD AUROC	All-severity OOD AUROC
Mahalanobis	0.0759	–	–	0.6680	0.7159
SWAG	0.2708	0.3030	0.4341	0.7165	0.7479
Laplace	0.1300	0.0606	0.0649	0.6956	0.7450

The post-hoc results confirm that the shared procedure covers base models, post-hoc wrappers, decomposition extraction, and corruption analysis. CIFAR-10 remains a technical validation stage rather than evidence about uncertainty components in medical data.

7 PathMNIST Experiments and Results

PathMNIST is the primary medical transfer dataset. The protocol remains fixed rather than being tuned for the medical task, following the benchmark principle of comparing methods under shared conditions and separate uncertainty probes (Mucsányi et al., 2024). This experiment transfers the recovered CIFAR-10 workflow to a nine-class colorectal histopathology dataset rather than reproducing a published PathMNIST result (Yang et al., 2023).

7.1 Experimental Protocol

All PathMNIST training-time methods use the same base recipe unless a method requires a specific loss or wrapper. The shared configuration uses a `timm/resnet_50` backbone, 224×224 inputs, AdamW, weight decay $2 \cdot 10^{-5}$, 90 epochs, batch size 128, 16 accumulation steps, ImageNet normalisation, and seed 42. The effective learning rate is 0.008 because the repository scales `lr-base=0.001` by the effective batch-size ratio. The classifier uses nine output classes.

The corruption axis uses the PathMNIST-C corruption files from MedMNIST-C (Di Salvo et al., 2024). These corruptions are treated as distribution-shift and

Table 8: PathMNIST experiment axes used in the thesis.

Axis	Intervention	Evaluation target
Clean full-data training	Train on the official training split and evaluate on clean validation/test data	Predictive performance, calibration, and correctness ranking
Data scarcity	Retrain with 50% and 10% of the training split	Response to limited training evidence
PathMNIST-C corruption	Evaluate trained checkpoints on corrupted PathMNIST-C images	Response to synthetic distribution shift and image degradation
Synthetic label noise	Retrain with 10%, 20%, and 40% wrong training labels	Response to corrupted training labels

image-degradation tests; they are not a pure intervention on epistemic uncertainty. The synthetic label-noise axis is a controlled probe of corrupted training labels. It is related to aleatoric uncertainty, but it does not reproduce natural image ambiguity or prove that an AU response is irreducible. The metrics are chosen to match these axes rather than to form one global score. Clean full-data runs report the checkpoint-selection metric, which is a method-specific correctness AUROC on the validation split. It tests whether the uncertainty score ranks wrong predictions above correct predictions, adding information not captured by accuracy alone. Data-scarcity and label-noise runs use changes in mean AU and EU because their scientific question is whether the intended component changes more than the off-target component when the intervention is applied. PathMNIST-C uses OOD AUROC because the target is binary at evaluation time: clean images should receive lower shift uncertainty than corrupted images. Total entropy is reported beside method-specific scores as a common baseline, but the interpretation follows the score each method is designed to produce.

7.2 Experimental Coverage

All reported PathMNIST classifiers use nine output classes. Results affected by an earlier class-count mismatch are excluded, and the analysis uses only outputs from the corrected configuration. The full-data and data-scarcity experiments provide the broadest PathMNIST evidence. The synthetic label-noise experiment covers 14 methods at each of the 10%, 20%, and 40% noise levels. The post-hoc analysis covers Deep Ensemble, Temperature Scaling, Mahalanobis, SWAG, DDU, and Laplace; numerical stabilization of the predictive covariance is used for Laplace. Post-hoc and label-noise results are reported separately from the main trainable-method table because they use different checkpoint and wrapper assumptions.

Several constraints shape the interpretation. HetClassNN is excluded from the full-data summary because its uncertainty output could not be validated after correcting

the class count. PostNet, Shallow Ensemble, and DUQ are represented by results from the corrected configuration. DDU's near-chance correctness-ranking AUROC makes it unsuitable for meaningful clean-test interpretation. PathMNIST-C is reported separately as a method-level OOD evaluation for the trainable methods.

7.3 Full-Data Training Results

Table 9 reports the interpretable full-data results. The metric is the best validation correctness-ranking metric used by the training script for checkpoint selection. It is not a claim that the listed method is the most accurate classifier; it asks whether the uncertainty score is effective for ranking likely errors on clean validation data. The table is a training summary; the later scarcity, label-noise, and post-hoc tables contain the source-probe comparisons.

Table 9: Interpretable full-data PathMNIST training results. "Best metric" is the highest method-specific validation correctness-ranking metric; "Best epoch" is the epoch at which it occurs.

Method	Best metric	Best epoch
CE baseline	0.9833	53
Correctness prediction	0.9869	62
Deep correctness prediction	0.9868	62
Loss prediction	0.9857	71
Deep loss prediction	0.9841	34
MC Dropout	0.9884	79
EDL	0.9974	81
SNGP	0.9878	44
HET	0.9875	71
HET-XL	0.9887	76
PostNet	0.9876	50
Shallow Ensemble	0.9842	66
DUQ	0.9857	40

The full-data runs produce finite checkpoint-selection metrics for most methods, but these metrics test error ranking rather than uncertainty-source separation. EDL reaches the highest checkpoint selection metric in the evaluated method set. The table uses the checked PostNet, Shallow Ensemble, and DUQ results. HetClassNN remains excluded because its corrected uncertainty output is not reliable enough for a meaningful comparison.

7.4 Uncertainty-Source Analysis

The PathMNIST analysis applies the evaluation procedure from Section 3.9. Data scarcity and synthetic label noise are reported through the direct component shifts

ΔAU and ΔEU using both the information-theoretic and Bregman decompositions. PathMNIST-C is evaluated through clean-versus-corrupted OOD AUROC and a summary of component shifts. These measures keep the intended-versus-off-target comparison defined in the theory chapter without introducing an additional composite metric.

Table 10: PathMNIST post-hoc uncertainty analysis. "Runs" is the number of evaluated configurations; numerical entries are means when more than one run is available. IT EU and Br EU denote information-theoretic and Bregman epistemic means; both decompositions share the reported AU estimate. A dash indicates that the method does not provide a measured epistemic component.

Method	Runs	Correctness AUROC	AU mean	IT EU	Br EU
Deep Ensemble	1	0.8678	0.4253	0.2343	0.2629
Temperature Scaling	5	0.8368	0.2137	–	–
Mahalanobis	5	0.8252	0.1513	–	–
SWAG	5	0.4385	0.1088	0.6886	4.8876
DDU	1	0.4302	2.1972	–	–
Laplace	5	0.8194	0.4245	0.0889	0.1783

The larger Bregman EU means for SWAG and Laplace reinforce that decomposition magnitudes are formula-dependent; they do not improve the corresponding correctness-ranking result by themselves.

Table 11: PathMNIST uncertainty shifts when the training fraction is reduced from 100% to 10%. The Bregman decomposition uses the same AU estimate as the IT decomposition, so only its EU shift is shown separately. A dash denotes an unavailable EU comparison for a method without multiple predictive samples.

Method	ΔAU	ΔEU_{IT}	ΔEU_{Br}
CE baseline	+0.1191	–	–
Correctness prediction	+0.1817	–	–
Deep correctness prediction	+0.1451	–	–
Loss prediction	+0.1791	–	–
Deep loss prediction	+0.1377	–	–
MC Dropout	+0.1496	−0.0042	−0.0053
EDL	+0.6467	+0.1340	+0.1633
SNGP	+0.2004	+0.0343	+0.0430
PostNet	+0.3878	−0.0048	−0.0060
HET	+0.1190	−0.0075	−0.0100
HET-XL	+0.2953	+0.1525	+1.0737
HetClassNN	+0.1628	−0.0024	−0.0038
Shallow Ensemble	+0.1376	+0.0316	+0.1220
DUQ	+0.6014	–	–

Note. The HET-XL Bregman EU values are finite for all 7180 test images, but the mean is influenced by a heavy upper tail (99th percentile 12.39, maximum 30.35). The magnitude should therefore be interpreted cautiously.

Table 12: PathMNIST uncertainty shifts between clean training and 40% symmetric training-label noise. The Bregman decomposition shares the AU estimate with the IT decomposition. A dash indicates that the method does not provide a measured epistemic component.

Method	ΔAU	ΔEU_{IT}	ΔEU_{Br}
CE baseline	+1.4874	–	–
Correctness prediction	+1.5229	–	–
Deep correctness prediction	+1.5981	–	–
Loss prediction	+1.6144	–	–
Deep loss prediction	+1.5062	–	–
MC Dropout	+1.4704	−0.0146	−0.0162
EDL	+1.2648	+0.2684	+0.3299
SNGP	+1.5750	+0.0063	+0.0063
HET	+1.4430	+0.0115	+0.0093
HET-XL	+1.4258	+0.0548	+0.0568
HetClassNN	+1.4988	−0.0132	−0.0151
Shallow Ensemble	+0.9368	+0.6061	+0.9306
DUQ	+1.4146	–	–
PostNet	+1.3570	−0.0120	−0.0144

On the common component-shift coordinate system in Figure 6, points above the diagonal respond more strongly through IT EU than AU; points below it respond more strongly through AU. The scarcity panel is AU-dominated for most methods. HET-XL is the exception under the Bregman decomposition, whose EU increase is much larger than both its AU increase and its information-theoretic EU increase. The label-noise panel shows a consistently large positive AU response with limited EU movement for most methods.

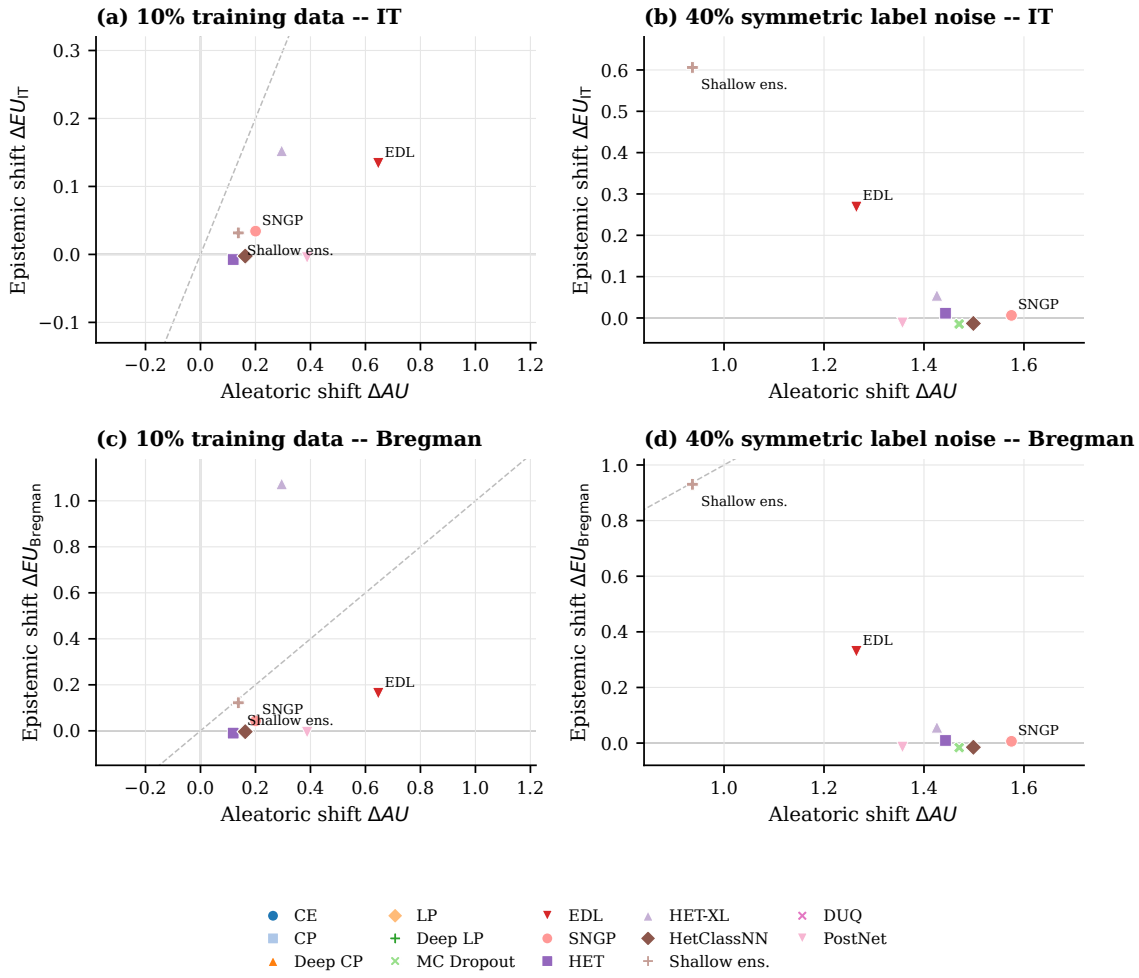


Figure 6: Paired information-theoretic and Bregman component shifts for PathMNIST at 10% training data and 40% symmetric training-label noise, each relative to the corresponding clean full-data model. The dashed diagonal marks equal AU and EU shifts.

The aggregate shifts conceal image- and method-level variation. Figure 7 compares identical test images across MC Dropout, SNGP, and Shallow Ensemble. The values are paired changes between the intervention model and its clean full-data counterpart for the same method and test index. They are interpreted within methods because raw uncertainty scales differ between methods.

(a) 40% label noise: AU-dominant



Index 3359
Adipose

Method	ΔAU	ΔEU_{IT}	ΔEU_{Br}
MC Dropout	+1.818	+0.002	+0.002
SNGP	+1.806	+0.005	+0.005
Shallow ens	+1.156	+0.554	+0.737

(b) 40% label noise: Cross-sensitive



Index 2919
Normal mucosa

Method	ΔAU	ΔEU_{IT}	ΔEU_{Br}
MC Dropout	+1.703	+0.002	+0.002
SNGP	+1.849	+0.012	+0.012
Shallow ens	+1.081	+0.544	+0.747

(c) 10% training data: EU-dominant



Index 2269
Normal mucosa

Method	ΔAU	ΔEU_{IT}	ΔEU_{Br}
MC Dropout	-0.000	-0.000	-0.000
SNGP	+0.692	+0.319	+0.322
Shallow ens	+0.112	+0.285	+2.112

(d) 10% training data: AU-dominant



Index 747
Mucus

Method	ΔAU	ΔEU_{IT}	ΔEU_{Br}
MC Dropout	+0.369	-0.001	-0.001
SNGP	+0.362	+0.029	+0.031
Shallow ens	+0.595	+0.013	+0.019

Figure 7: Example-level AU, information-theoretic EU, and Bregman EU shifts under 40% symmetric training-label noise and 10% training data. Bregman uses the same AU estimate, so AU is shown once. The same test index is evaluated by each intervention model and its clean full-data counterpart. Each panel is nearest the 98th percentile of its robust cross-method selection score.

Across 12 evaluated PathMNIST-C configurations, mean AU changes from 1.9954 on clean images to 1.9955 on corrupted images, a shift of only $+9.9 \cdot 10^{-5}$. Mean EU changes from 0.1649 to 0.1645, a shift of $-4.7 \cdot 10^{-4}$. This aggregate comparison does not support a strong decomposition-shift conclusion; the method-level OOD ranking in Table 13 gives the more direct distribution-shift evaluation.

Table 13: Method-level OOD AUROC for the fixed official PathMNIST-C corruption set. "Primary" uses each method's designated uncertainty score; total entropy provides a common comparison.

Method	Primary score	Primary OOD AUROC	Total-entropy OOD AUROC
CE baseline	$1 - \max p(y \mid x)$	0.3023	0.3410
Correctness prediction	Error probability	0.3763	0.3410
Deep correctness prediction	Error probability	0.7581	0.3419
Loss prediction	Predicted loss	0.6237	0.3410
Deep loss prediction	Predicted loss	0.2419	0.3419
MC Dropout	Expected confidence gap	0.3533	0.3442
EDL	Dempster–Shafer score	0.1088	0.3400
SNGP	$1 - \max p(y \mid x)$	0.5045	0.5023
HET	Expected confidence gap	0.4919	0.4885
HET-XL	Expected confidence gap	0.4891	0.4863
HetClassNN	Expected confidence gap	0.3657	0.3522
DUQ	DUQ score	0.5486	0.5624

The primary score in Table 13 is the method-specific uncertainty score used for OOD ranking; total entropy is included as a common predictive-uncertainty comparison. The AUROC is computed with corrupted images as the positive class, so a useful shift score should be larger on corrupted images than on clean images. The corrected fixed-corruption results are method-specific rather than uniformly positive. Deep Correctness Prediction gives the strongest OOD ranking (0.7581), followed by Loss Prediction (0.6237). DUQ is only moderately above chance, while SNGP and the heteroscedastic variants are close to chance. The baseline, MC Dropout, HetClassNN, Deep Loss Prediction, and EDL scores are below chance, meaning that their chosen uncertainty score is not higher on the fixed corruptions than on the clean images. PathMNIST-C allows only a narrow conclusion: some task-trained auxiliary heads rank this corruption mixture above clean images, but the common predictive-entropy and decomposition-style scores do not reliably assign higher uncertainty to shifted images under this fixed PathMNIST-C protocol.

The additional severity-axis experiment applies eight PathMNIST-C corruption types at all five available severity levels. The purpose is not to classify an image as clean or corrupted as a separate medical task. Instead, corruption creates a controlled distribution shift. If predictive entropy responds to that shift, corrupted images should usually receive higher entropy than clean images. OOD AUROC measures this ordering across all clean–corrupted pairs. An AUROC of 0.5 means that predictive entropy does no better than chance, while an AUROC of 0.53 means that a randomly selected corrupted image receives higher entropy than a randomly selected clean image in about 53% of comparisons.

Figure 8 reports the difference from the chance value in percentage points. Even the largest difference is only about three points, and no method improves monotonically as corruption severity increases. Predictive entropy shows little response to the PathMNIST-C corruptions as a distribution-shift probe.

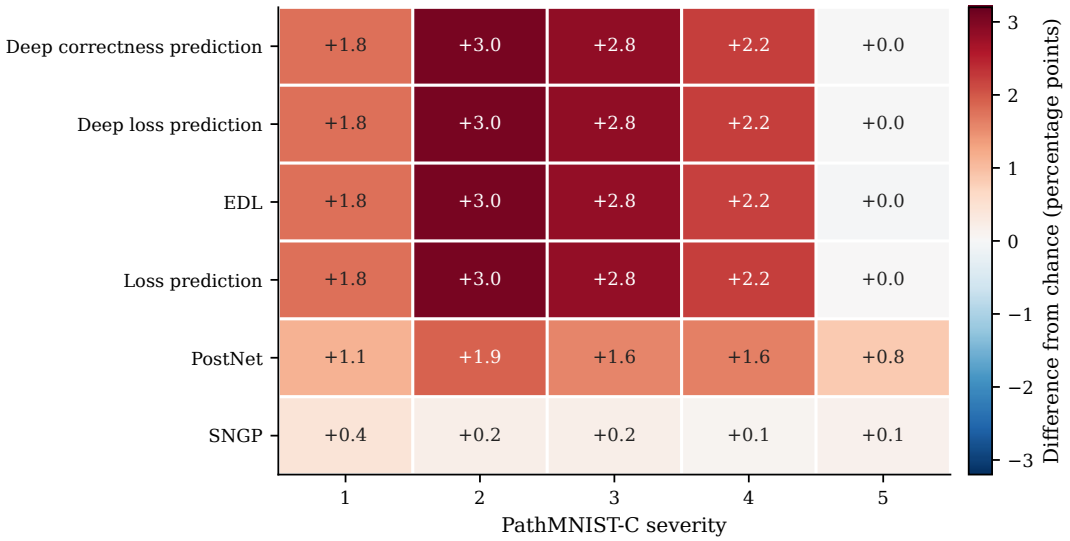


Figure 8: Response of predictive entropy to increasing PathMNIST-C corruption severity, averaged across eight corruption types. Each cell gives the OOD AUROC minus the chance value of 0.5, in percentage points. For example, +3.0 corresponds to an AUROC of 0.53: a corrupted image receives higher entropy than a clean image in about 53% of pairwise comparisons. Values near zero indicate that predictive entropy does not reliably react to the corruption shift. The nearly identical rows reflect nearly identical predictive-entropy rankings.

The decomposition results give the same negative pattern. For Deep Correctness Prediction, Loss Prediction, and Deep Loss Prediction, IT EU is constant at chance while IT AU follows predictive entropy. EDL's IT AU rises from 0.4962 at severity 1 to 0.5508 at severity 5, but its IT EU falls from 0.5158 to 0.4286; this opposing movement is not evidence of stable epistemic shift detection. PostNet's IT EU increases only from 0.5088 to 0.5191, and SNGP remains close to chance. Bregman EU also stays close to chance for every method and severity (between 0.4967 and 0.5039). Thus, extending the evaluation from one to five severity levels does not overturn the fixed-corruption conclusion: these decomposition scores do not provide a robust, severity-sensitive distribution-shift signal on average.

Figures 9–11 show the complete information-theoretic response trajectories. Label noise produces the expected ordering under this probe: AU increases, remains positive, and exceeds EU in all 24 method–noise-level comparisons with both components. This is the only probe with a consistent intended ordering across all paired comparisons, although EDL and Shallow Ensemble still show notable EU movement. This ordering shows that the quantity labelled AU is more responsive to synthetic label corruption; it does not by itself establish separation of natural aleatoric uncertainty. Scarcity gives the opposite result: 15 of 16 method–fraction comparisons are AU-dominated, including all eight comparisons at 10% training data. For PathMNIST-C, EU AUROC is generally close to chance and does not move consistently into the intended region as severity increases. The figure thus shows that the

consistent label-noise ordering does not generalize to epistemic and distribution-shift probes.

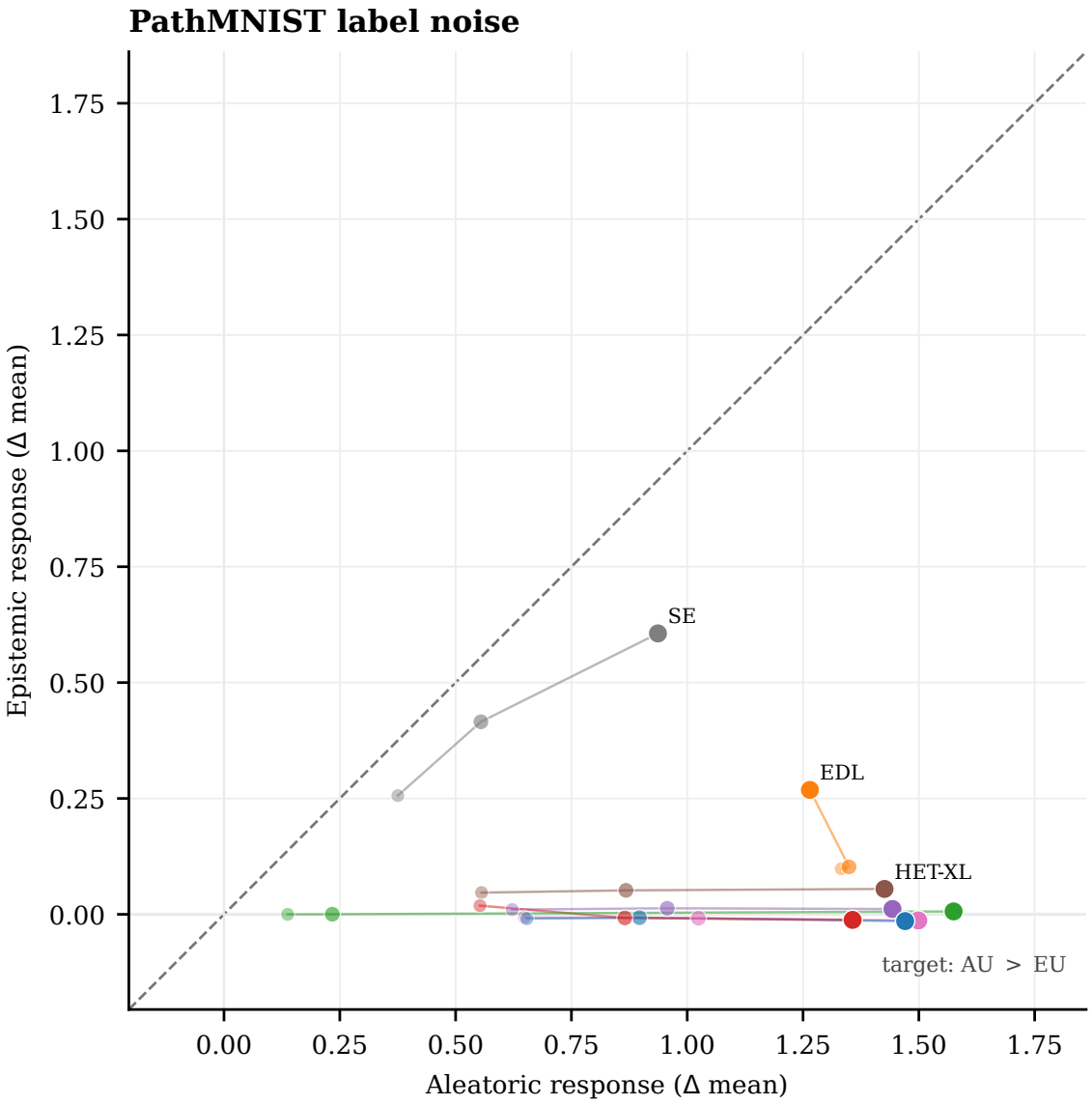


Figure 9: Information-theoretic AU and EU responses under 10%, 20%, and 40% symmetric training-label noise. Larger, darker points represent stronger noise. The ordering expected under this proxy is $\Delta AU > \Delta EU$; it is not direct evidence of natural image ambiguity.

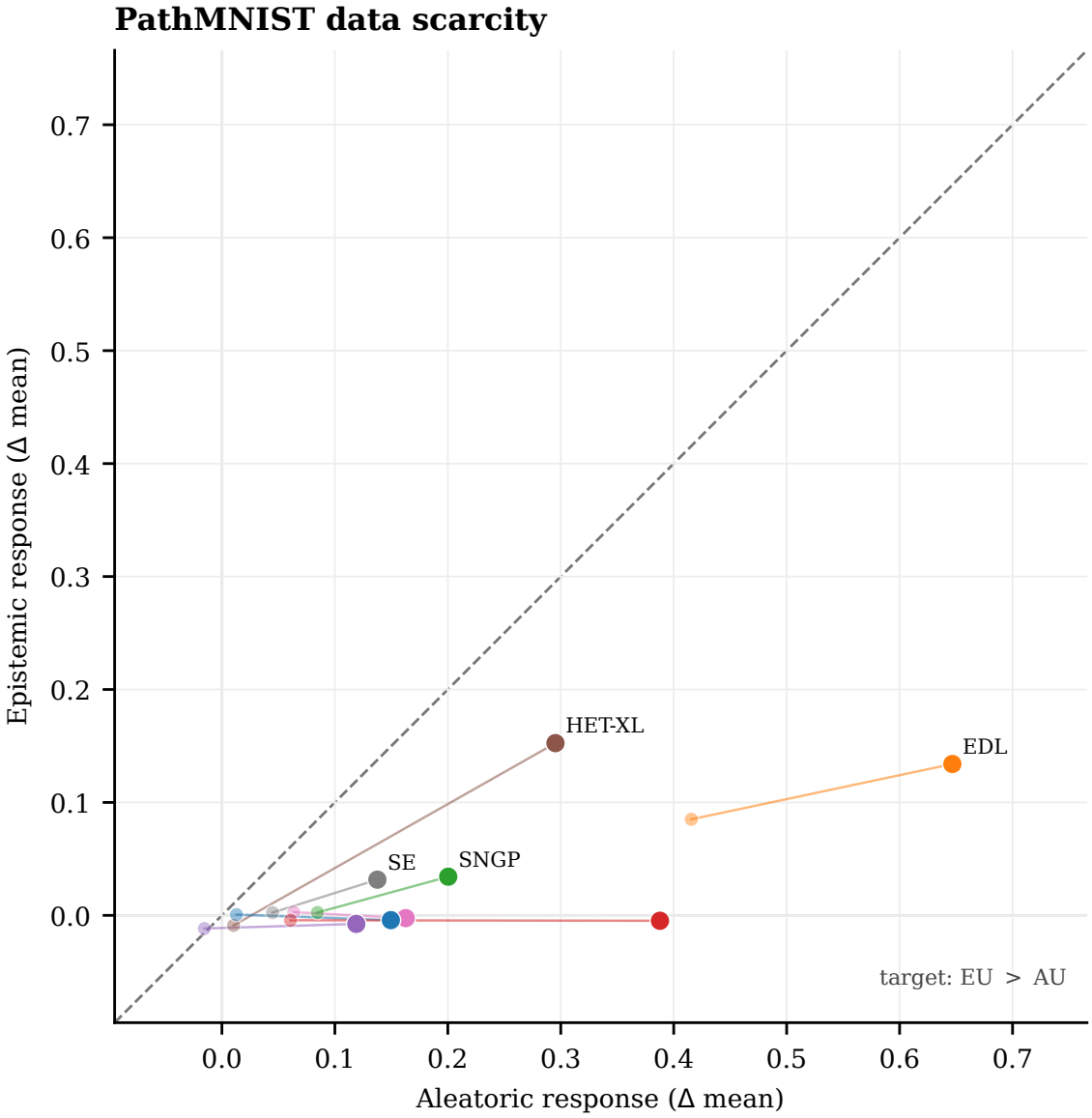


Figure 10: Information-theoretic AU and EU responses with 50% and 10% of the PathMNIST training data. Larger, darker points represent stronger scarcity. The ordering expected under this epistemic proxy has a positive ΔEU that exceeds ΔAU.

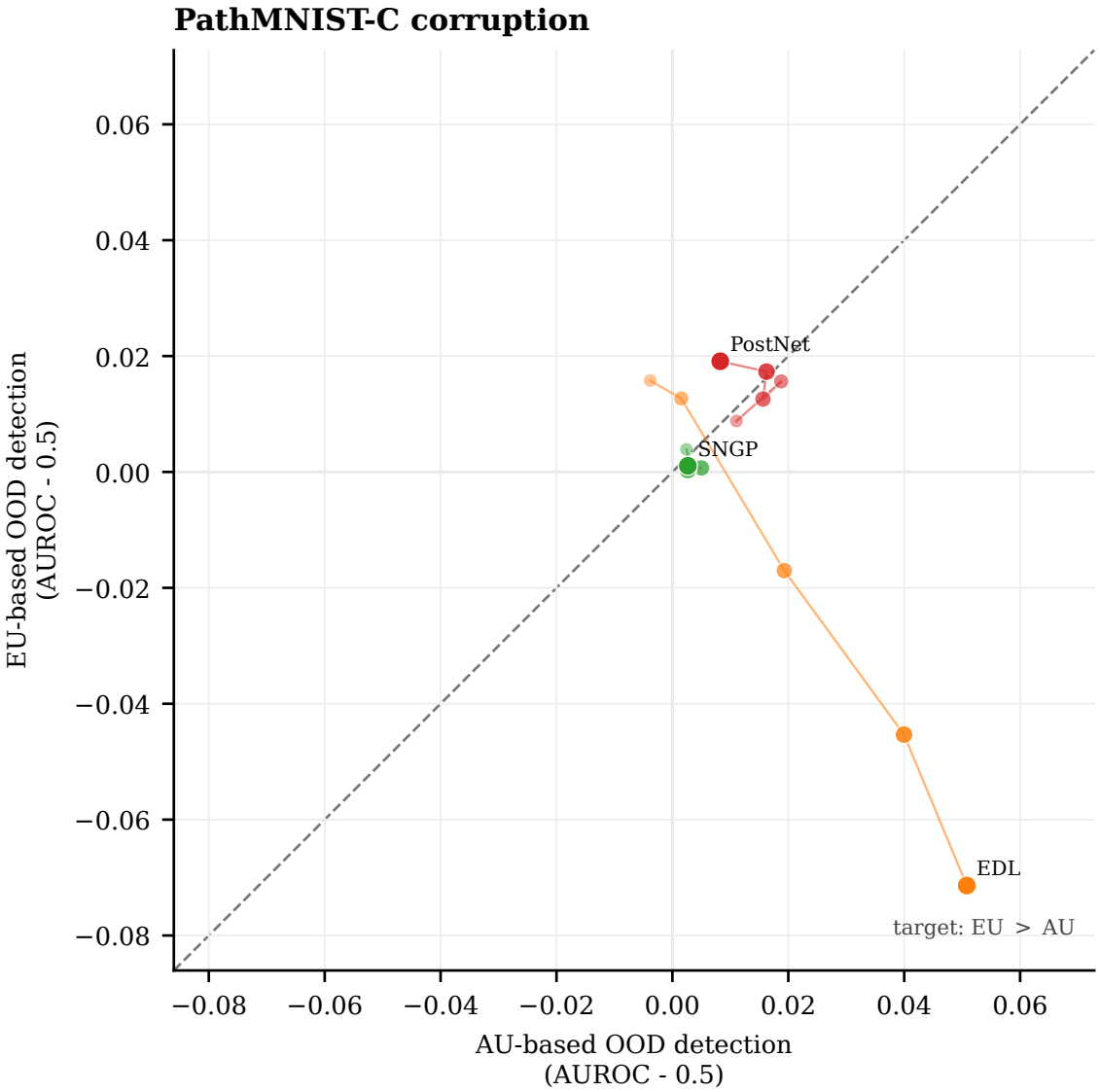


Figure 11: Information-theoretic AU and EU OOD-detection responses across PathMNIST-C severities 1–5. Larger, darker points represent stronger corruption. An epistemic-oriented shift response would have higher EU than AU AUROC; values above zero denote above-chance detection. Image degradation may, however, affect both components.

7.5 Interpretation of PathMNIST Results

Under the common transfer protocol, most trainable methods and several post-hoc wrappers produce uncertainty estimates that can be compared across probes. Only results using the validated nine-class configuration enter the analysis. The consolidated label-noise experiment covers 14 methods. At 40% label noise, every method has a positive AU shift, ranging from +0.9368 for Shallow Ensemble to +1.6144 for Loss Prediction (Table 12). Among methods with a measured epistemic component, IT EU remains within $|\Delta EU| < 0.06$ except for EDL (+0.2684) and Shallow Ensemble (+0.6061). The Bregman decomposition preserves this overall pattern but assigns an even larger EU shift to Shallow Ensemble (+0.9306). Synthetic label corruption is reflected mainly in AU across this set of methods, with method-specific cross-sensitivity rather than clean separation (Figure 9); the controlled intervention does not represent natural diagnostic ambiguity.

The post-hoc results are mixed. Deep Ensemble, Temperature Scaling, and Mahalanobis give correctness-ranking AUROC values well above chance on clean PathMNIST. Laplace also gives a correctness-ranking AUROC well above chance with covariance stabilization, but its averaged IT decomposition is AU-dominated and the member-level rank correlation between AU and EU is high. It is useful for error ranking here, but provides little evidence of clean component separation. SWAG produces large EU but weak correctness ranking, and DDU is not interpretable because its ranking metric is near chance. The complete data-scarcity comparison is more cautious. When the training fraction falls to 10%, HET-XL has the largest positive IT EU shift, followed by EDL, SNGP, and Shallow Ensemble, but each method's AU shift is larger. Most remaining methods have a small, zero, or negative EU shift while AU changes more strongly (Figure 10). Bregman assigns HET-XL a much larger EU shift that exceeds its AU shift, whereas the other methods remain AU-dominated. This formula-dependent exception does not provide consistent epistemic specialization across methods. The PathMNIST-C tensor shift is similarly small on average, and the corrected method-level AUROCs show that distribution-shift sensitivity is concentrated in only a few auxiliary uncertainty scores; the decomposition trajectories in Figure 11 likewise remain close to chance or move away from the ordering expected under the epistemic-oriented probe. These results show that the benchmark protocol can be applied to PathMNIST and that a small number of scores rank specific targets. They do not support a claim that the evaluated methods cleanly disentangle aleatoric, epistemic, and distribution-shift uncertainty in the medical setting.

8 MHIST Experiments and Results

MHIST complements the PathMNIST transfer by providing a rater-disagreement proxy for ambiguity: seven pathologist votes for each image. The experiments transfer the recovered CIFAR-10 settings to the binary MHIST task without a new hyperparameter search. They use the Wide ResNet 26-10 backbone at 32×32 pixels,

CIFAR-10 normalization, the recovered optimization schedules, and majority-vote hard labels for training; the individual votes are kept for the ambiguity analysis.

8.1 Experimental Scope

The primary experiment evaluates 13 configurations at each data fraction: 100%, 50%, and 10% of the MHIST training split. Post-hoc methods reuse five cross-entropy member checkpoints for each split. Supplementary experiments extend the comparison to the remaining evaluated methods. Repeated SNGP entries differ in whether spectral normalization is enabled. Appendix B lists the transferred parameters. The MHIST release provides train and test partitions but no separate validation partition. Validation-style evaluations use the official test partition. The reported final and best AUROC values should be read as benchmark results under a common protocol on the released split, not as a claim that hyperparameters or stopping epochs were selected on an independent validation set. The main comparisons use final values and the disentanglement summaries; best epochs are reported to show the run trajectory. Because no validated medical OOD protocol is available for MHIST, its results are interpreted through rater disagreement and data scarcity rather than as a distribution-shift benchmark.

Mahalanobis is omitted from the MHIST result tables because the CIFAR-style post-hoc detector requires a validated calibration or detector-fitting set. A pure class-distance score could be computed, but it would change the method definition and would no longer be directly comparable to the benchmarked Mahalanobis variant.

8.2 Primary MHIST Results

The tables report final accuracy, final correctness-prediction AUROC, best AUROC, and the epoch with the best AUROC on the released evaluation partition. The AUROC value uses each method's assigned uncertainty score, so it measures error-ranking usefulness rather than a common raw uncertainty scale. Correctness AUROC is used here because the first question is whether uncertainty helps identify mistakes at all; source-specific claims are reserved for the later rater-disagreement and data-scarcity analyses.

Mean final accuracy falls from 0.7720 with full data to 0.7512 with 50% and 0.6736 with 10%. Mean final AUROC falls from 0.7316 to 0.7043 and 0.6554. The scarcity intervention therefore harms both classification and error ranking, although the loss differs across methods.

Table 14: MHIST results with 100% of the training split.

Method	Final acc.	Final AUROC	Best AUROC	Best epoch
CE baseline	0.7584	0.7376	0.7779	58
Correctness prediction	0.7533	0.7299	0.7555	171
MC Dropout	0.7718	0.7431	0.7826	23
SNGP (SN off)	0.7718	0.7040	**0.8062**	69
EDL	**0.8127**	0.7533	0.7862	64
DDU	0.7707	0.7559	0.7821	31
PostNet	0.7697	0.6955	0.7573	119
Loss prediction	0.7707	0.6482	0.7268	103
HET	0.7984	0.7566	0.7802	106
SNGP (SN on)	0.7472	0.7342	0.7602	93
Shallow ensemble	0.7799	**0.7691**	0.7810	68
HET-XL	0.7758	0.7378	0.7718	24
HetClassNN	0.7554	0.7454	0.7633	24

Table 15: MHIST results with 50% of the training split.

Method	Final acc.	Final AUROC	Best AUROC	Best epoch
CE baseline	0.7451	0.7238	0.7771	49
Correctness prediction	0.7441	0.6889	0.7643	19
MC Dropout	0.7462	0.7198	0.7728	53
SNGP (SN off)	0.7134	0.7073	0.7546	89
EDL	0.7707	0.7265	0.7496	21
DDU	0.7718	0.7050	0.7724	14
PostNet	0.7523	0.6665	0.7435	40
Loss prediction	0.7513	0.6734	0.7721	38
HET	0.7677	0.7093	0.7636	50
SNGP (SN on)	0.7185	0.7103	0.7492	57
Shallow ensemble	**0.7748**	0.6903	0.7648	52
HET-XL	0.7564	**0.7399**	**0.7811**	38
HetClassNN	0.7533	0.6947	0.7556	57

Table 16: MHIST results with 10% of the training split.

Method	Final acc.	Final AUROC	Best AUROC	Best epoch
CE baseline	0.6592	0.6921	0.7164	97
Correctness prediction	0.6981	0.6578	0.7414	116
MC Dropout	0.6377	0.6747	0.7227	85
SNGP (SN off)	0.6387	0.6783	0.6881	235
EDL	0.7042	0.6819	0.7105	39
DDU	0.6909	0.6726	**0.7318**	63
PostNet	0.6745	0.6408	0.6804	170
Loss prediction	**0.7073**	0.5281	0.6283	104
HET	0.6776	**0.6942**	0.7142	70
SNGP (SN on)	0.6264	0.5664	0.6173	209
Shallow ensemble	0.7021	0.6751	0.7010	117
HET-XL	0.6397	0.6660	0.6832	121
HetClassNN	0.7001	0.6928	0.7098	151

8.3 Post-Hoc and Supplementary Results

Deep Ensemble, Temperature Scaling, SWAG, and Laplace are evaluated at all three training fractions. Laplace uses a Kronecker Hessian, GLM prediction, 1000 predictive Monte Carlo samples, 50 cross-validation samples, and an 80-point prior-precision grid. The following two tables report the full-data ambiguity analysis. Changes across training fractions enter the separate data-scarcity analysis in Section 8.5.

Table 17: Full-data post-hoc MHIST ambiguity results with the information-theoretic decomposition. ρ_A and ρ_E are Spearman correlations between AU or EU and normalized rater-vote entropy. A dash denotes an unavailable epistemic component.

Method	Mean AU	Mean EU	ρ_A	ρ_E
Deep Ensemble	0.1995	0.0957	0.1409	0.1711
Laplace	0.2961	0.0121	0.1357	0.1168
Temperature Scaling	0.5567	–	0.1353	–
SWAG	0.0248	0.0933	0.1774	0.1701

Table 18: Full-data post-hoc MHIST ambiguity results with the Bregman decomposition. ρ_A and ρ_E are Spearman correlations with normalized rater-vote entropy. A dash denotes an unavailable epistemic component.

Method	Mean AU	Mean EU	ρ_A	ρ_E
Deep Ensemble	0.1995	0.2252	0.1409	0.1567
Laplace	0.2961	0.0125	0.1357	0.1175
Temperature Scaling	0.5567	–	0.1353	–
SWAG	0.0248	0.2216	0.1774	0.1695

Table 19: Supplemental MHIST method results. AUROC refers to correctness-prediction AUROC using each method's uncertainty score. A dash in "Best epoch" denotes evaluation without additional training.

Training data	Method	Final acc.	Final AUROC	Best AUROC	Best epoch
100%	Deep correctness prediction	0.7769	0.7334	0.7641	16
100%	Deep loss prediction	0.7687	0.7529	0.7905	23
100%	DUQ	0.7503	0.6951	0.7605	60
100%	Fast Deep Ensemble	0.6571	0.8001	0.8001	–
50%	Deep correctness prediction	0.7134	0.6957	0.7197	73
50%	Deep loss prediction	0.7308	0.6609	0.7372	41
50%	DUQ	0.7349	0.6954	0.7382	27
50%	Fast Deep Ensemble	0.6438	0.7939	0.7939	–
10%	Deep correctness prediction	0.6837	0.6689	0.7276	117
10%	Deep loss prediction	0.6510	0.6230	0.6729	64
10%	DUQ	0.5824	0.4837	0.6269	142
10%	Fast Deep Ensemble	0.6356	0.7369	0.7369	–

Fast Deep Ensemble is evaluated on all three splits. Because it is applied without additional training, its table entries report test accuracy and correctness AUROC,

with a dash in the epoch column. The DUQ 10% result contains one non-finite decomposition entry among 977 MHIST test samples; this observation is excluded pairwise, leaving 976 samples for the corresponding analysis.

8.4 Rater-Disagreement Analysis

The rater-disagreement analysis tests whether the quantity labelled aleatoric uncertainty follows an image-level ambiguity proxy. The target is the normalized entropy of the seven pathologist votes. Spearman correlation is used as the primary measure because the relevant question is ordinal: do images with stronger expert disagreement receive higher uncertainty, even if different methods use different numerical scales? On the full training split, the largest information-theoretic aleatoric correlations with rater entropy are EDL ($\rho_A = 0.2358$), Shallow Ensemble ($\rho_A = 0.2210$), HET ($\rho_A = 0.2171$), PostNet ($\rho_A = 0.2170$), SNGP with spectral normalization ($\rho_A = 0.2108$), and DUQ ($\rho_A = 0.1939$). These values show measurable alignment with rater disagreement, but the correlations remain modest.

The main caution is cross-sensitivity. For several methods, the epistemic component also correlates with rater entropy. For example, EDL has $\rho_A = 0.2358$ and $\rho_E = 0.2365$ on the full split, and HET has $\rho_A = 0.2171$ and $\rho_E = 0.2180$. This pattern is consistent with the methods responding to general image difficulty rather than separating label ambiguity from model uncertainty.

Of the 13 full-data methods with both components, only six have $\rho_A > \rho_E$; seven show the reverse ordering. Even the favourable correlations are modest. The ambiguity experiment shows cross-sensitivity rather than general separation.

Among methods with both components, PostNet has the widest full-data gap in the intended direction, although its aleatoric correlation remains modest.

8.5 Data-Scarcity Analysis

Data scarcity tests the epistemic axis by comparing the full-data model with models trained on 50% and 10% of the training split. This comparison is made within each method rather than between raw method scales, because AU and EU magnitudes are not calibrated to a shared unit across methods. The ordering expected under this probe has a positive ΔEU that is larger than the corresponding ΔAU. In the 10% setting, HET, HET-XL, SNGP without spectral normalization, and Laplace meet this comparison under the information-theoretic decomposition. For most other methods, AU changes as much as, or more than, EU.

Across both reduced-data levels, only 9 of 26 comparisons combine a positive EU shift with an EU shift larger than AU; 17 fail at least one of these conditions. At 10% training data, 5 of 13 comparisons meet both conditions. Selective epistemic responses are thus the minority for these decomposition outputs.

Figures 12 and 13 separate the two MHIST probes. The rater-ambiguity figure compares AU and EU correlations for full-data models only. Most correlations remain close, so positive ambiguity correlations alone do not establish aleatoric specificity. The scarcity figure instead compares the changes caused by reducing the training set. HET and HET-XL move into the region expected under the scarcity probe, while several other methods remain close to the diagonal or respond more strongly through AU.

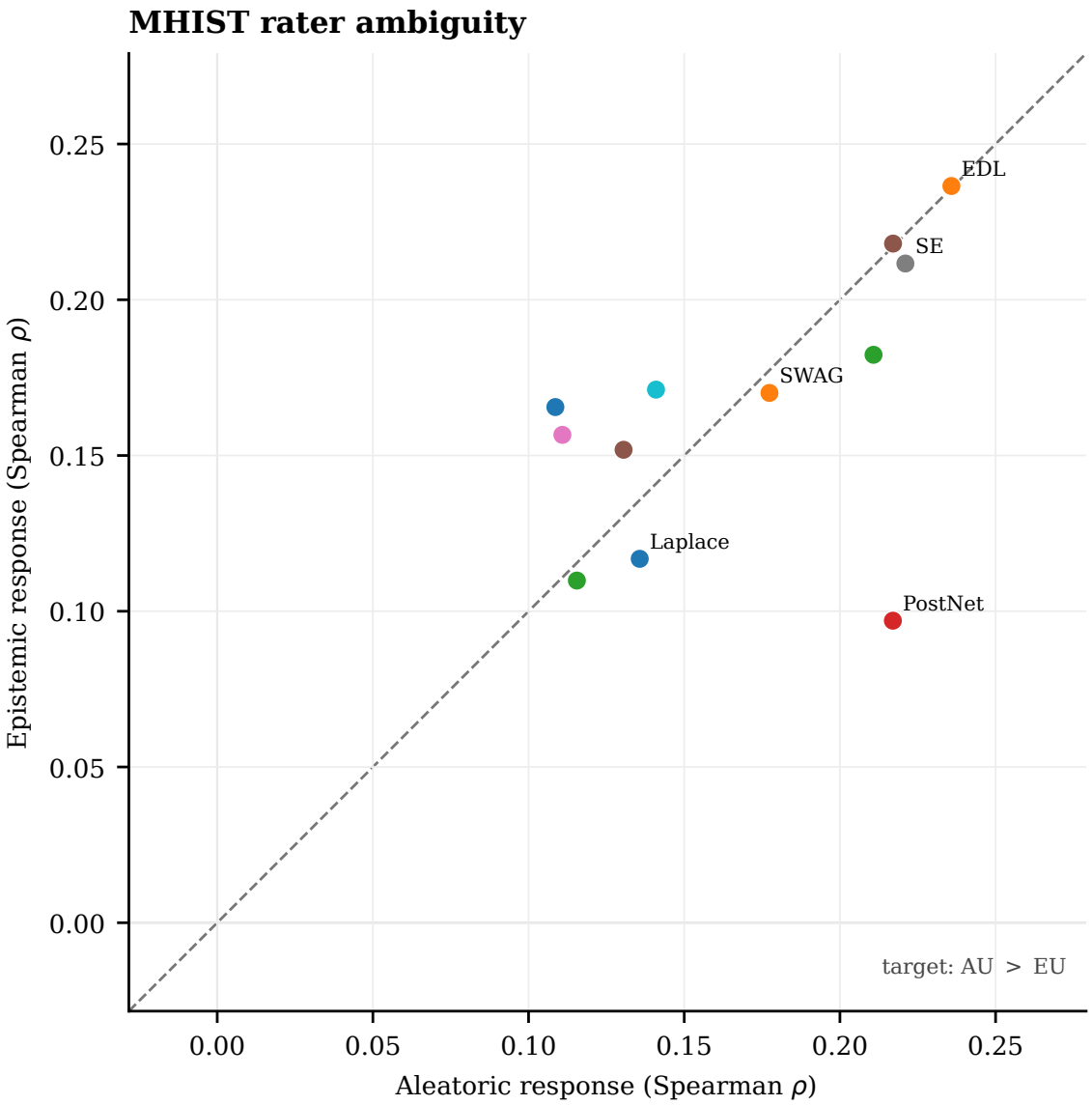


Figure 12: Information-theoretic AU and EU correlations with rater-vote entropy for the full-data MHIST models. Aleatoric specificity requires $\rho_A > \rho_E$.

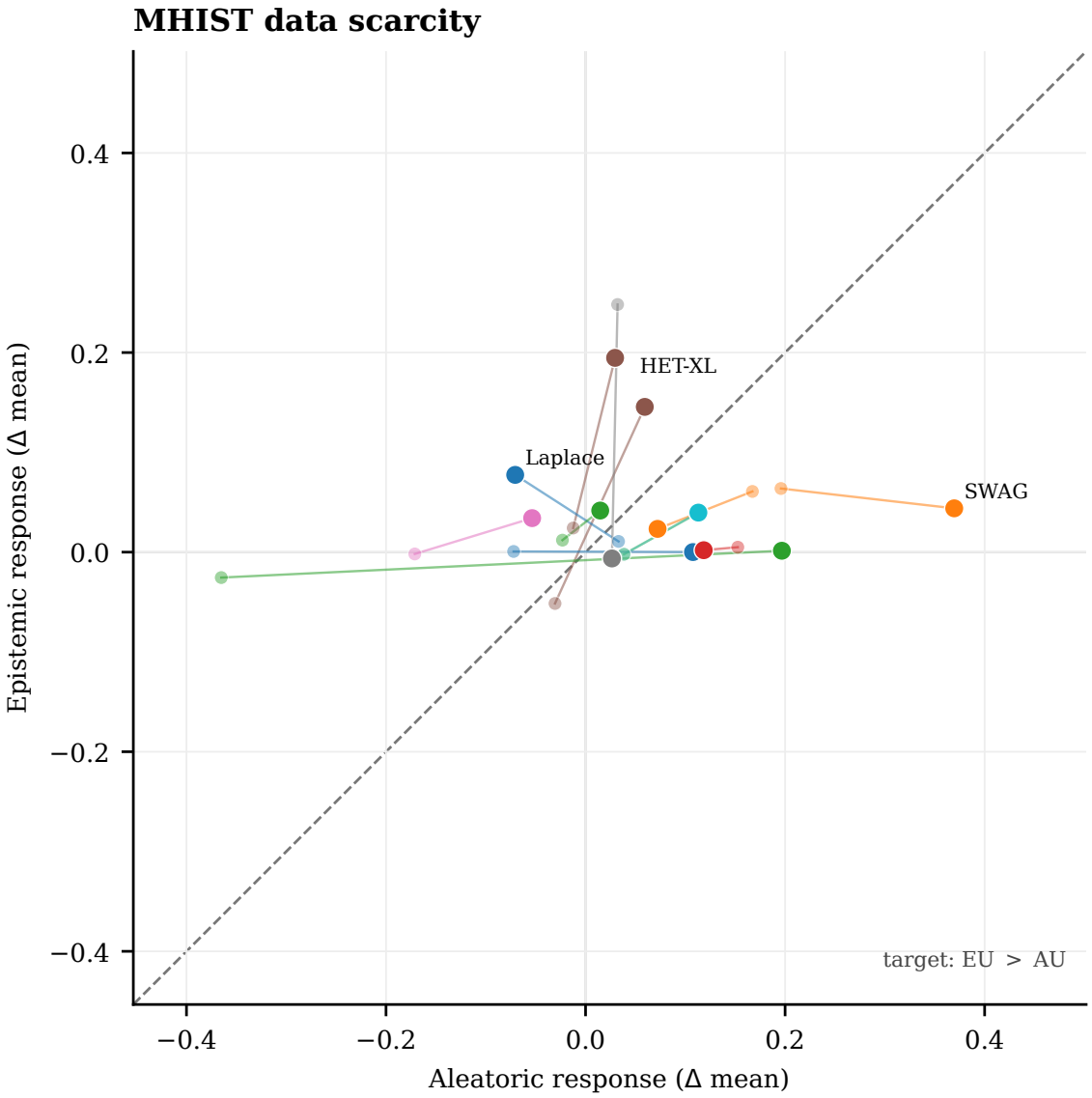


Figure 13: Information-theoretic AU and EU shifts with 50% and 10% of the MHIST training data relative to full-data training. Larger, darker points denote stronger scarcity. Epistemic specificity requires a positive ΔEU that exceeds ΔAU.

8.6 Integrated Disentanglement Analysis

Following Section 3.9, MHIST compares each intended response with its off-target component. Tables 20 and 21 report the paired comparisons directly: ρ_A and ρ_E for rater disagreement, and ΔAU_{10} and ΔEU_{10} for data scarcity.

Table 20: Selected MHIST component comparisons from the information-theoretic decomposition. ρ_A and ρ_E are full-data correlations with rater-vote entropy; ΔAU_{10} and ΔEU_{10} compare 10% training data with full-data training.

Method	ρ_A	ρ_E	ΔAU_{10}	ΔEU_{10}
PostNet	0.2170	0.0970	0.1185	0.0019
HET	0.2171	0.2180	0.0296	0.1947
HET-XL	0.1305	0.1518	0.0594	0.1456
SNGP (SN off)	0.1155	0.1099	0.0147	0.0417
Laplace	0.1357	0.1168	-0.0705	0.0774
EDL	0.2358	0.2365	0.0722	0.0233
Shallow ens.	0.2210	0.2116	0.0264	-0.0063
Deep ens.	0.1409	0.1711	0.1131	0.0396
MC Dropout	0.1086	0.1655	0.1077	0.0001
SWAG	0.1774	0.1701	0.3695	0.0439

In the information-theoretic table, PostNet has the largest rater-disagreement gap because ρ_A exceeds ρ_E, although the gap remains modest. Its 10% scarcity run changes AU more than EU. HET and HET-XL show the largest EU shifts under scarcity because ΔEU_{10} exceeds ΔAU_{10}, while their rater-disagreement signal is shared across AU and EU. Laplace shows the expected ordering for both probes, but both effects are small. SWAG is numerically stable, but its 10% scarcity response is dominated by AU ($\Delta AU_{10} = 0.3695$ versus $\Delta EU_{10} = 0.0439$), contrary to the ordering expected under the scarcity probe.

The Bregman formula re-evaluates the same rater-disagreement and scarcity targets. Mucsányi et al. (2024) report that the decomposition itself can affect the observed separation of aleatoric and epistemic uncertainty.

The Bregman decomposition keeps the main rater-disagreement pattern: PostNet has the largest observed gap between ρ_A and ρ_E, while EDL, HET, and the ensemble-style rows show similar AU and EU correlations. Compared with the information-theoretic table, Bregman assigns much larger ΔEU_{10} values to HET and HET-XL. This makes their scarcity response look larger, but it also shows that the measured size of epistemic separation depends on the chosen decomposition formula.

The two decompositions agree on the main rater-disagreement result: no method has a strong aleatoric correlation while keeping the epistemic correlation close to zero. PostNet has the largest observed gap between ρ_A and ρ_E in both tables, but its scarcity response is AU-dominated. HET and HET-XL have the strongest Bregman

Table 21: Selected MHIST component comparisons from the Bregman decomposition. The columns have the same meaning as in the information-theoretic table.

Method	ρ_A	ρ_E	ΔAU_{10}	ΔEU_{10}
PostNet	0.2170	0.0946	0.1185	0.0023
HET	0.2171	0.2167	0.0296	0.5074
HET-XL	0.1305	0.1534	0.0594	0.6128
SNGP (SN off)	0.1155	0.1100	0.0147	0.0909
Laplace	0.1357	0.1175	-0.0705	0.1143
EDL	0.2358	0.2347	0.0722	0.0330
Shallow ens.	0.2210	0.2129	0.0264	-0.0061
Deep ens.	0.1409	0.1567	0.1131	-0.0582
MC Dropout	0.1086	0.1654	0.1077	0.0001
SWAG	0.1774	0.1695	0.3695	-0.0252

ΔEU_{10} shifts, and HET also has the strongest information-theoretic epistemic shift, yet their rater-disagreement correlations remain almost identical for AU and EU. Laplace shows the expected ordering for both probes, but both effects are small. Neither formula yields clean component separation, and the few method-dependent responses change in size.

Several methods provide only one uncertainty signal by design. CE baseline and temperature scaling produce one calibrated or uncalibrated predictive distribution, so they can yield confidence or entropy but no model-disagreement component. Correctness prediction, loss prediction, and their deep variants train an auxiliary head to predict error or loss; their scalar target is predictive risk rather than an aleatoric–epistemic split. DDU, DUQ, and Mahalanobis-style distance methods mainly score feature-space atypicality or training support; this is useful for epistemic or OOD-related questions, but it does not provide a separate label-noise component. For these methods, correctness AUROC, calibration, OOD detection, or correlation with rater disagreement can still be reported. They should not be read as direct evidence for a two-component disentanglement claim.

8.7 MHIST Findings

MHIST provides only partial source-specific responses. Correlations with rater disagreement are modest, and scarcity produces a positive, EU-dominant information-theoretic response for only a subset of methods (Figures 12 and 13).

9 Discussion

The experiments do not show consistent uncertainty disentanglement in medical image classification. CIFAR-10 provides an implementation check before transfer, PathMNIST provides controlled scarcity, label-noise, and corruption experiments, and MHIST adds rater disagreement as an ambiguity proxy.

9.1 Main Findings

Several methods rank unreliable predictions well without separating their uncertainty sources. MHIST illustrates the difference: EDL and Shallow Ensemble have relatively high rater-disagreement correlations, but their epistemic components also correlate with the same target (Figure 12). Such responses indicate general image difficulty rather than selective aleatoric sensitivity.

The available analyses do not establish that AU and EU are strongly correlated on every individual image because that correlation was not measured throughout. They do show that both components often respond to the same probe, or that the off-target component responds more strongly. In MHIST, only 6 of 13 full-data comparisons have $\rho_A > \rho_E$ for rater ambiguity. Under scarcity, only 9 of 26 comparisons combine $\Delta EU > 0$ with $\Delta EU > \Delta AU$. The assigned AU and EU labels therefore do not reliably correspond to component behaviour under the medical probes.

Some MHIST results nevertheless follow the expected direction. PostNet gives the largest observed gap between aleatoric and epistemic correlations with rater disagreement among the methods with both AU and EU components, but it does not show an epistemic scarcity response. HET and HET-XL show stronger ΔEU than ΔAU under data scarcity, especially under the Bregman decomposition, but their rater-disagreement correlations are shared between AU and EU. The information-theoretic contrast between these probes is visible in Figures 12 and 13. Laplace shows the expected ordering on both probes, but both effects are small. Equal, off-target, or wrong-direction responses remain more common than stable specialization.

PathMNIST supports broader controlled training, scarcity, and corruption experiments, but its released labels do not expose image-level human disagreement. MHIST is smaller and binary, but its seven pathologist votes provide a rater-disagreement proxy. Together, the datasets broaden the available probes without directly observing the source of uncertainty.

PathMNIST shows its clearest source-specific pattern under synthetic label noise. By contrast, scarcity remains predominantly AU-dominated with the information-theoretic decomposition, and Bregman reverses this ordering for only one method. Under corruption, the decomposition scores stay near chance or move away from the intended epistemic ordering. Figures 9, 10, and 11 show the corresponding trajectories. The label-noise response therefore does not generalize to the epistemic probes or establish separation of natural diagnostic ambiguity.

Figure 14 places all experiments with paired AU and EU responses on a common visual coordinate system. Each panel compares the aleatoric response on the horizontal axis with the epistemic response on the vertical axis. The diagonal therefore marks equal component responses: aleatoric-oriented probes favour points below it, whereas epistemic-oriented and distribution-shift probes favour points above it. In either case, the intended component must also show a positive response; position relative to the diagonal alone is insufficient. The response itself remains probe-specific: mean changes are used for the interventions, correlations for rater ambiguity, and AUROC relative to chance for corruption detection. Where a probe has multiple intervention or severity levels, lines connect them and larger, darker points indicate stronger stress. The MHIST ambiguity panel reports full-data correlations. The full-data CIFAR-10 reproduction is an implementation check without a paired AU–EU intervention and therefore has no coordinate in this figure. Most points lie near the diagonal or on the off-target side. Points in the wrong region show that the intended component did not dominate; points below zero in the OOD panels rank shifted inputs opposite to the intended direction.

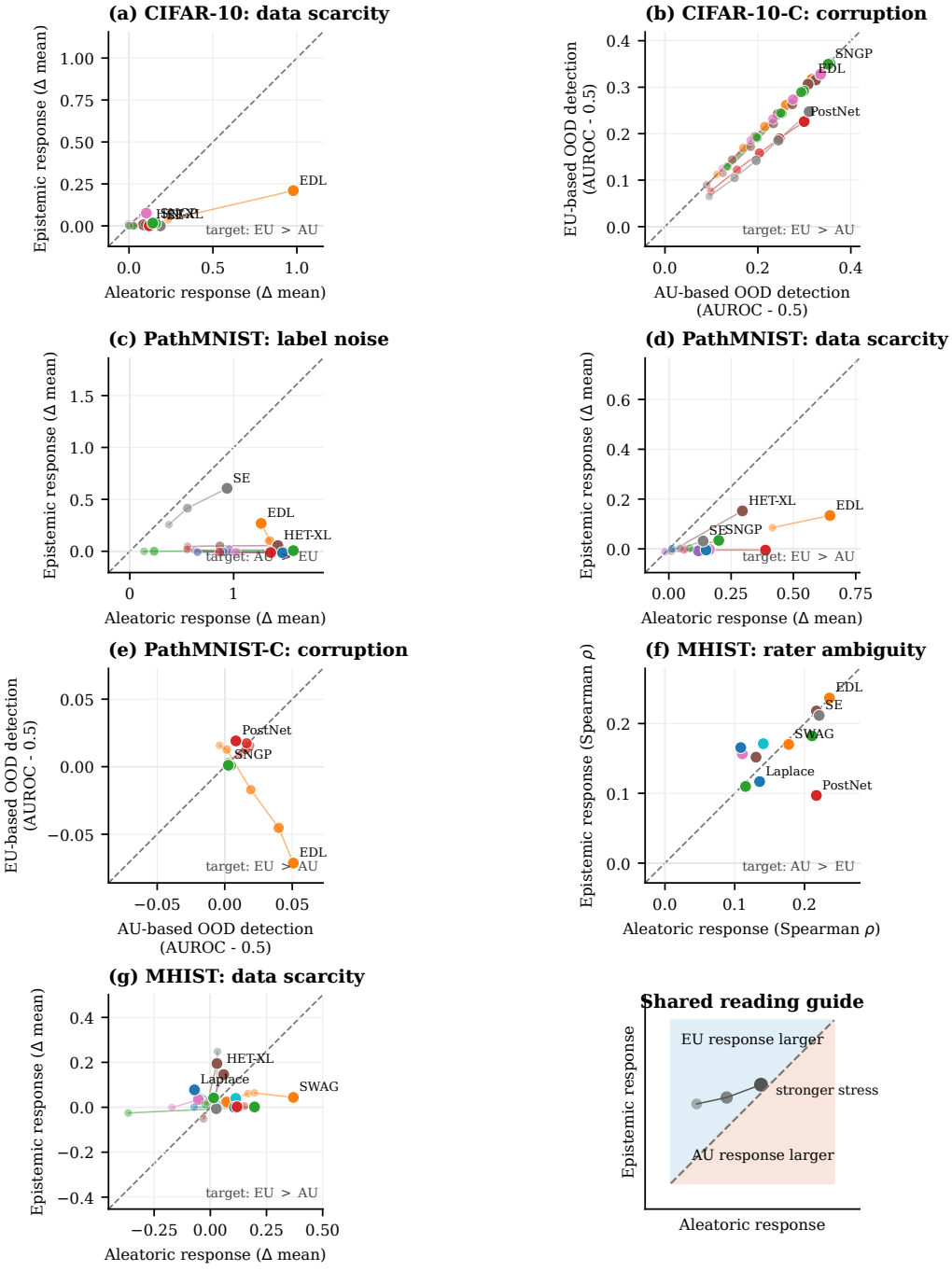


Figure 14: Experiments with paired AU and EU results shown using the information-theoretic decomposition. Each colour represents a method. Lines connect intervention or severity levels where several levels exist; larger, darker points represent stronger stress. MHIST rater ambiguity is shown only for full-data models. The horizontal coordinate is the AU response and the vertical coordinate is the EU response. For label noise and rater ambiguity, the expected proxy response lies below the diagonal; for data scarcity and corruption, it lies above the diagonal. These regions encode the tested ordering, not ground truth for the uncertainty source. Axis ranges differ because changes in mean uncertainty, correlations, and AUROC differences have different scales. Abbreviations: MCD, MC Dropout; SE, Shallow Ensemble; HCNN, HetClassNN.

The PathMNIST-C corruption experiments also require method-specific interpretation. On the fixed official corruption set, Deep Correctness Prediction and Loss Prediction provide the highest observed OOD AUROC values, whereas most predictive-entropy, evidential, sampling, and heteroscedastic scores are weak or near chance. The scores that rank clean-test errors or respond to label noise do not automatically capture this shift. Across six methods and eight corruption types, mean predictive-entropy OOD AUROC remains near chance and does not increase monotonically with severity. The IT and Bregman components likewise provide no consistent severity-sensitive epistemic signal; the IT trajectories in Figure 11 show no consistent movement into the intended region. Greater corruption strength alone does not make these decompositions reliable shift detectors.

9.2 Sensitivity to the Decomposition Formula

The information-theoretic and Bregman decompositions are compared as a sensitivity analysis rather than as competing objectives. The comparison covers every experiment for which paired decomposition results are available. Because the experiments use different scales and targets, pooling their raw values into one mean would not be meaningful. The comparison instead considers three properties within each probe: whether the formulas rank methods or conditions similarly, whether the response has the same direction, and whether they support the same probe-level interpretation. For scarcity, the interpretation is whether the EU shift exceeds the AU shift; for label noise, whether the AU shift exceeds the EU shift; for rater disagreement, whether $\rho_A > \rho_E$; and for OOD detection, whether EU AUROC is above chance. Changes smaller than 10^{-6} are treated as zero in the direction comparison.

Table 22: Agreement between the information-theoretic (IT) and Bregman (Br) decompositions across paired experimental conditions. ρ_{rank} is the Spearman correlation between IT and Br EU estimates. "Direction" is the proportion with matching response direction, and "Interpretation" is the proportion yielding the same probe-specific comparison.

Probe	n	ρ_{rank}	Direction	Interpretation
PathMNIST scarcity	8	0.976	100.0%	87.5%
PathMNIST label noise	8	1.000	100.0%	100.0%
MHIST scarcity	13	0.504	76.9%	100.0%
MHIST rater disagreement	13	0.901	100.0%	84.6%
CIFAR-10 scarcity	8	1.000	100.0%	87.5%
PathMNIST-C EU OOD AUROC	15	0.479	80.0%	80.0%
CIFAR-10C EU OOD AUROC	24	0.997	100.0%	100.0%

For the intervention experiments, formula choice rarely reverses the recorded component ordering. The PathMNIST label-noise ordering is identical for all eight methods with a measured EU component, and the scarcity interpretation agrees for seven of

eight PathMNIST methods with a measured EU component, all 13 MHIST methods, and seven of eight CIFAR-10 configurations. The method ordering is also very similar except for MHIST scarcity, where the lower rank correlation reflects larger Bregman EU changes for methods such as HET and HET-XL. Rater-disagreement correlations are similarly ordered, although two of 13 finite comparisons cross the $\rho_A = \rho_E$ boundary.

The corruption experiments separate stable ordering from numerical agreement near chance. On CIFAR-10C, IT and Bregman EU give almost the same ordering and the same above-chance interpretation for all 24 comparisons with a measured EU. On PathMNIST-C, however, the rank correlation is only 0.479 and the chance-side interpretation agrees for 12 of 15 method–severity pairs. This disagreement does not indicate that one formula detects the shift reliably: the median absolute distances from chance are only 0.013 for IT EU and 0.002 for Bregman EU. It instead shows that weak deviations near chance are formula-sensitive.

The two formulas agree more closely on qualitative conclusions than on raw EU magnitudes. They usually preserve the interpretation of the medical interventions, but Bregman can increase EU greatly for individual sampling or heteroscedastic methods. Changing the formula rarely resolves the lack of separation. Agreement between formulas makes a source-specific interpretation less dependent on one decomposition, but it does not validate that interpretation. When the formulas disagree, the scale and distance from chance or zero should be inspected rather than selecting the more favorable decomposition after observing the result.

9.3 Comparison with Prior Work

The findings are consistent with Mucsányi et al. (2024), whose benchmark argues that specialized uncertainty estimates should be evaluated on specialized tasks and that common decomposition formulas do not automatically produce disentangled aleatoric and epistemic estimates. The CIFAR-10 reproduction supports this interpretation: correctness-ranking behaviour can be reproduced, but decomposition values and OOD performance require separate task-specific interpretation. The MHIST results then show the same problem in a medical setting: rater disagreement and data scarcity do not yield one method that cleanly separates both uncertainty sources. The cross-experiment comparison in Table 22 adds that the two formulas usually agree on intervention direction but can differ in magnitude, method ordering, or unstable near-chance OOD results.

This conclusion is also consistent with studies that question whether standard entropy and mutual-information decompositions are sufficient. Wimmer et al. (2023) argue that conditional entropy and mutual information can violate desirable properties as aleatoric and epistemic measures, while de Jong et al. (2024) show that classification uncertainty disentanglement remains imperfect even under controlled manipulations of data size, OOD classes, and label noise. The MHIST result fits that picture: changing the decomposition from IT to Bregman affects the ΔEU re-

sponse under scarcity for some methods, but it does not turn the rater-disagreement signal into a clean two-component separation.

Medical uncertainty studies lead to a similar caution. Lakara and Valdenegro-Toro (2022) report that epistemic uncertainty is useful for OOD detection in medical image translation, while aleatoric uncertainty reflects input noise and should not be expected to serve the same role. Baur et al. (2025) likewise find method- and architecture-dependent uncertainty behaviour in chest X-ray classification, with ensembles performing reliably in their setting but decomposition-based scores still requiring careful interpretation. The medical experiments here follow the same broad pattern: usefulness depends on the task used to probe the score.

9.4 Implications for Medical Image Classification

For medical image classification, the practical implication is that uncertainty scores should not be reported as if they had one universal meaning. A high score on a histology image may indicate label ambiguity, limited model knowledge, distribution shift, or simply a likely error. These possible meanings could lead to different actions, such as expert review, additional data collection, or quality control, but these experiments do not validate any such decision rule. Medical-imaging reviews and clinical prediction reporting guidance support a broader reporting principle: uncertainty estimates should be presented together with their evaluation setting, intended use, and limitations (Lambert et al., 2024; Zou et al., 2023; Huang et al., 2024; Collins et al., 2024).

Probe-specific reporting follows from these distinctions. Correctness AUROC should be interpreted as error-ranking usefulness. Rater-disagreement correlation measures association with a diagnostic-ambiguity proxy, not aleatoric uncertainty itself. Scarcity shifts measure response to reduced training information, while also including possible optimization and sampling effects. Corruption AUROC should be interpreted as clean-versus-corrupted ranking under a synthetic shift, not as a pure measure of epistemic uncertainty. No single metric is sufficient to claim clinical reliability or full uncertainty disentanglement.

9.5 Limitations

PathMNIST coverage differs between trainable and post-hoc methods, and not every method has the same corruption evaluation. The PathMNIST-C severity axis covers five synthetic corruption strengths, but these remain simulated shifts rather than acquisition-site, scanner, or patient-population changes. PathMNIST is a controlled test environment, not evidence that the findings will hold under real medical shifts or clinical use.

The medical transfer also uses fixed hyperparameters rather than medical-dataset-specific tuning. This design tests the recovered benchmark settings rather than the best possible version of each method. Some methods may underperform because the

fixed CIFAR-style settings are not ideal for histopathology. This choice makes comparisons more controlled, but it limits claims about the best achievable performance of each method.

The uncertainty targets themselves are proxies. Rater disagreement is an available aleatoric proxy, but it is not a direct observation of aleatoric uncertainty: experts can disagree because of image ambiguity, annotation practice, class definition, or limited information. Data scarcity approximates epistemic uncertainty, but reducing the training split can also change optimization and class representation. Corruptions approximate distribution shift, but they do not cover all clinically relevant staining, scanner, site, or preparation differences. These limitations constrain the interpretation of each proxy.

Most method–condition cells are based on one completed training run. Repeated seeds are not available for the full intervention matrix, so the reported counts describe the observed runs rather than the probability that a method would show the same ordering after retraining. The frequent wrong-direction responses weigh against consistent separation in this evaluation, but small gaps near the diagonal may change under another seed. No confidence intervals are attached to the cross-method success counts. The reported Spearman correlations and AUROCs are likewise interpreted descriptively. Image-level resampling alone would not capture variation from retraining the models, so model-level uncertainty would require repeated training runs.

9.6 Outlook

The main open question is why AU and EU often respond to the same intervention, or in the opposite order from the intended one. The first step should be to repeat the scarcity, label-noise, ambiguity, and corruption experiments across several seeds. This would show which wrong-direction responses are stable and which small differences are due to training variation. Reporting confidence intervals would also prevent minor gaps near the equal-response diagonal from being interpreted as separation.

Further experiments should then vary one uncertainty source while controlling the others more closely. Repeated expert labels could provide an ambiguity target, matched removal of training cases could vary training support, and data from different hospitals, scanners, staining protocols, or patient populations could provide medical distribution shifts. Combining these factors in a controlled design would reveal whether a component remains selective when two sources occur together. It would also help distinguish genuine source sensitivity from a general response to difficult images.

The disagreement between some information-theoretic and Bregman results also suggests that changing the decomposition formula alone is unlikely to provide a general solution. A useful next comparison would test the standard AU–EU split against separate, source-specific estimators, such as rater-ambiguity prediction, feature-space

support, and acquisition-shift detection. Each output would need its own validation target rather than receiving an aleatoric or epistemic interpretation from its name alone (Mucsányi et al., 2024).

Only after stable source-specific behaviour has been established should the signals be evaluated in clinical workflows. A multi-site study could examine whether they support different actions, such as additional expert review for ambiguous cases, quality control for shifted inputs, or data collection for regions with weak training support. Such a study would require repeated runs, confidence intervals, subgroup analyses, and external validation before claims about clinical usefulness could be made (Collins et al., 2024).

10 Conclusion

The study asked whether uncertainty methods separate aleatoric ambiguity, epistemic missing evidence, and distribution shift when a recovered benchmark protocol is transferred to medical image data. The methods often rank difficult cases usefully, but do not consistently separate the underlying uncertainty sources.

The CIFAR-10 reproduction provides the implementation check for transferring the protocol. On PathMNIST, synthetic label noise produces predominantly aleatoric responses, whereas data scarcity does not yield consistently epistemic-dominant behaviour. PathMNIST-C likewise provides little consistent decomposition-based shift signal. MHIST shows only partial separation between rater disagreement and scarcity. The information-theoretic and Bregman decompositions usually agree qualitatively, so changing the formula does not resolve the broader lack of source separation. Because repeated seeds are not available for the complete experiment matrix, these findings describe observed runs rather than population-level method effects.

For medical image classification, no single method is supported as a globally best choice. An uncertainty report should state what a score is intended to detect, which probe evaluates that interpretation, and which sources remain entangled. The experiments support probe-specific usefulness, but not clean aleatoric–epistemic disentanglement across the full method set.

A PathMNIST Method Parameters

This appendix records the parameters that materially affect reproduction of the PathMNIST experiments. Runtime-specific settings such as local data directories, checkpoint file names, download flags, and cluster execution details are omitted. Unless Tables 24 and 25 state otherwise, the shared defaults in Table 23 apply. In the override tables, the implementation line gives the method selector and its training or evaluation objective; "epochs 0" denotes a post-hoc evaluation without backbone retraining.

Table 23: Shared PathMNIST defaults.

Setting	Value
Dataset	`hard/pathmnist`; `hard/pathmnist-c` for the corruption-based OOD evaluation
Backbone	`timm/resnet_50`
Image preprocessing	Image size `224`; normalization mean `0.485,0.456,0.406`; normalization standard deviation `0.229,0.224,0.225`
Optimizer	`adamw` with weight decay `2e-5`
Base learning rate	`0.001`
Batching	Batch size `128`; accumulation steps `16`; effective batch size `2048`
Training length	`90` epochs for directly trained methods
Seed	`42`
Data scarcity variants	Train subsets `1.0`, `0.5`, and `0.1`; validation and test sets remain fixed

Table 24: PathMNIST overrides for directly trained methods.

Method and implementation	Method-specific settings
CE baseline `ce-baseline / cross-entropy`	Shared defaults only; see Section 5.2.1.
Correctness Prediction `correctness-prediction`	Uncertainty loss weight 0.01; see Section 5.3.1.
Loss Prediction `loss-prediction`	Uncertainty loss weight 0.01; detached auxiliary-loss target; see Section 5.3.2.
Deep Correctness Prediction `deep-correctness-prediction / correctness-prediction`	Uncertainty loss weight 0.01; auxiliary feature taps `^(act1|layer[1-4])$`; see Section 5.3.3.
Deep Loss Prediction `deep-loss-prediction / loss-prediction`	Uncertainty loss weight 0.01; detached auxiliary-loss target; auxiliary feature taps `^(act1|layer[1-4])$`; see Section 5.3.4.
Shallow Ensemble `shallow-ensemble / bma-cross-entropy`	10 heads; see Section 5.4.3.
MC Dropout `mc-dropout / cross-entropy`	Dropout probability 0.05; 10 Monte Carlo samples; see Section 5.4.4.
DUQ `duq`	RBF length scale 0.1; EMA momentum 0.999; gradient-penalty weight 0.75; batch size 32; accumulation steps 64; see Section 5.5.2.
PostNet `postnet / uce`	Latent dimension 6; 256 hidden features; 6 density components; UCE regularization factor 10^{-5}; see Section 5.5.4.
SNGP `sngp / cross-entropy`	Spectral normalization: 1 iteration, bound 6; 1024 ORF random features; kernel scale 1; output bias 0; covariance momentum -1; ridge penalty 1; GP input dimension 128; see Section 5.6.2.
HET `het-xl / bma-cross-entropy`	HET compatibility flag `use-het`; 10 Monte Carlo samples; temperature 1.5; see Section 5.7.1.
HET-XL `het-xl / bma-cross-entropy`	Matrix rank 15; 10 Monte Carlo samples; temperature 1.5; see Section 5.7.2.
HetClassNN `hetclassnn / bma-cross-entropy`	Dropout probability 0.05; 10 Monte Carlo samples; 1000 integral Monte Carlo samples; see Section 5.7.3.
Evidential Deep Learning `edl`	Start epoch 0; scaler 1; evidence activation `exp`; see Section 5.7.4.

Table 25: Post-hoc and checkpoint-initialized PathMNIST method overrides.

Method and implementation	Method-specific settings
Temperature Scaling `temperature-scaling / calibration`	Cross-entropy checkpoint; no backbone retraining; epochs 0; see Section 5.2.2.
Deep Ensemble `deep-ensemble / ensemble`	Five independently trained CE member checkpoints; epochs 0; see Section 5.4.1.
Fast Deep Ensemble `fast-deep-ensemble / ensemble`	Five available member checkpoints; epochs 0; see Section 5.4.2.
SWAG `swag / fine-tuning`	Pretrained checkpoint; 40 fine-tuning epochs; low-rank covariance with maximum rank 20; 4 checkpoint samples per epoch; see Section 5.4.5.
DDU `ddu / density estimation`	Cross-entropy checkpoint; epochs 0; spectral normalization with 1 iteration and bound 6; spectral-normalized batch normalization; see Section 5.5.1.
Mahalanobis `mahalanobis / feature statistics`	Cross-entropy checkpoint; epochs 0; input perturbation magnitude 0.001; feature taps `^(act1|layer[1-4])$`; see Section 5.5.3.
Laplace `laplace / approximation`	Cross-entropy checkpoint; epochs 0; GLM predictive type; Kronecker-factored Hessian; 10 Monte Carlo samples; 50 cross-validation samples; see Section 5.6.1.

B CIFAR-10-to-MHIST Configuration Transfer

The MHIST experiments transfer selected CIFAR-10 configurations from the original benchmark. Dataset-specific fields are changed to MHIST with two classes; local paths and execution flags are omitted.

Table 26: Settings shared by the repeated MHIST transfer configurations.

Setting	Value
Backbone and input	Wide ResNet 26-10; image size `32`; CIFAR-10 normalization
Training constants	Batch size `128`; seed `42`; Nesterov momentum `0.9`
Augmentation	Horizontal flip probability `0.5`; padding `2`
Warmup and decay	One warmup epoch; warmup learning rate `0.005`; multistep decay rate `0.2`

Table 27: CIFAR-10 configurations transferred to the MHIST experiments. LR is learning rate, MC samples is the number of Monte Carlo samples, and SN denotes spectral normalization. Shared settings are shown in Table 26.

Thesis label	Method	Epochs	LR	Weight decay	MC samples	Schedule	Method-specific parameters
SNGP (SN off)	`sngp`	250	0.0647731871	0.0014056843	1000	Multistep 75,150,200	1024 random features; ORF random features; GP kernel scale 1; GP ridge penalty 1; GP covariance momentum -1; spectral-normalization bound 6, iteration 1; spectral normalization disabled.
SNGP (SN on)	`sngp`	250	0.1053200118	0.0012217236	1000	Multistep 75,150,200	Same GP settings as SNGP (SN off), but spectral normalization enabled.
HET	`het-xl` with `use-het`	200	0.0875579735	0.0002782346	1000	Multistep 60,120,160	`bma-cross-entropy`; matrix rank 6; temperature 1.3; HET compatibility flag enabled.
HET-XL	`het-xl`	200	0.0952229144	0.0031745703	1000	Multistep 60,120,160	`bma-cross-entropy`; matrix rank 6; temperature 1.3; HET compatibility flag disabled.

C Software Availability

The experiment code, configurations, analysis scripts, and reproducibility instructions are available in the following repository:

Repository name: `untangle-medical-uncertainty`
Repository URL:
`https://github.com/FloWgl/untangle-medical-uncertainty`
Author and maintainer: Florian Weigl

The repository extends the original *untangle* benchmark implementation associated with Mucsányi et al. (2024).

D Selected Glossary

This appendix defines supporting terms that recur in tables, result comments, and method descriptions. Core concepts that are introduced in the main text, such as aleatoric and epistemic uncertainty, the information-theoretic and Bregman decompositions, AUROC, calibration metrics, and the individual uncertainty methods are not repeated here.

Auxiliary head An additional output branch attached to a neural network and trained for a side task, such as predicting whether the classifier will be correct or predicting the loss on an input.

Checkpoint A saved model state from training. Post-hoc methods and ensemble methods reuse checkpoints instead of always training a new model from scratch.

Confidence score A scalar summary of how certain a classifier appears to be. A common example is the maximum predicted class probability; low confidence is often treated as high uncertainty.

Diagnostic ambiguity Ambiguity in the medical interpretation of an image. In this thesis it is approximated by pathologist disagreement in MHIST or by controlled label-noise interventions in PathMNIST.

Feature representation An intermediate vector produced by a neural network before the final classifier layer. Distance and density methods use this representation to decide whether an input looks familiar.

Finite sample count The number of examples with valid numeric uncertainty values after excluding non-finite entries such as `NaN` or infinity.

Latent space A learned representation space in which images are encoded as feature vectors rather than raw pixels. Some methods estimate uncertainty by modelling class evidence or density in this space.

Maximum softmax probability The largest class probability after the softmax transformation. Its complement, $1 - \max_k p(y = k \mid x)$, is a simple uncertainty baseline.

Set of methods The uncertainty methods included in a given experiment or result table.

Non-finite value A numerical value that is undefined in ordinary arithmetic, such as `NaN` or infinity. Such entries are excluded before summary statistics are computed.

Post-hoc method A method applied after a base classifier has already been trained. Examples in the thesis include temperature scaling, Mahalanobis, SWAG, DDU, and Laplace variants.

Prior precision A Laplace-related hyperparameter that controls the strength of the prior around the fitted model parameters.

Severity The strength of a corruption in a corruption benchmark. Here, the official PathMNIST-C NPZ files are treated as one fixed corruption set, not as a one-to-five severity sequence.

Wrapper A method layer that adds uncertainty estimation around an existing classifier. Post-hoc methods often act as wrappers because they reuse trained checkpoint weights.

Bibliography

Simon Baur, Wojciech Samek, and Jackie Ma. Benchmarking uncertainty and its disentanglement in multi-label chest x-ray classification. *arXiv preprint arXiv:2508.04457*, 2025. URL https://arxiv.org/abs/2508.04457.

Glenn W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950. doi: 10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2.

Gavin C. Cawley and Nicola L. C. Talbot. On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11:2079–2107, 2010. URL https://www.jmlr.org/papers/v11/cawley10a.html.

Bertrand Charpentier, Daniel Zügner, and Stephan Günnemann. Posterior network: Uncertainty estimation without ood samples via density-based pseudo-counts. In *Advances in Neural Information Processing Systems*, 2020. URL https://proceedings.neurips.cc/paper/2020/hash/0eac690d7059a8de4b48e90f14510391-Abstract.html.

Mark Collier, Basil Mustafa, Effrosyni Kokiopoulou, Rodolphe Jenatton, and Jesse Berent. Correlated input-dependent label noise in large-scale image classification. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1551–1560, 2021. URL https://openaccess.thecvf.com/content/CVPR2021/html/Collier_Correlated_Input-Dependent_Label_Noise_in_Large-Scale_Image_Classification_CVPR_2021_paper.html.

Gary S. Collins, Paula Dhiman, Constanza L. Andaur Navarro, Jie Ma, Lotty Hooft, Johannes B. Reitsma, Patricia Logullo, Andrew L. Beam, Lu Peng, Ben Van Calster, Maarten van Smeden, Richard D. Riley, and Karel G. M. Moons. Tripod+ai statement: Updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*, 385:e078378, 2024. doi: 10.1136/bmj-2023-078378.

Charles Corbière, Nicolas Thome, Avner Bar-Hen, Matthieu Cord, and Patrick Pérez. Addressing failure prediction by learning model confidence. In *Advances in Neural Information Processing Systems*, 2019. URL https://proceedings.neurips.cc/paper/2019/hash/757f843a169cc678064d9530d12a1881-Abstract.html.

Jesse Davis and Mark Goadrich. The relationship between precision-recall and roc curves. In *Proceedings of the 23rd International Conference on Machine Learning*, pages 233–240, 2006. doi: 10.1145/1143844.1143874.

Ivo Pascal de Jong, Andreea Ioana Sburlea, and Matias Valdenegro-Toro. How disentangled are your classification uncertainties? *arXiv preprint arXiv:2408.12175*, 2024. URL https://arxiv.org/abs/2408.12175.

Francesco Di Salvo, Sebastian Doerrich, and Christian Ledig. Medmnist-c: Comprehensive benchmark and improved classifier robustness by simulating realistic image corruptions. *arXiv preprint arXiv:2406.17536*, 2024. doi: 10.48550/arXiv.2406.17536. URL https://arxiv.org/abs/2406.17536.

Tom Fawcett. An introduction to roc analysis. *Pattern Recognition Letters*, 27(8): 861–874, 2006. doi: 10.1016/j.patrec.2005.10.010.

Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *International Conference on Machine Learning*, 2016. URL https://proceedings.mlr.press/v48/gal16.html.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning*, 2017. URL https://proceedings.mlr.press/v70/guo17a.html.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2016. URL https://openaccess.thecvf.com/content_cvpr_2016/html/He_Deep_Residual_Learning_CVPR_2016_paper.html.

Dan Hendrycks and Thomas Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. In *International Conference on Learning Representations*, 2019. URL https://openreview.net/forum?id=HJz6tiCqYm.

Ling Huang, Su Ruan, Yucheng Xing, and Mengling Feng. A review of uncertainty quantification in medical image analysis: Probabilistic and non-probabilistic methods. *Medical Image Analysis*, page 103223, 2024. doi: 10.1016/j.media.2024.103223. URL https://doi.org/10.1016/j.media.2024.103223.

D. R. Jaravaza, J. M. Rigby, J. P. Houghton, and O. P. Houghton. Hyperplastic polyp or sessile serrated lesion? the contribution of serial sections to reclassification. *Diagnostic Pathology*, 15(1):140, 2020. doi: 10.1186/s13000-020-01057-0.

Jakob Nikolas Kather, Johannes Krisam, Pornpimol Charoentong, Tom Luedde, Esther Herpel, Cleo-Aron Weis, Timo Gaiser, Alexander Marx, Nektarios A. Valous, Dyke Ferber, Lina Jansen, Constantino C. Reyes-Aldasoro, Inka Zörnig, Dirk Jäger, Hermann Brenner, Jenny Chang-Claude, Michael Hoffmeister, and Niels Halama. Predicting survival from colorectal cancer histology slides using deep learning: A retrospective multicenter study. *PLOS Medicine*, 16(1):e1002730, 2019. doi: 10.1371/journal.pmed.1002730. URL https://doi.org/10.1371/journal.pmed.1002730.

Alex Kendall and Yarin Gal. What uncertainties do we need in bayesian deep learning for computer vision? In *Advances in Neural Information Processing Systems*, 2017. URL https://proceedings.neurips.cc/paper/2017/hash/2650d6089a6d640c5e85b2b88265dc2b-Abstract.html.

Kumud Lakara and Matias Valdenegro-Toro. Disentangled uncertainty and out of distribution detection in medical generative models. *arXiv preprint arXiv:2211.06250*, 2022. URL https://arxiv.org/abs/2211.06250.

Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems*, 2017. URL https://proceedings.neurips.cc/paper/2017/hash/9ef2ed4b7fd2c810847ffa5fa85bce38-Abstract.html.

Benjamin Lambert, Florence Forbes, Alan Tucholka, Senan Doyle, Harmonie Dehaene, and Michel Dojat. Trustworthy clinical ai solutions: A unified review of uncertainty quantification in deep learning models for medical image analysis. *Artificial Intelligence in Medicine*, page 102830, 2024. doi: 10.1016/j.artmed.2024.102830. URL https://doi.org/10.1016/j.artmed.2024.102830.

Kimin Lee, Kibok Lee, Honglak Lee, and Jinwoo Shin. A simple unified framework for detecting out-of-distribution samples and adversarial attacks. In *Advances in Neural Information Processing Systems*, 2018. URL https://proceedings.neurips.cc/paper/2018/hash/abdeb6f575ac5c6676b747bca8d09cc2-Abstract.html.

Jeremiah Zhe Liu, Zi Lin, Shreyas Padhy, Dustin Tran, Tania Bedrax-Weiss, and Balaji Lakshminarayanan. Simple and principled uncertainty estimation with deterministic deep learning via distance awareness. In *Advances in Neural Information Processing Systems*, 2020. URL https://proceedings.neurips.cc/paper/2020/hash/543e83748234f7cbab21aa0ade66565f-Abstract.html.

Wesley J. Maddox, Timur Garipov, Pavel Izmailov, Dmitry Vetrov, and Andrew Gordon Wilson. A simple baseline for bayesian uncertainty in deep learning. In *Advances in Neural Information Processing Systems*, 2019. URL https://proceedings.neurips.cc/paper/2019/hash/118921efba23fc329e6560b27861f0c2-Abstract.html.

Bálint Mucsányi, Michael Kirchhof, and Seong Joon Oh. Benchmarking uncertainty disentanglement: Specialized uncertainties for specialized tasks. In *Advances in Neural Information Processing Systems*, volume 37, pages 50972–51038, 2024. doi: 10.52202/079017-1614. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/5afa9cb1e917b898ad418216dc726fbd-Abstract-Datasets_and_Benchmarks_Track.html.

Jishnu Mukhoti, Andreas Kirsch, Joost van Amersfoort, Philip H. S. Torr, and Yarin Gal. Deep deterministic uncertainty: A new simple baseline. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023. URL https://openaccess.thecvf.com/content/CVPR2023/html/Mukhoti_Deep_Deterministic_Uncertainty_A_New_Simple_Baseline_CVPR_2023_paper.html.

Mahdi Pakdaman Naeini, Gregory F. Cooper, and Milos Hauskrecht. Obtaining well calibrated probabilities using bayesian binning. In *Proceedings of the Twenty-Ninth AAAI Conference on Artificial Intelligence*, pages 2901–2907, 2015. URL https://ojs.aaai.org/index.php/AAAI/article/view/9602.

Curtis G. Northcutt, Lu Jiang, and Isaac L. Chuang. Confident learning: Estimating uncertainty in dataset labels. *Journal of Artificial Intelligence Research*, 70:1373–1411, 2021. doi: 10.1613/jair.1.12125. URL https://jair.org/index.php/jair/article/view/12125.

Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, D. Sculley, Sebastian Nowozin, Joshua V. Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model's uncertainty? evaluating predictive uncertainty under dataset shift. In *Advances in Neural Information Processing Systems*, 2019. URL https://proceedings.neurips.cc/paper/2019/hash/8558cb408c1d76621371888657d2eb1d-Abstract.html.

Hippolyt Ritter, Aleksandar Botev, and David Barber. A scalable laplace approximation for neural networks. In *International Conference on Learning Representations*, 2018. URL https://openreview.net/forum?id=Skdvd2xAZ.

Murat Sensoy, Lance Kaplan, and Melih Kandemir. Evidential deep learning to quantify classification uncertainty. In *Advances in Neural Information Processing Systems*, 2018. URL https://proceedings.neurips.cc/paper/2018/hash/a981f2b708044d6fb4a71a1463242520-Abstract.html.

Matias Valdenegro-Toro and Daniel Saromo. A deeper look into aleatoric and epistemic uncertainty disentanglement. In *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pages 1508–1516, 2022. doi: 10.1109/CVPRW56347.2022.00157. URL https://arxiv.org/abs/2204.09308.

Joost van Amersfoort, Lewis Smith, Yee Whye Teh, and Yarin Gal. Uncertainty estimation using a single deep deterministic neural network. In *International Conference on Machine Learning*, 2020. URL https://proceedings.mlr.press/v119/van-amersfoort20a.html.

Jerry Wei, Arief Suriawinata, Bing Ren, Xiaoying Liu, Mikhail Lisovsky, Louis Vaickus, Charles Brown, Michael Baker, Naofumi Tomita, Lorenzo Torresani, Jason Wei, and Saeed Hassanpour. A petri dish for histopathology image analysis. In *Artificial Intelligence in Medicine*, pages 11–24. Springer, 2021. URL https://arxiv.org/abs/2101.12355.

Lisa Wimmer, Yusuf Sale, Paul Hofman, Bernd Bischl, and Eyke Hüllermeier. Quantifying aleatoric and epistemic uncertainty in machine learning: Are conditional entropy and mutual information appropriate measures? In *Proceedings of the Thirty-Ninth Conference on Uncertainty in Artificial Intelligence*, volume 216 of *Proceedings of Machine Learning Research*, pages 2282–2292, 2023. URL https://proceedings.mlr.press/v216/wimmer23a.html.

Jiancheng Yang, Rui Shi, Donglai Wei, Zequan Liu, Lin Zhao, Bilian Ke, Hanspeter Pfister, and Bingbing Ni. Medmnist v2 - a large-scale lightweight benchmark for 2d and 3d biomedical image classification. *Scientific Data*, 10(1):41, 2023. doi: 10.1038/s41597-022-01721-8. URL `https://www.nature.com/articles/s41597-022-01721-8`.

Donggeun Yoo and In So Kweon. Learning loss for active learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019. URL `https://openaccess.thecvf.com/content_CVPR_2019/html/Yoo_Learning_Loss_for_Active_Learning_CVPR_2019_paper.html`.

Ke Zou, Zhihao Chen, Xuedong Yuan, Xiaojing Shen, Meng Wang, and Huazhu Fu. A review of uncertainty estimation and its application in medical imaging. *Meta-Radiology*, 1:100003, 2023. doi: 10.1016/j.metrad.2023.100003. URL `https://doi.org/10.1016/j.metrad.2023.100003`.

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